

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
C	0000897412	PRODUCTION RELEASED	2010-04-26

SCHEM MLB_LD0 K87
SCRATCH0
04/26/2010

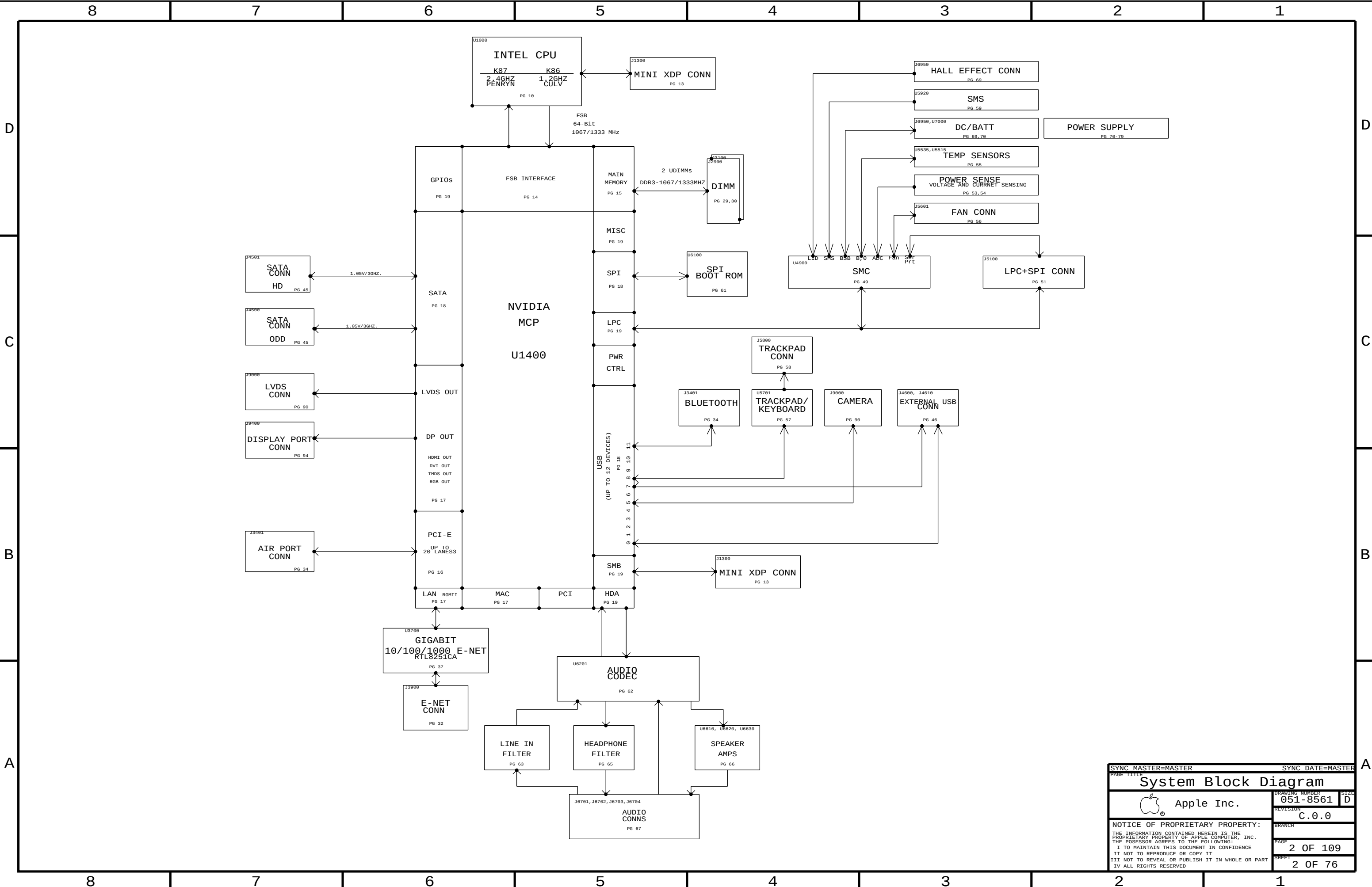
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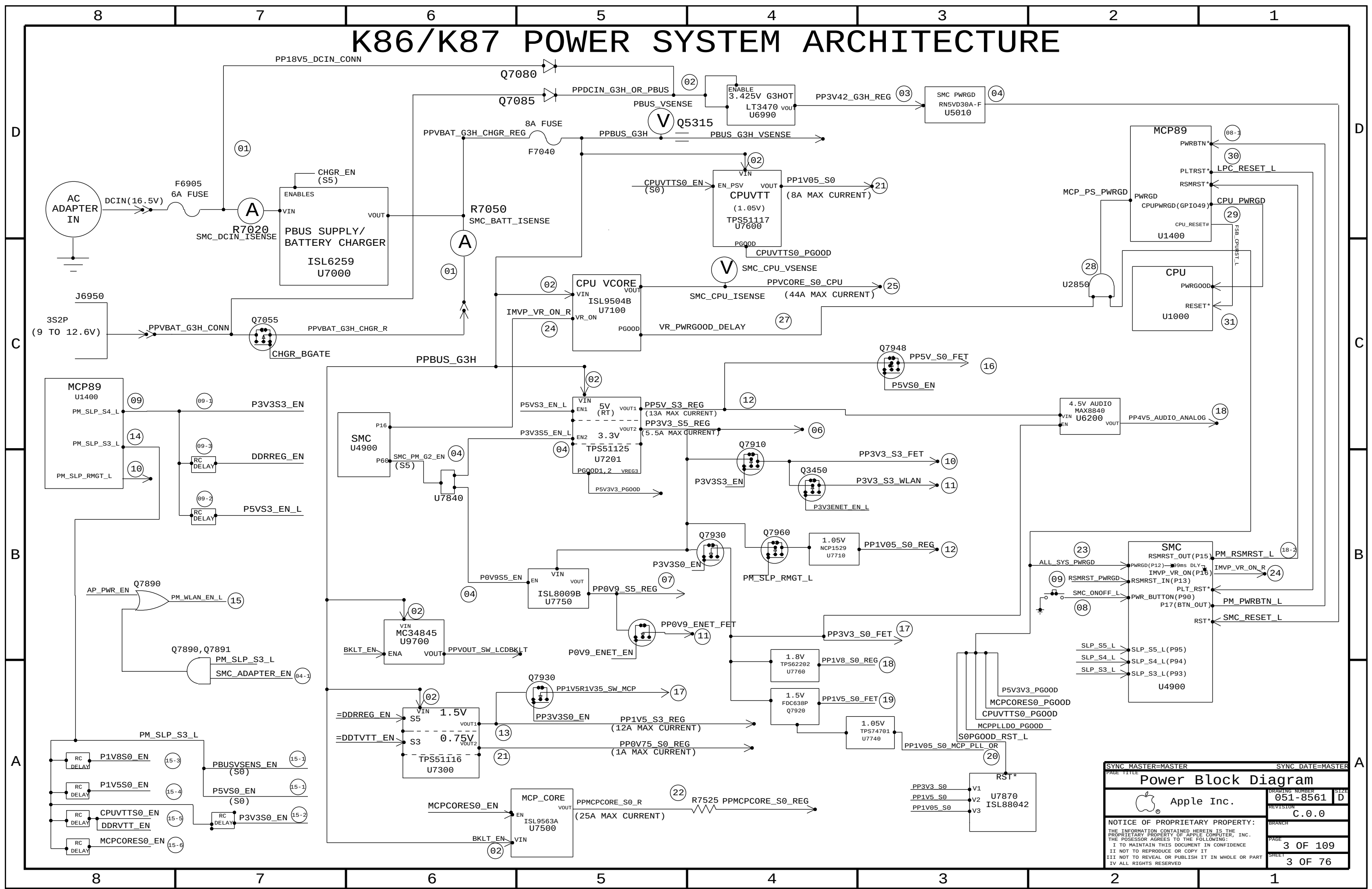
Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-8561	1	SCHEM, MLB_LDO, K87	SCH	CRITICAL	
820-2877	1	PCBF, MLB_LDO, K87	PCB	CRITICAL	

DRAWING TITLE		DRAWING NUMBER		SIZE	
SCHEM, MLB_LDO, SCRATCHO, K87		051-8561		D	
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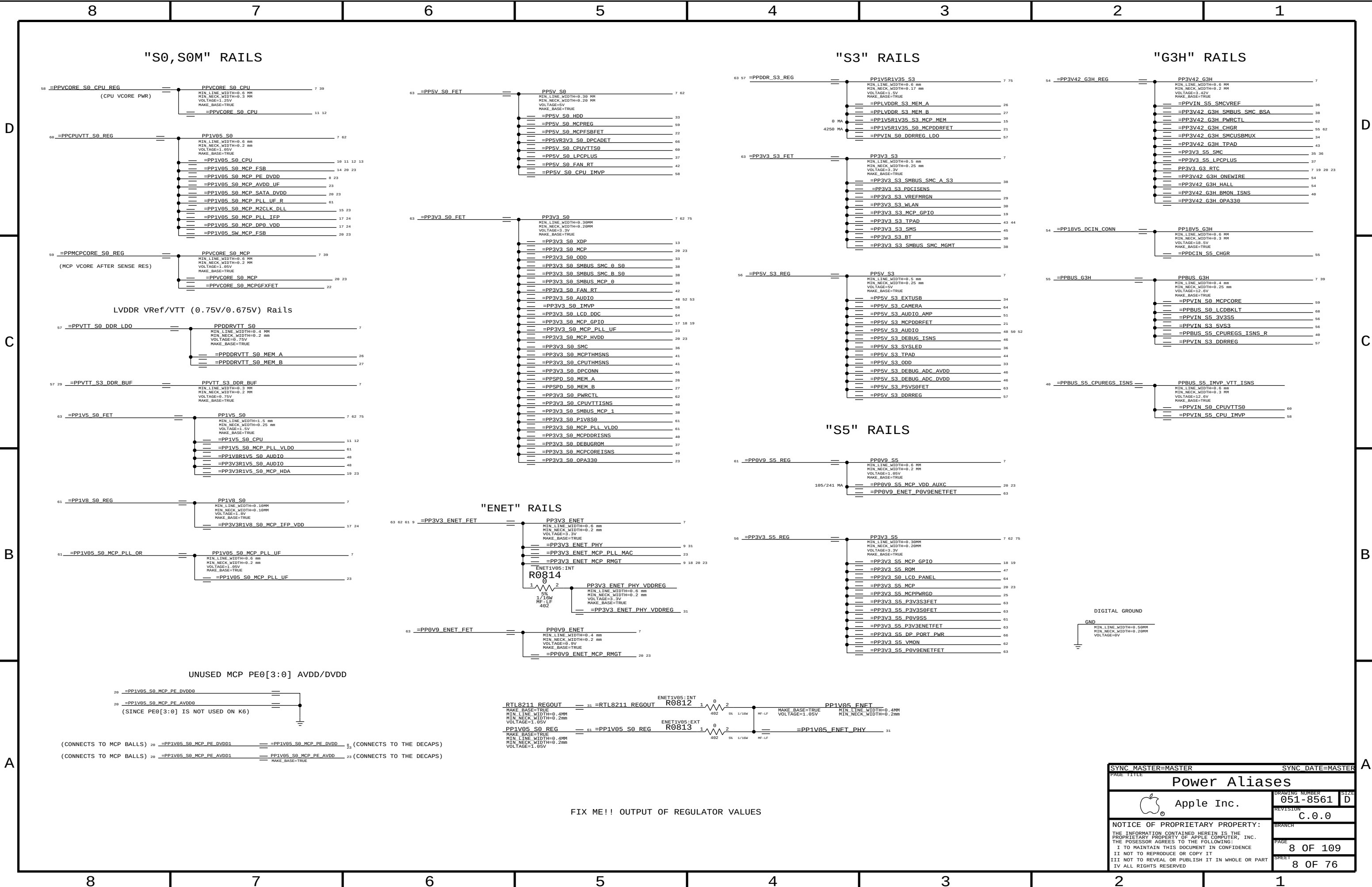
K86/K87 POWER SYSTEM ARCHITECTURE



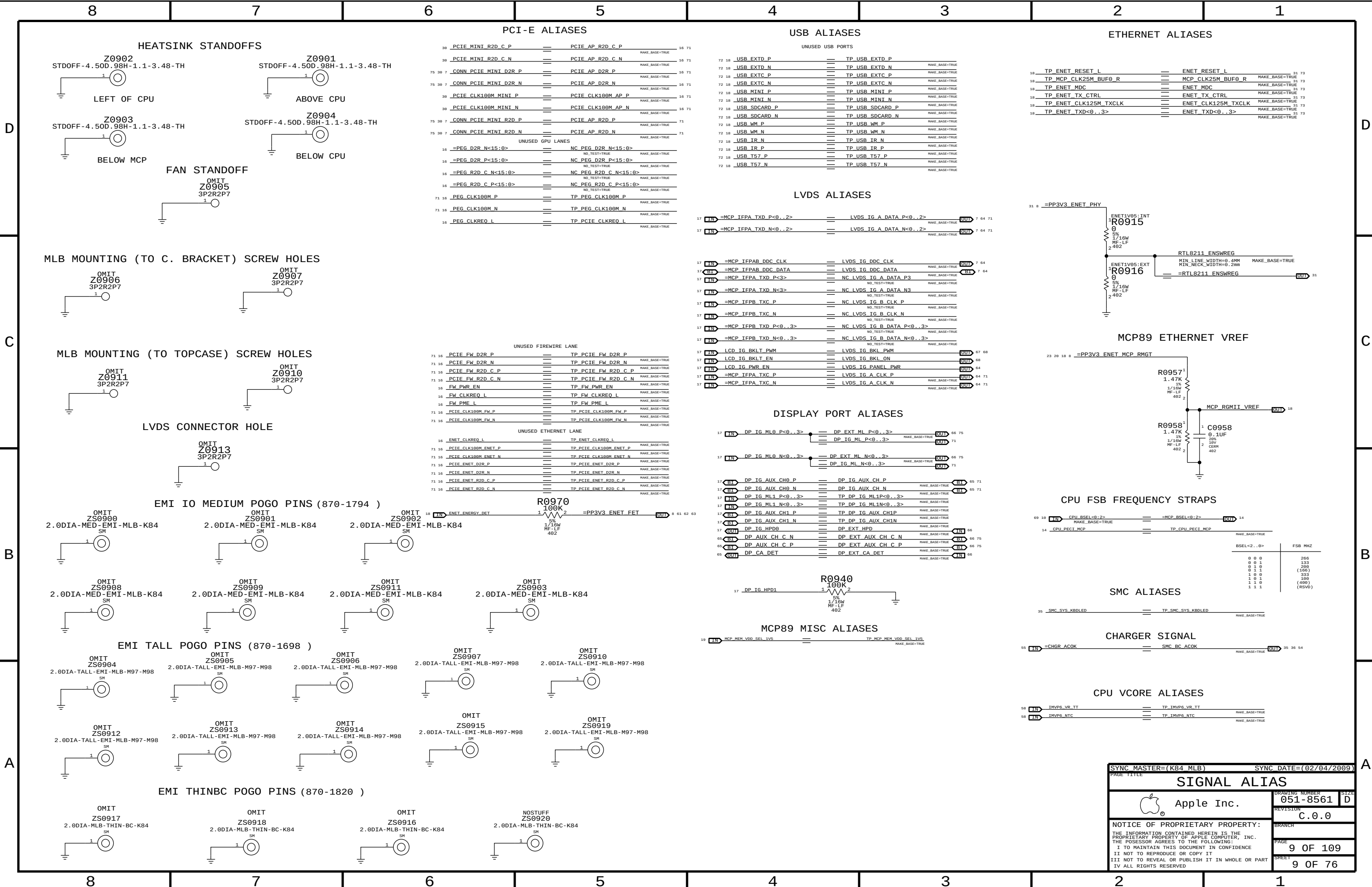
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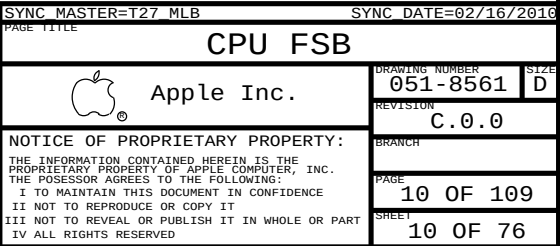
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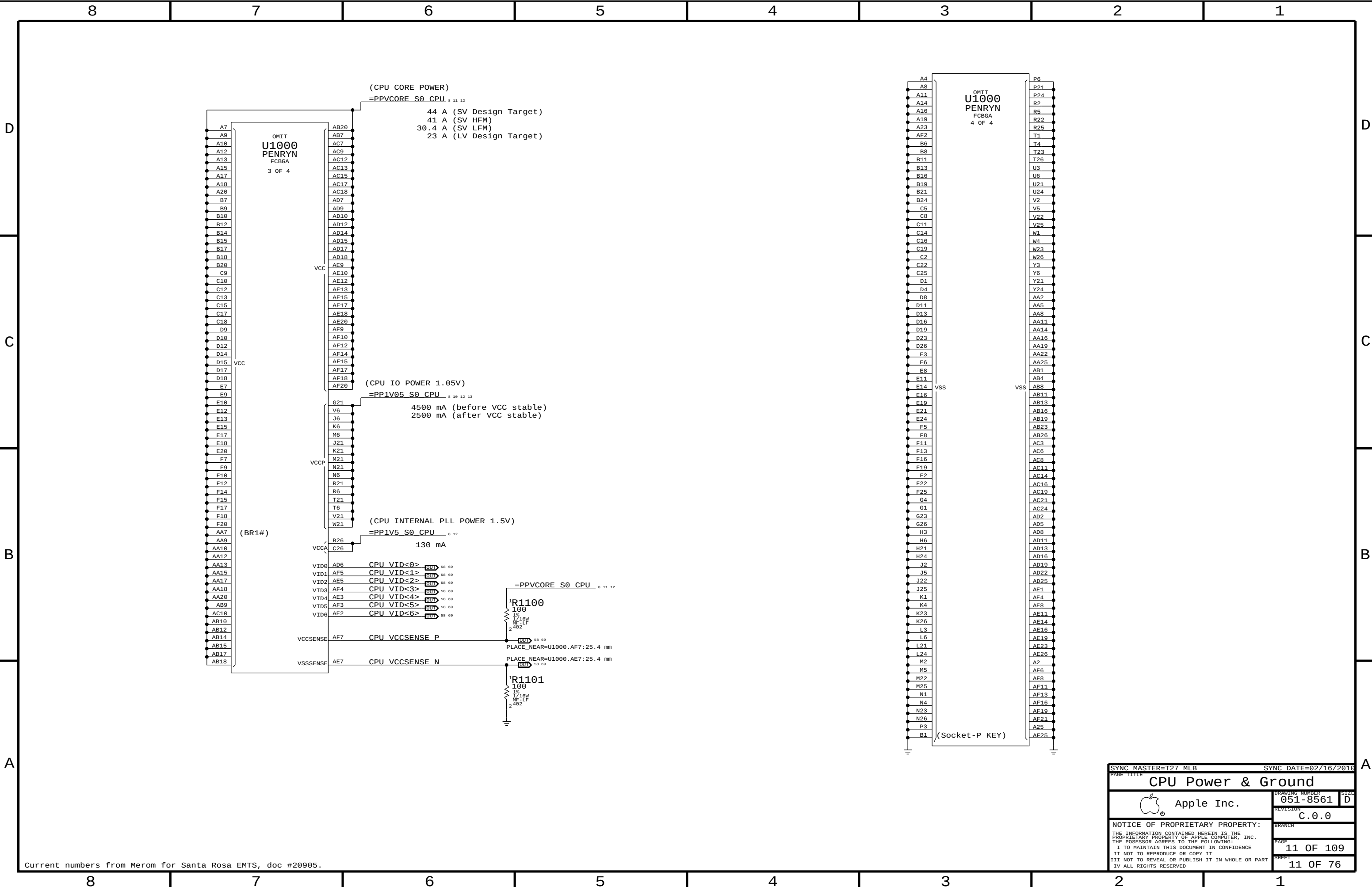


SYNC MASTER=(K84 MLB)

SYNC DATE=(02/04/2009)

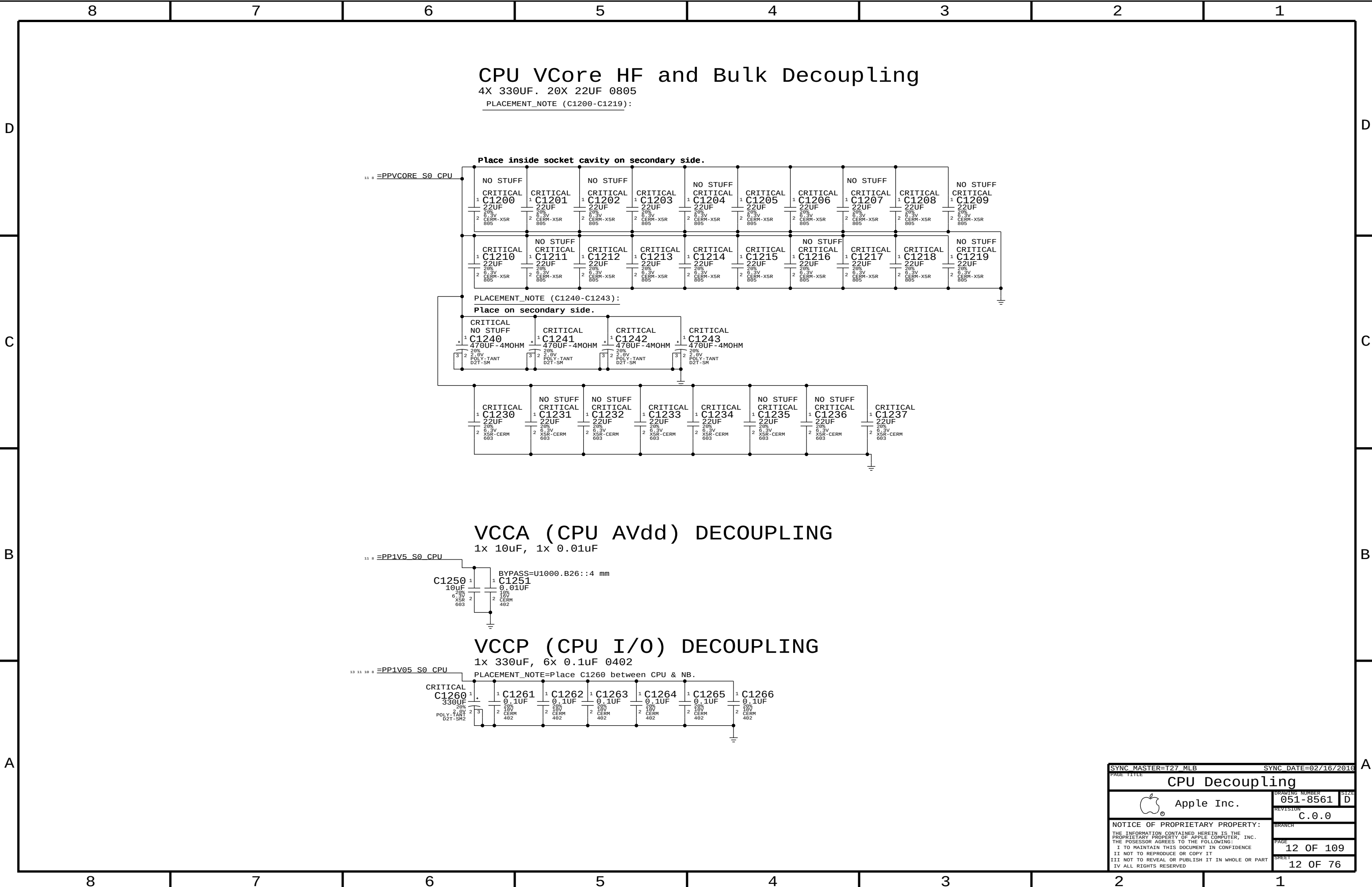
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SIGNAL ALIAS		
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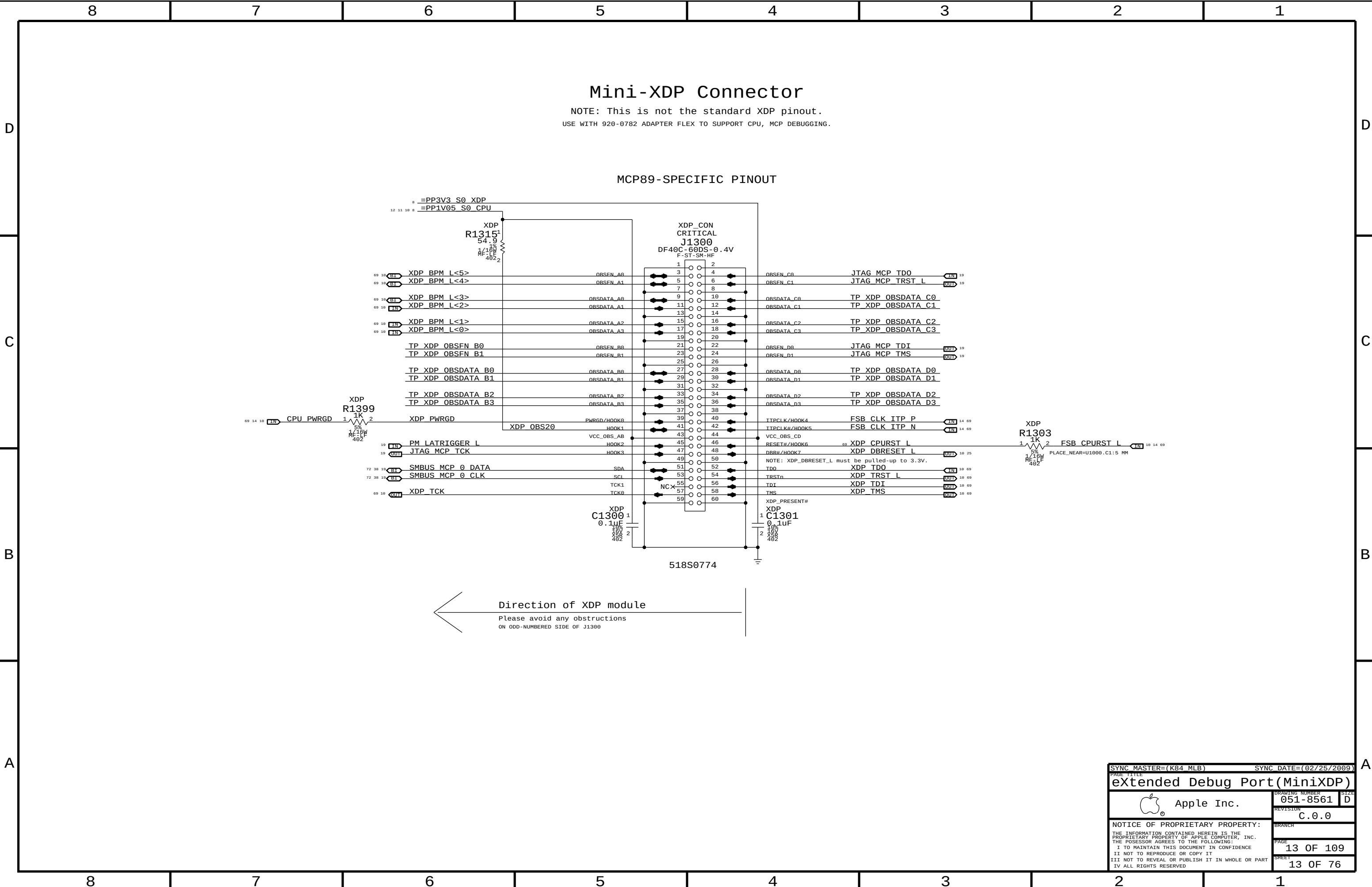


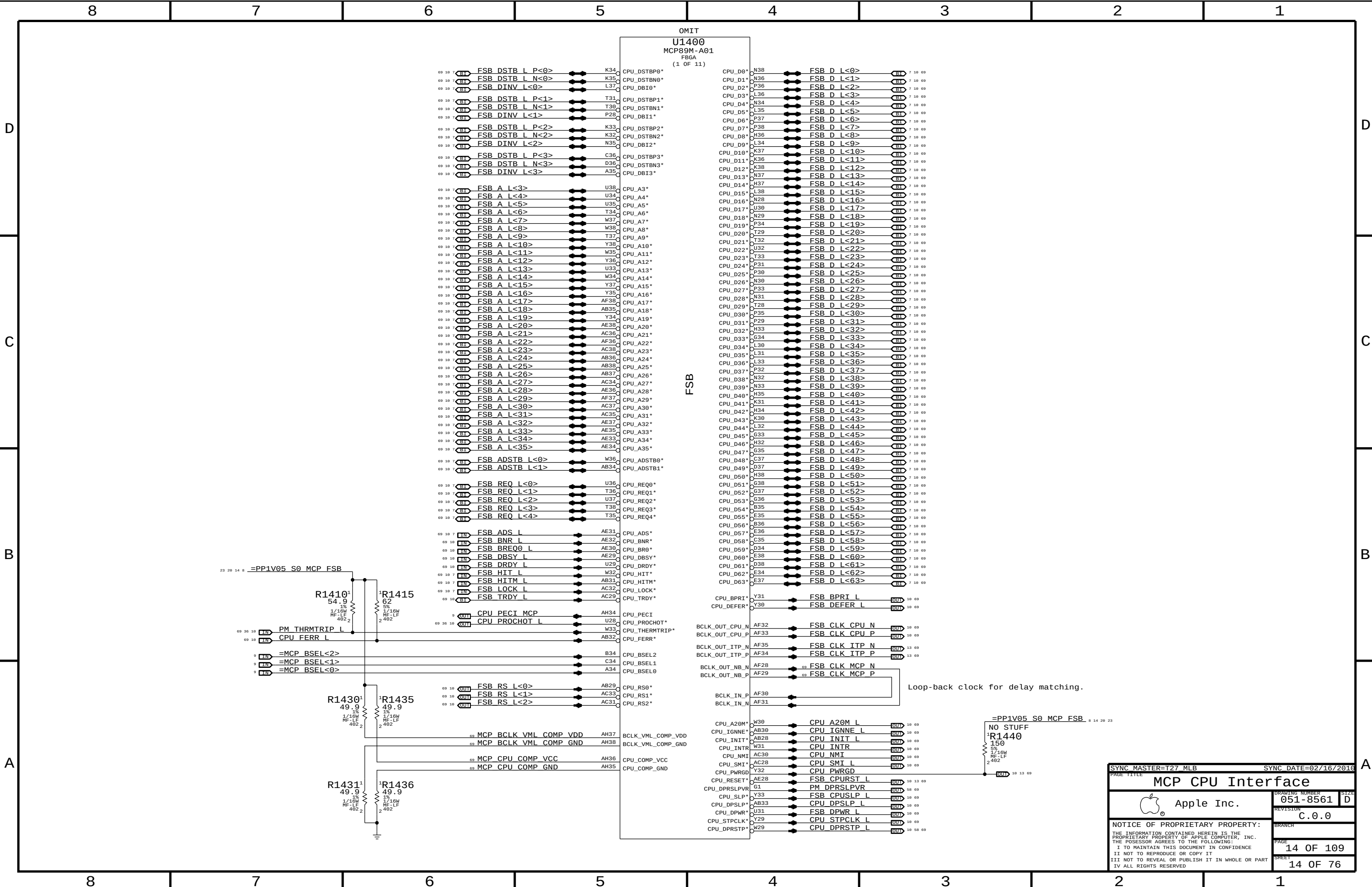


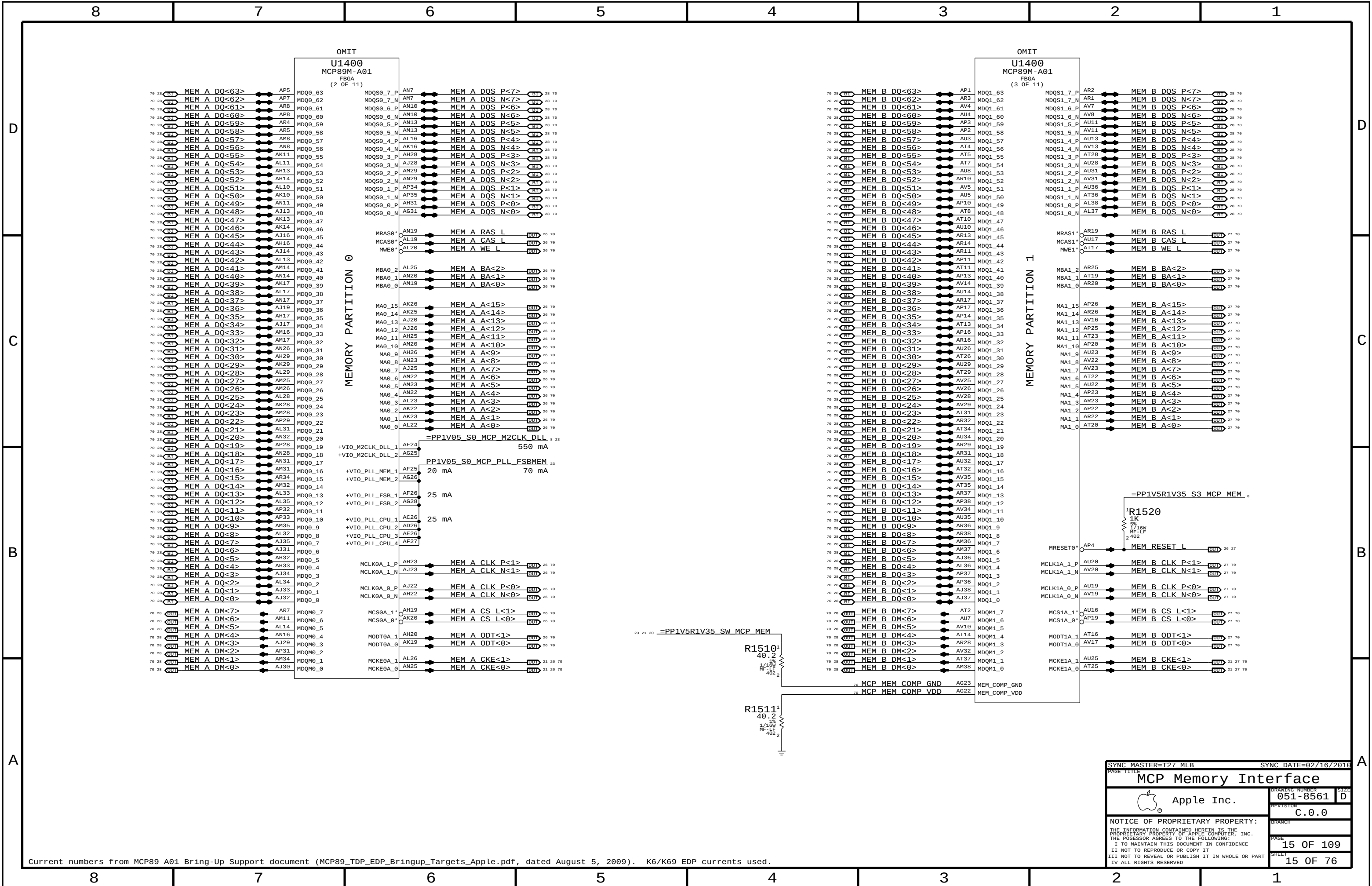
Current numbers from Merom for Santa Rosa EMTS, doc #20905.

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PAGE TITLE			
CPU Power & Ground			
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PAGE TITLE		MCP Memory Interface	
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D

C

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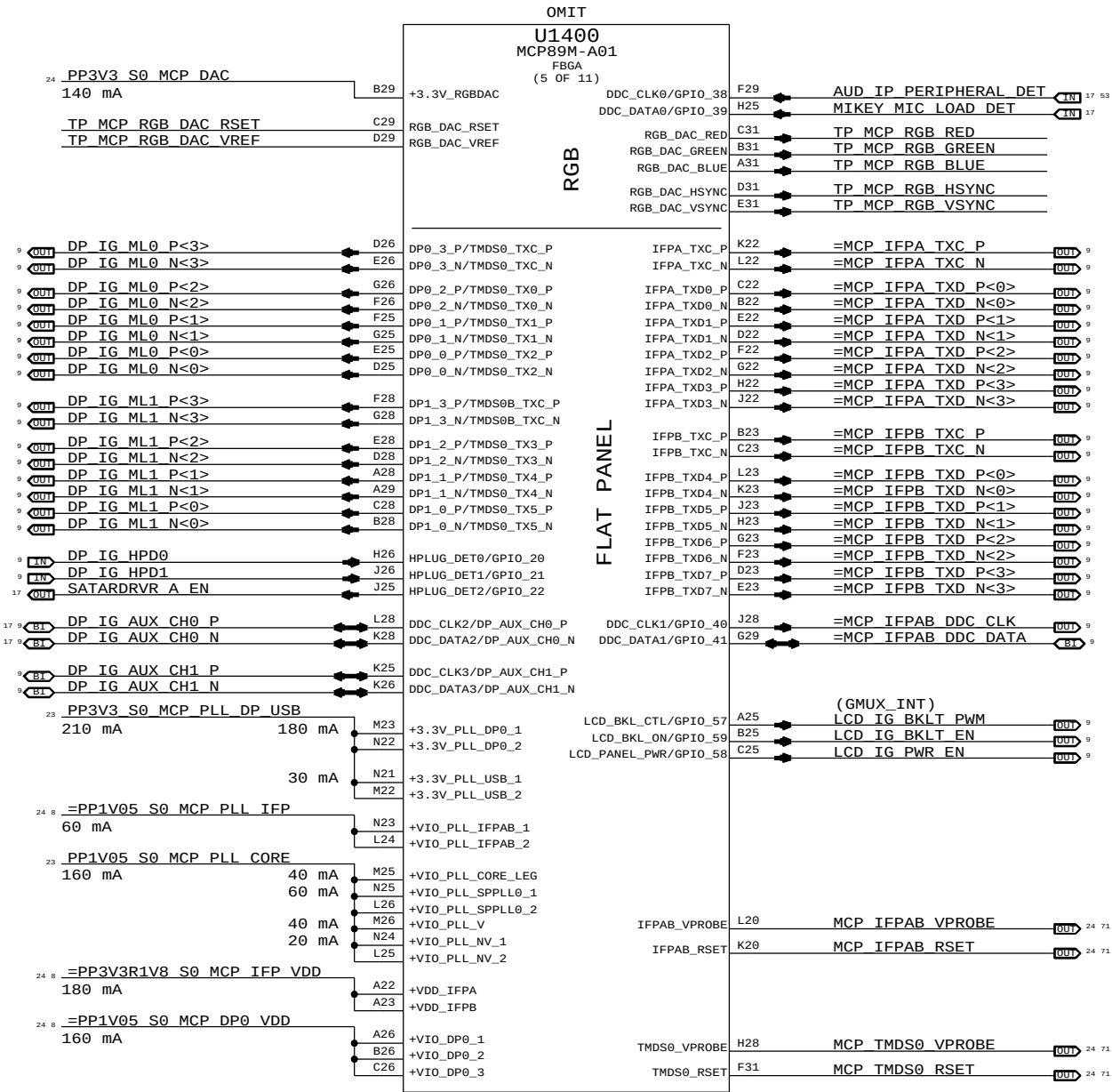
A

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B

A



NOTE: 100K pull-downs required if
HPLUG_DET0/HPLUG_DET1 are not used.

RGB DAC Disable:
Okay to float all RGB_DAC signals.
DDC_CLK0/DDC_DATA0 pull-ups still required (or use as GPIOs).
Connect +3.3V_RGBDAC pin to GND.

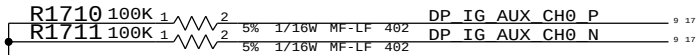
NOTE: No Composite/S-Video/Component Video support on MCP89

Interface Mode		
MCP Signal	TMDS/HDMI	LVDS
=MCP_IFPA_TXC_P/N	TMDS_IG_TXC_P/N	LVDS_IG_A_CLK_P/N
=MCP_IFPA_TXD_P/N<0>	TMDS_IG_TXD_P/N<0>	LVDS_IG_A_DATA_P/N<0>
=MCP_IFPA_TXD_P/N<1>	TMDS_IG_TXD_P/N<1>	LVDS_IG_A_DATA_P/N<1>
=MCP_IFPA_TXD_P/N<2>	TMDS_IG_TXD_P/N<2>	LVDS_IG_A_DATA_P/N<2>
=MCP_IFPA_TXD_P/N<3>	(UNUSED)	LVDS_IG_A_DATA_P/N<3>
=MCP_IFPB_TXC_P/N	(UNUSED)	LVDS_IG_B_CLK_P/N
=MCP_IFPB_TXD_P/N<0>	TMDS_IG_TXD_P/N<3>	LVDS_IG_B_DATA_P/N<0>
=MCP_IFPB_TXD_P/N<1>	TMDS_IG_TXD_P/N<4>	LVDS_IG_B_DATA_P/N<1>
=MCP_IFPB_TXD_P/N<2>	TMDS_IG_TXD_P/N<5>	LVDS_IG_B_DATA_P/N<2>
=MCP_IFPB_TXD_P/N<3>	(UNUSED)	LVDS_IG_B_DATA_P/N<3>
=MCP_IFPAB_DDC_CLK	TMDS_IG_DDC_CLK	LVDS_IG_DDC_CLK
=MCP_IFPAB_DDC_DATA	TMDS_IG_DDC_DATA	LVDS_IG_DDC_DATA

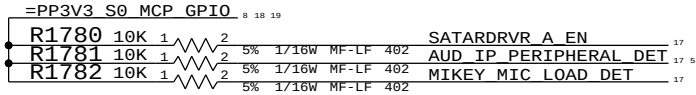
LVDS: Power +VDD_IFPx at 1.8V
TMDS: Power +VDD_IFPx at 3.3V
NOTE: TMDS/HDMI not supported on IFPA/B for MCP89 A01.

DDC Mode Pull-downs

NOTE: DP_AUX_CH1 also requires pull-downs if used for dual-mode DisplayPort (DP++). If unused no pulls are necessary, if used for TMDS/HDMI only then only pull-ups are necessary.

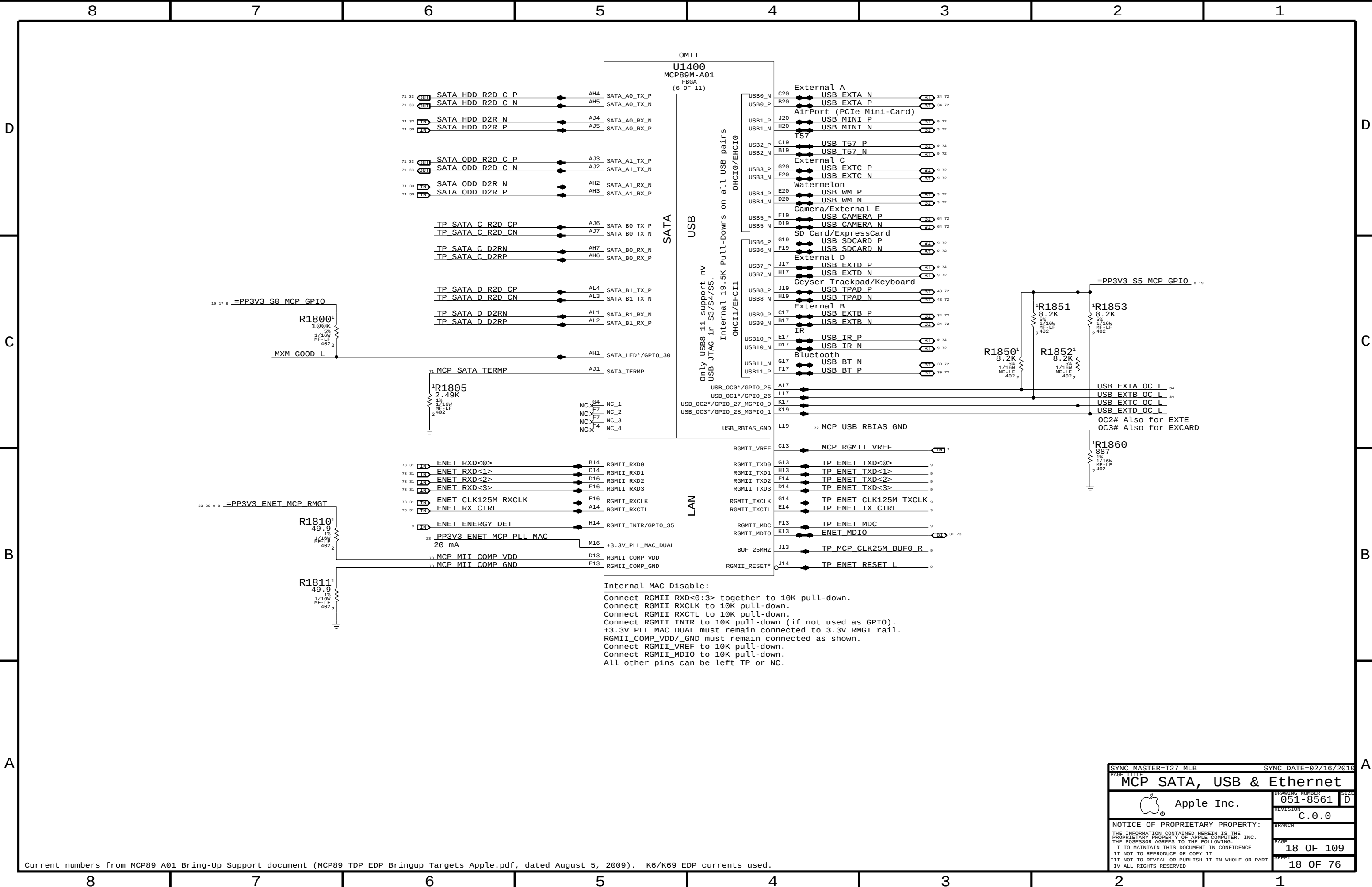


GPIO Pull-Ups



Current numbers from MCP89 A01 Bring-Up Support document (MCP89_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

SYNC MASTER=T27 MLB		SYNC DATE=02/16/2010	
PAGE TITLE			
MCP Graphics			
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Current numbers from MCP89 A01 Bring-Up Support document (MCP89_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

SYNC MASTER=T27_MLB

SYNC DATE=02/16/2016

MCP SATA, USB & Ethernet

Apple Inc.

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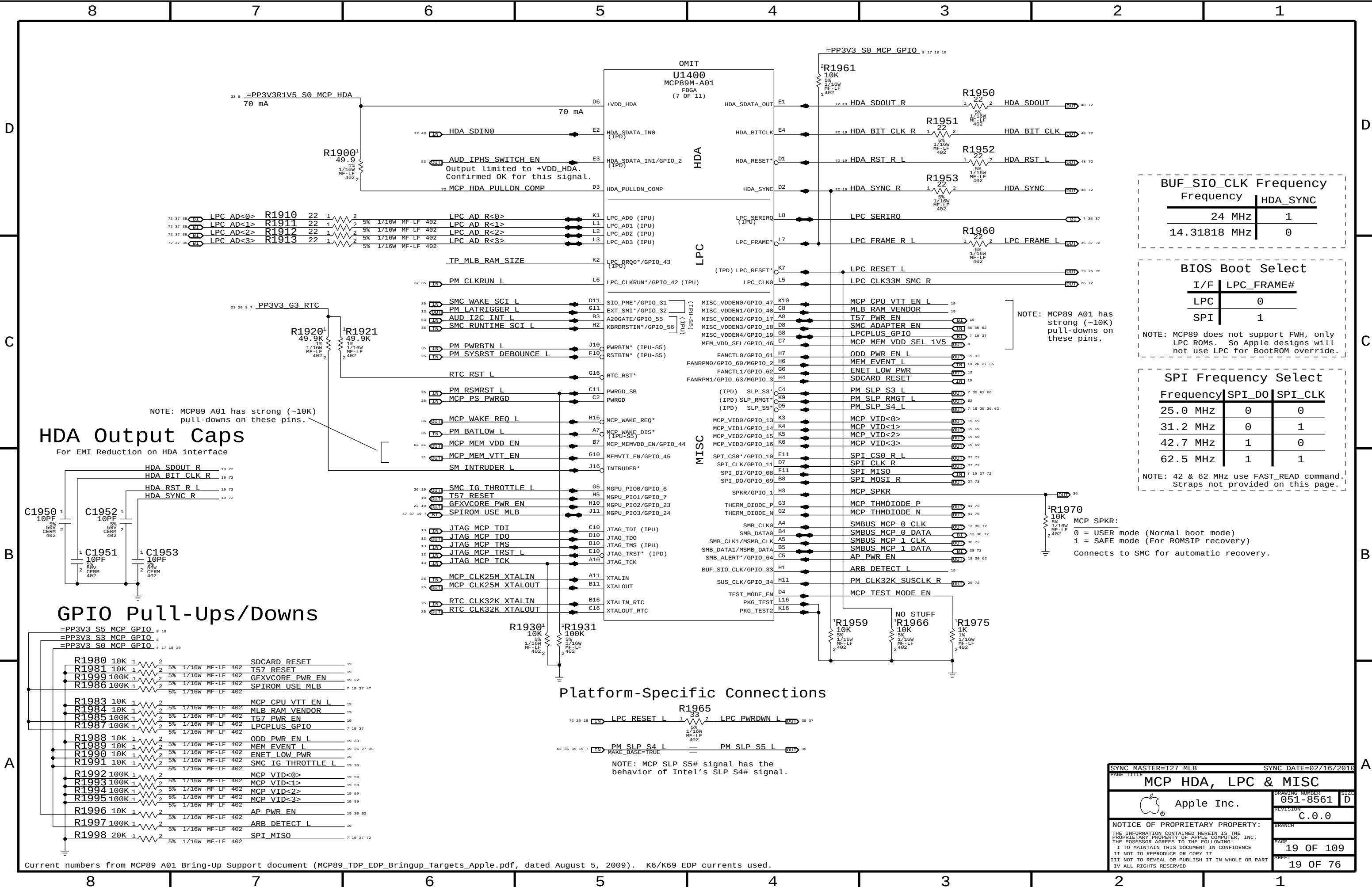
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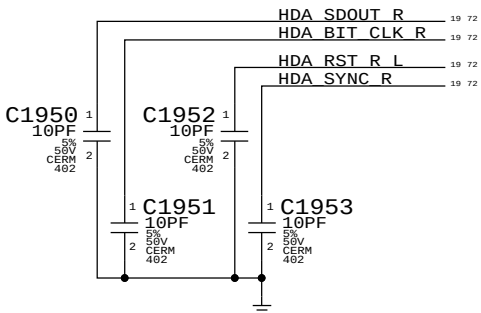
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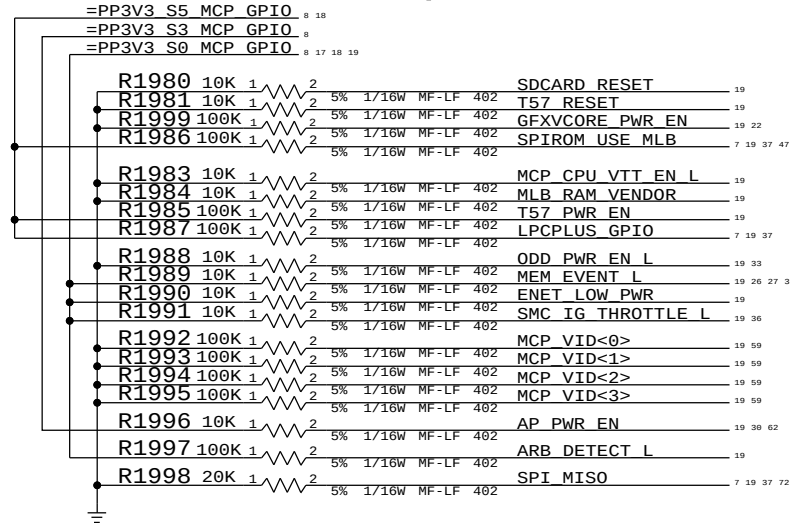


HDA Output Caps

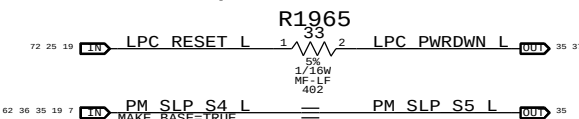
For EMI Reduction on HDA interface



GPIO Pull-Ups/Downs



Platform-Specific Connections



NOTE: MCP SLP_S5# signal has the behavior of Intel's SLP_S4# signal.

BUF_SIO_CLK Frequency	
Frequency	HDA_SYNC
24 MHz	1
14.31818 MHz	0

BIOS Boot Select

I/F	LPC_FRAME#
LPC	0
SPI	1

NOTE: MCP89 does not support FWH, only LPC ROMs. So Apple designs will not use LPC for BootROM override.

SPI Frequency Select

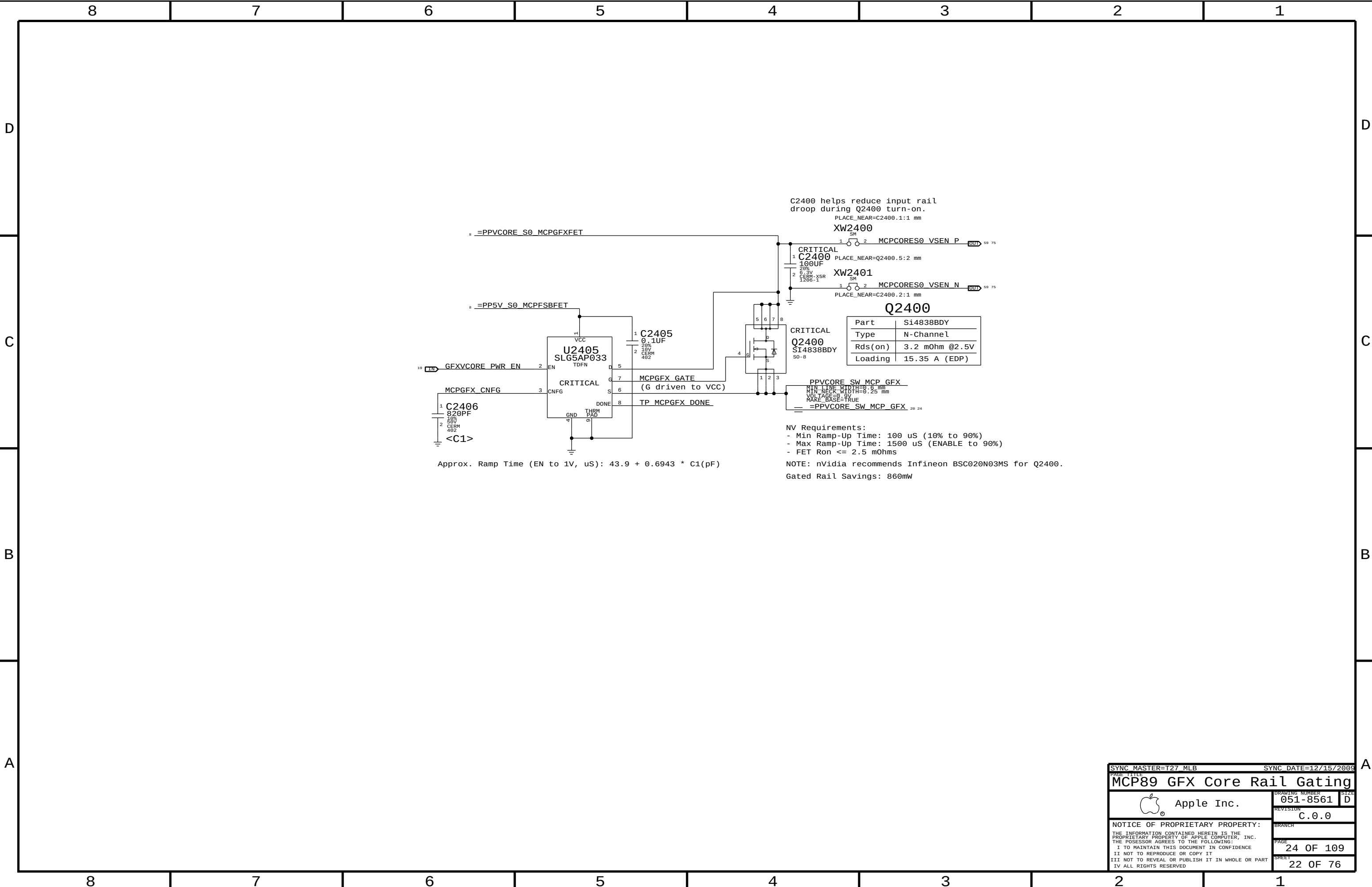
Frequency	SPI_DO	SPI_CLK
25.0 MHz	0	0
31.2 MHz	0	1
42.7 MHz	1	0
62.5 MHz	1	1

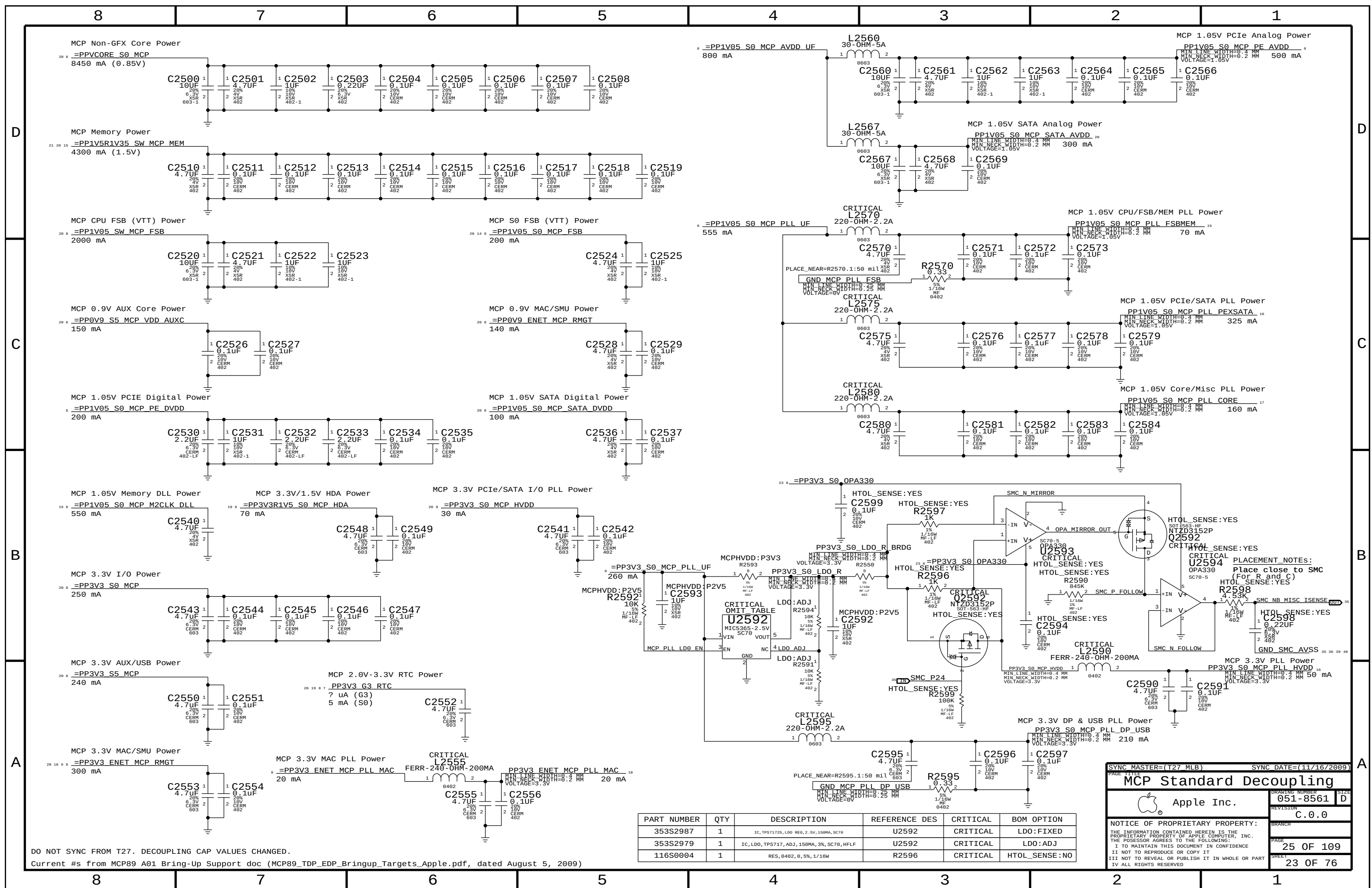
NOTE: 42 & 62 MHz use FAST_READ command. Straps not provided on this page.

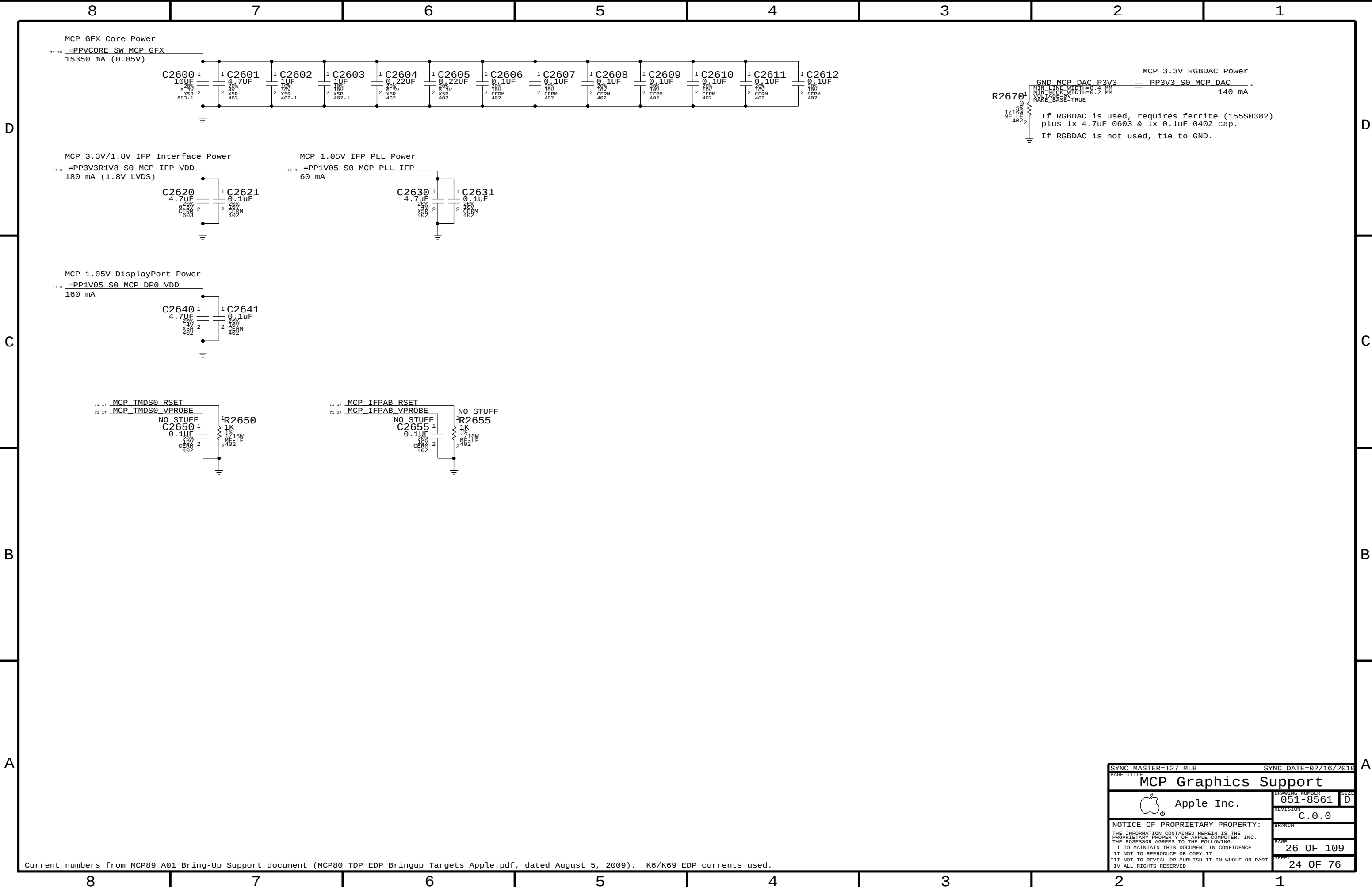
NOTE: MCP89 A01 has strong (~10K) pull-downs on these pins.

MCP_SPKR:
0 = USER mode (Normal boot mode)
1 = SAFE mode (For ROMSIP recovery)
Connects to SMC for automatic recovery.

SYNC MASTER=T27 MLB		SYNC DATE=02/16/2016	
PAGE TITLE			
MCP HDA, LPC & MISC		DRAWING NUMBER	051-8561
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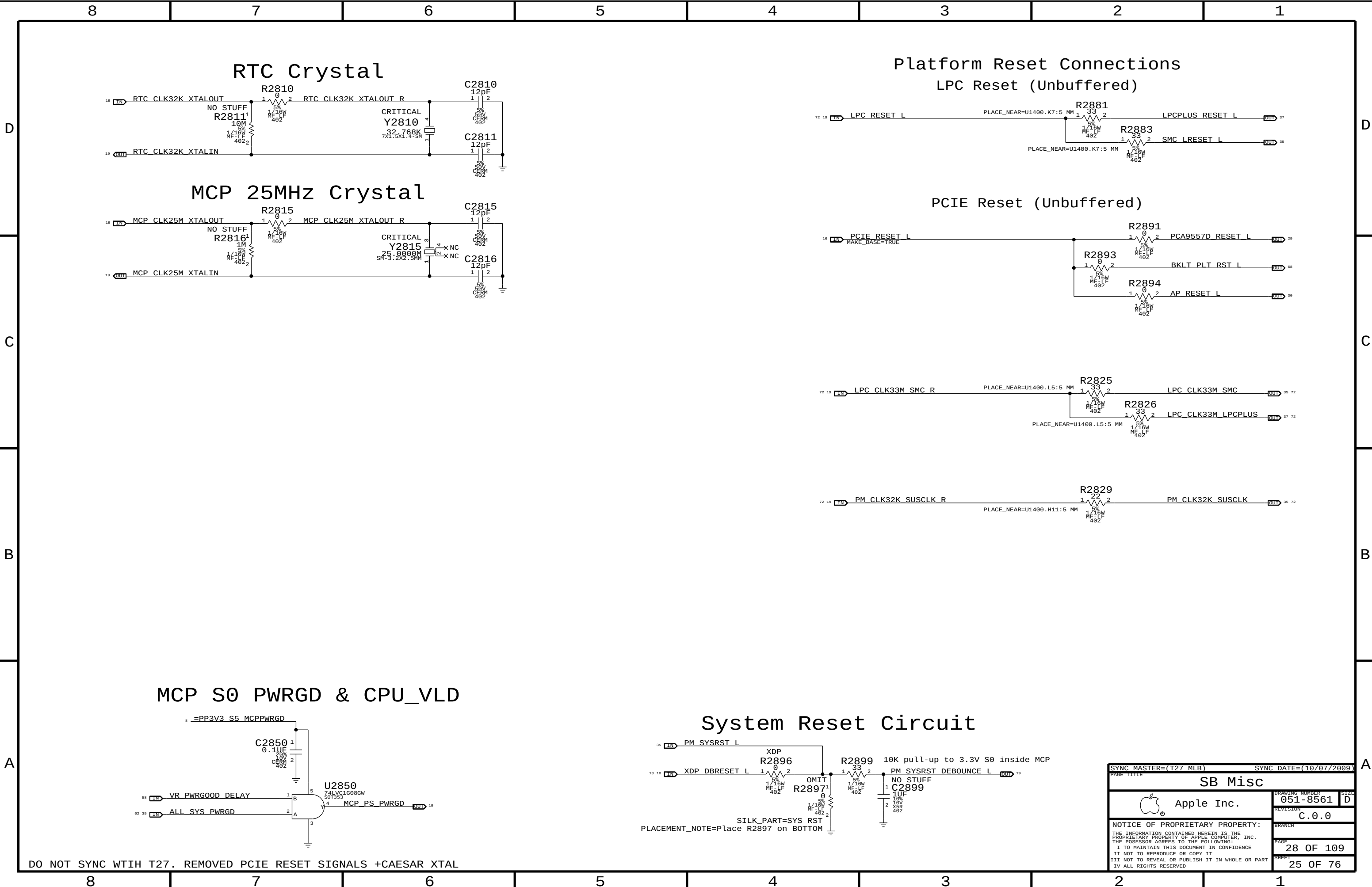




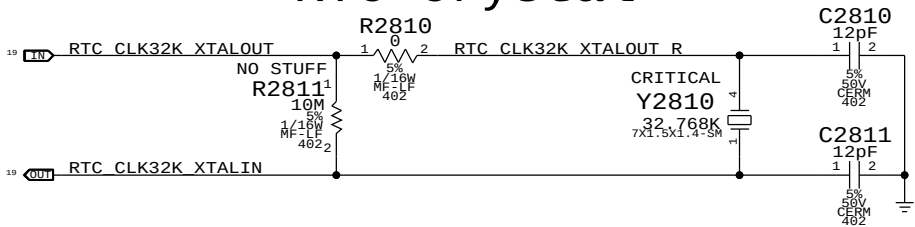


Current numbers from MCP89 A01 Bring-Up Support document (MCP80_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

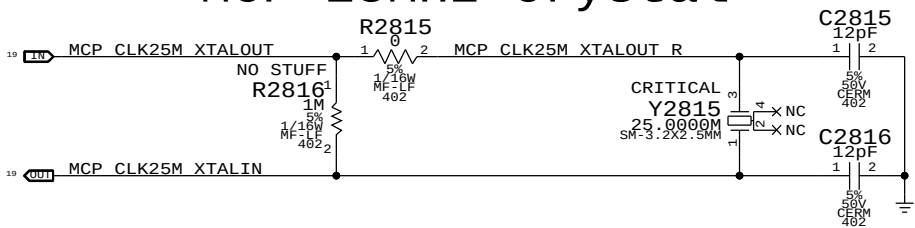
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 Apple Inc.		DRAWING NUMBER	051-8561
		REVISION	C.0.0
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RTC Crystal

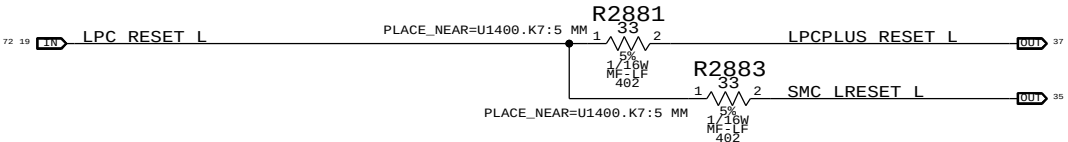


MCP 25MHz Crystal

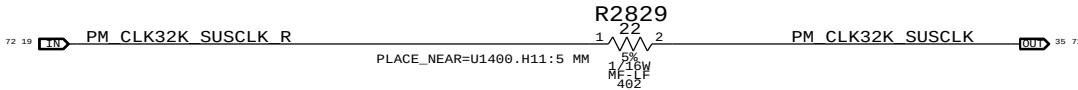
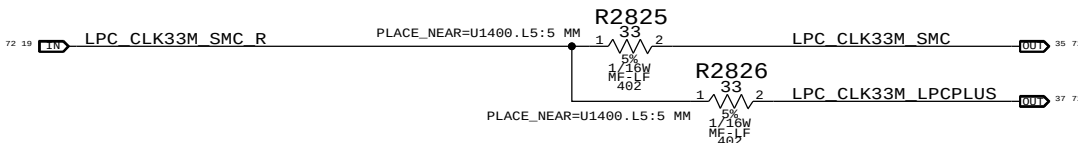
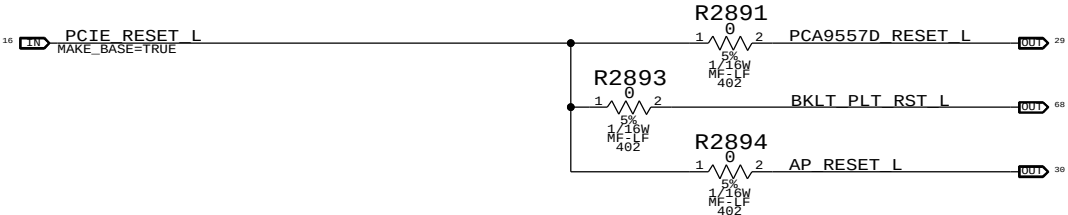


Platform Reset Connections

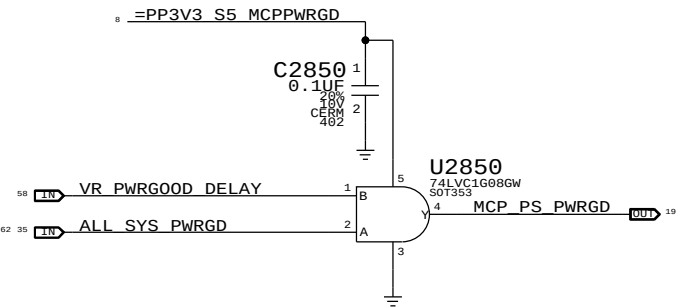
LPC Reset (Unbuffered)



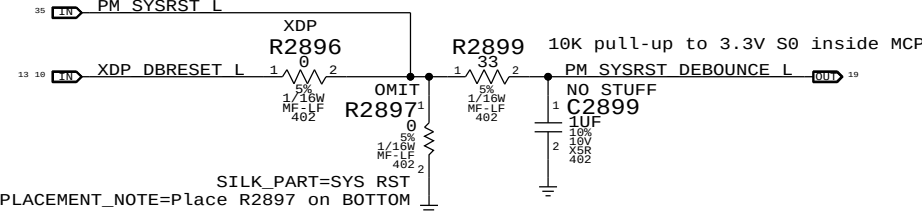
PCIE Reset (Unbuffered)



MCP S0 PWRGD & CPU_VLD



System Reset Circuit



DO NOT SYNC WITH T27. REMOVED PCIE RESET SIGNALS +CAESAR XTAL

PAGE TITLE		PAGE TITLE	
DRAWING NUMBER		DRAWING NUMBER	
051-8561		051-8561	
REVISION		REVISION	
C.0.0		C.0.0	
BRANCH		BRANCH	
PAGE		PAGE	
28 OF 109		28 OF 109	
SHEET		SHEET	
25 OF 76		25 OF 76	

Page Notes

Power aliases required by this page:

- =PPLVDDR_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

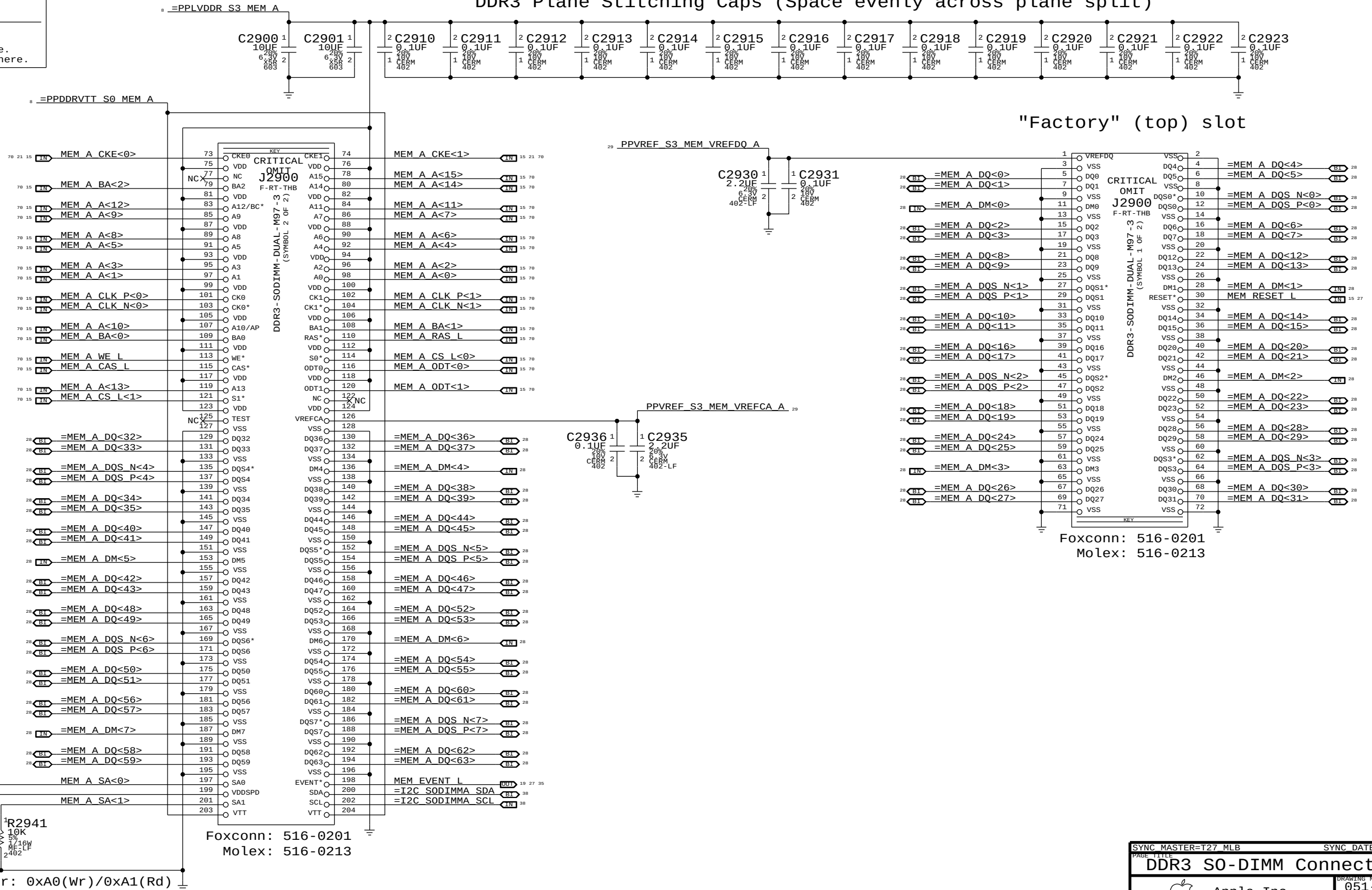
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)

NOTE: J3100 is OMITted on this page.
Proper APN(s) required elsewhere.

DDR3 Plane Stitching Caps (Space evenly across plane split)



SYNC MASTER=T27_MLB		SYNC DATE=02/16/2016	
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DDR3 S0-DIMM Connector A			
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Power aliases required by this page:

- =PPLVDDR_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

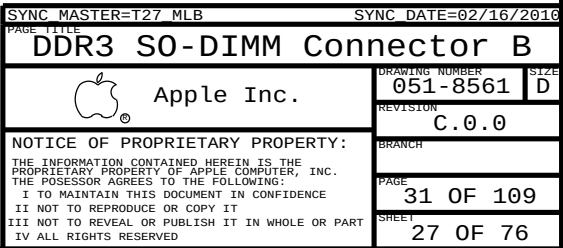
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- =I2C_SODIMMB_SDA

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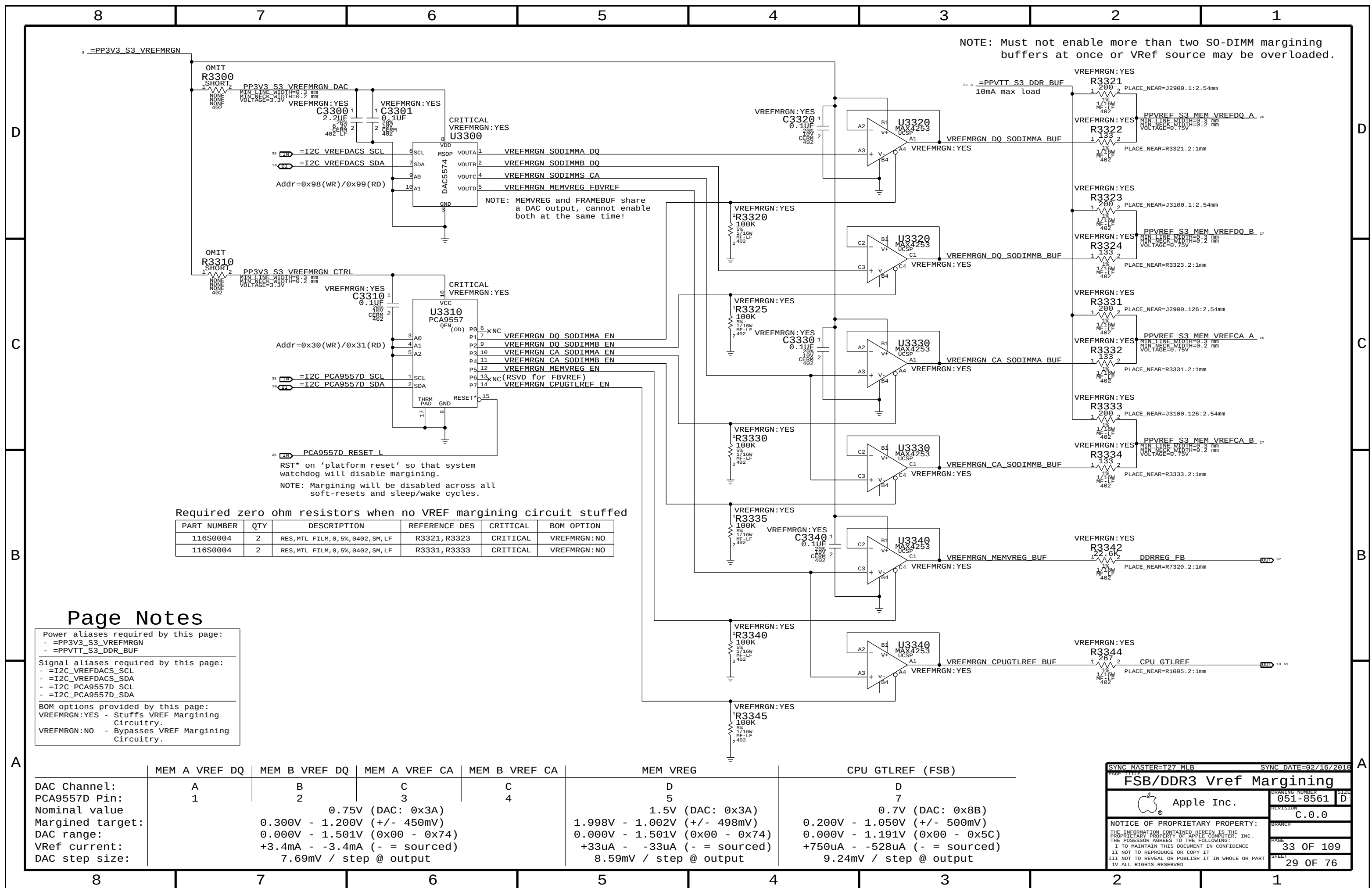
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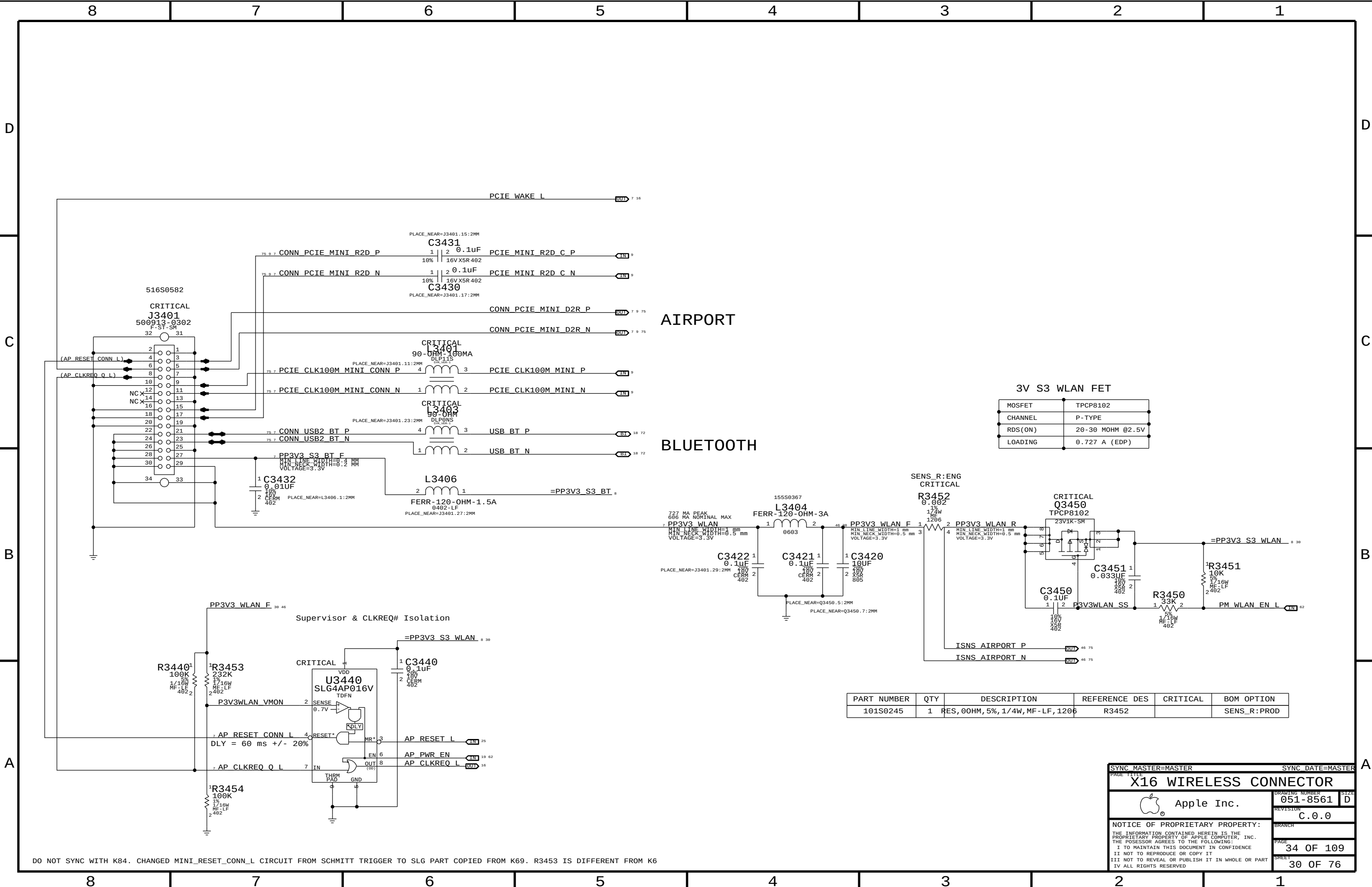
NOTE: J3100 is OMITTED on this page.
Proper APN(s) required elsewhere.



8		7		6		5		4		3		2		1			
D	Host Partition A 0 -> SO-DIMM A 0														D		
	70 15	81	MEM A DQS P<0>	MAKE_BASE=TRUE	==	MEM A DQS P<0>	81	26	70 15	81	MEM B DQS P<0>	MAKE_BASE=TRUE	==	MEM B DQS P<0>		81	27
	70 15	81	MEM A DQS N<0>	MAKE_BASE=TRUE	==	MEM A DQS N<0>	81	26	70 15	81	MEM B DQS N<0>	MAKE_BASE=TRUE	==	MEM B DQS N<0>		81	27
	70 15	81	MEM A DM<0>	MAKE_BASE=TRUE	==	MEM A DM<0>	81	26	70 15	81	MEM B DM<0>	MAKE_BASE=TRUE	==	MEM B DM<0>		81	27
	70 15	81	MEM A DQ<0>	MAKE_BASE=TRUE	==	MEM A DQ<0>	81	26	70 15	81	MEM B DQ<0>	MAKE_BASE=TRUE	==	MEM B DQ<0>		81	27
	70 15	81	MEM A DQ<1>	MAKE_BASE=TRUE	==	MEM A DQ<1>	81	26	70 15	81	MEM B DQ<1>	MAKE_BASE=TRUE	==	MEM B DQ<1>		81	27
	70 15	81	MEM A DQ<2>	MAKE_BASE=TRUE	==	MEM A DQ<2>	81	26	70 15	81	MEM B DQ<2>	MAKE_BASE=TRUE	==	MEM B DQ<2>		81	27
	70 15	81	MEM A DQ<3>	MAKE_BASE=TRUE	==	MEM A DQ<3>	81	26	70 15	81	MEM B DQ<3>	MAKE_BASE=TRUE	==	MEM B DQ<3>		81	27
	70 15	81	MEM A DQ<4>	MAKE_BASE=TRUE	==	MEM A DQ<4>	81	26	70 15	81	MEM B DQ<4>	MAKE_BASE=TRUE	==	MEM B DQ<4>		81	27
	70 15	81	MEM A DQ<5>	MAKE_BASE=TRUE	==	MEM A DQ<5>	81	26	70 15	81	MEM B DQ<5>	MAKE_BASE=TRUE	==	MEM B DQ<5>		81	27
C	HOST PARTITION A 1 -> SO-DIMM A 1														C		
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	70 15	81	MEM A DQS N<1>	MAKE_BASE=TRUE	==	MEM A DQS N<1>	81	26	70 15	81	MEM B DQS N<1>	MAKE_BASE=TRUE	==	MEM B DQS N<2>		81	27
	70 15	81	MEM A DM<1>	MAKE_BASE=TRUE	==	MEM A DM<1>	81	26	70 15	81	MEM B DM<1>	MAKE_BASE=TRUE	==	MEM B DM<2>		81	27
	70 15	81	MEM A DQ<8>	MAKE_BASE=TRUE	==	MEM A DQ<12>	81	26	70 15	81	MEM B DQ<8>	MAKE_BASE=TRUE	==	MEM B DQ<17>		81	27
	70 15	81	MEM A DQ<9>	MAKE_BASE=TRUE	==	MEM A DQ<8>	81	26	70 15	81	MEM B DQ<9>	MAKE_BASE=TRUE	==	MEM B DQ<16>		81	27
	70 15	81	MEM A DQ<10>	MAKE_BASE=TRUE	==	MEM A DQ<15>	81	26	70 15	81	MEM B DQ<10>	MAKE_BASE=TRUE	==	MEM B DQ<19>		81	27
	70 15	81	MEM A DQ<11>	MAKE_BASE=TRUE	==	MEM A DQ<10>	81	26	70 15	81	MEM B DQ<11>	MAKE_BASE=TRUE	==	MEM B DQ<23>		81	27
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	70 15	81	MEM A DQS N<2>	MAKE_BASE=TRUE	==	MEM A DQS N<3>	81	26	70 15	81	MEM B DQS N<2>	MAKE_BASE=TRUE	==	MEM B DQS N<3>		81	27
	70 15	81	MEM A DM<2>	MAKE_BASE=TRUE	==	MEM A DM<3>	81	26	70 15	81	MEM B DM<2>	MAKE_BASE=TRUE	==	MEM B DM<3>		81	27
	70 15	81	MEM A DQ<16>	MAKE_BASE=TRUE	==	MEM A DQ<24>	81	26	70 15	81	MEM B DQ<16>	MAKE_BASE=TRUE	==	MEM B DQ<29>		81	27
	70 15	81	MEM A DQ<17>	MAKE_BASE=TRUE	==	MEM A DQ<30>	81	26	70 15	81	MEM B DQ<17>	MAKE_BASE=TRUE	==	MEM B DQ<25>		81	27
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	70 15	81	MEM A DQ<19>	MAKE_BASE=TRUE	==	MEM A DQ<27>	81	26	70 15	81	MEM B DQ<19>	MAKE_BASE=TRUE	==	MEM B DQ<30>		81	27
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	70 15	81	MEM A DQ<21>	MAKE_BASE=TRUE	==	MEM A DQ<28>	81	26	70 15	81	MEM B DQ<21>	MAKE_BASE=TRUE	==	MEM B DQ<24>		81	27
A	HOST PARTITION A 3 -> SO-DIMM A 2														A		
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	70 15	81	MEM A DQS N<3>	MAKE_BASE=TRUE	==	MEM A DQS N<2>	81	26	70 15	81	MEM B DQS N<3>	MAKE_BASE=TRUE	==	MEM B DQS N<1>		81	27
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PAGE TITLE S0-DIMM Pinswaps																	
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AIRPORT

BLUETOOTH

3V S3 WLAN FET	
MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	20-30 MOHM @2.5V
LOADING	0.727 A (EDP)

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
101S0245	1	RES, 00HM, 5%, 1/4W, MF-LF, 1206	R3452		SENS_R:PROD

SYNC MASTER=MASTER

SYNC DATE=MASTER

X16 WIRELESS CONNECTOR

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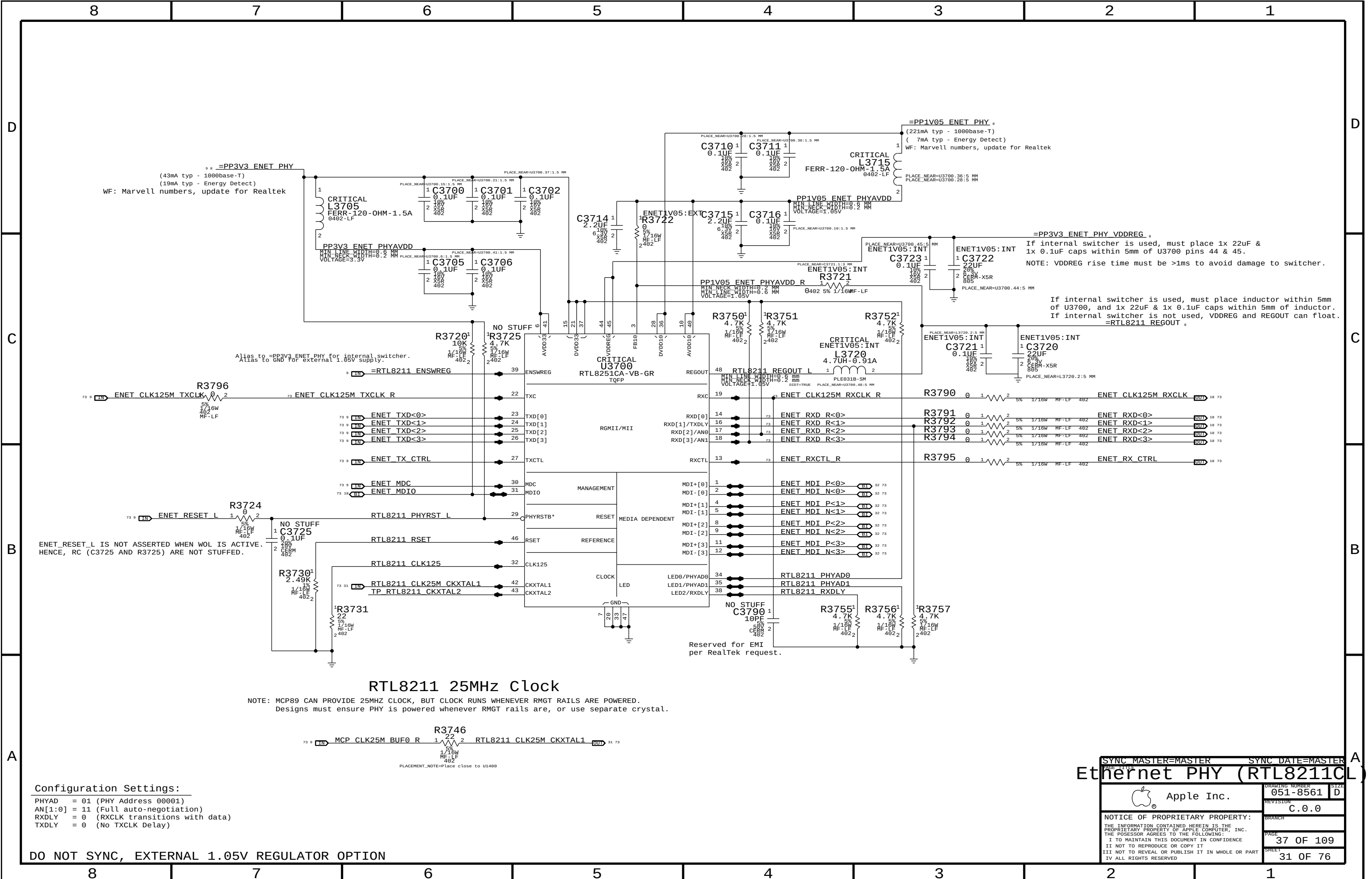
REVISION
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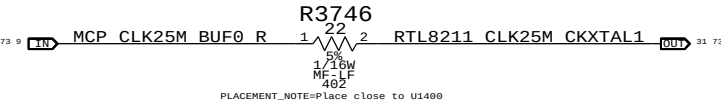
SHEET
30 OF 76

DO NOT SYNC WITH K84. CHANGED MINI_RESET_CONN_L CIRCUIT FROM SCHMITT TRIGGER TO SLG PART COPIED FROM K69. R3453 IS DIFFERENT FROM K6



RTL8211 25MHz Clock

NOTE: MCP89 CAN PROVIDE 25MHZ CLOCK, BUT CLOCK RUNS WHENEVER RMGT RAILS ARE POWERED.
Designs must ensure PHY is powered whenever RMGT rails are, or use separate crystal.



- Configuration Settings:
- PHYAD = 01 (PHY Address 00001)
 - AN[1:0] = 11 (Full auto-negotiation)
 - RXDLY = 0 (RXCLK transitions with data)
 - TXDLY = 0 (No TXCLK Delay)

DO NOT SYNC, EXTERNAL 1.05V REGULATOR OPTION

SYNC MASTER=MASTER

SYNC DATE=MASTER

Ethernet PHY (RTL8211C)

 Apple Inc.

DRAWING NUMBER
051-8561

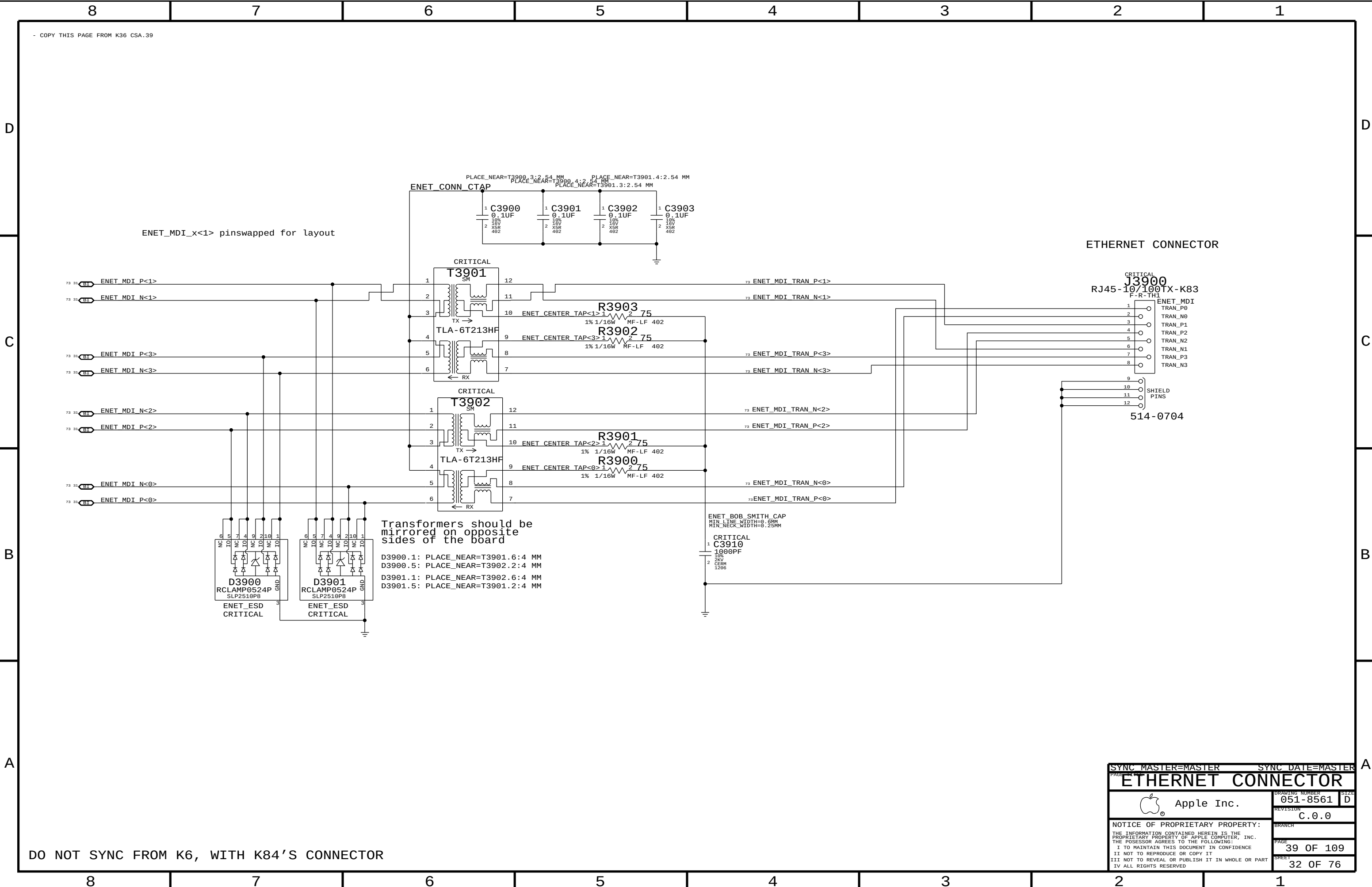
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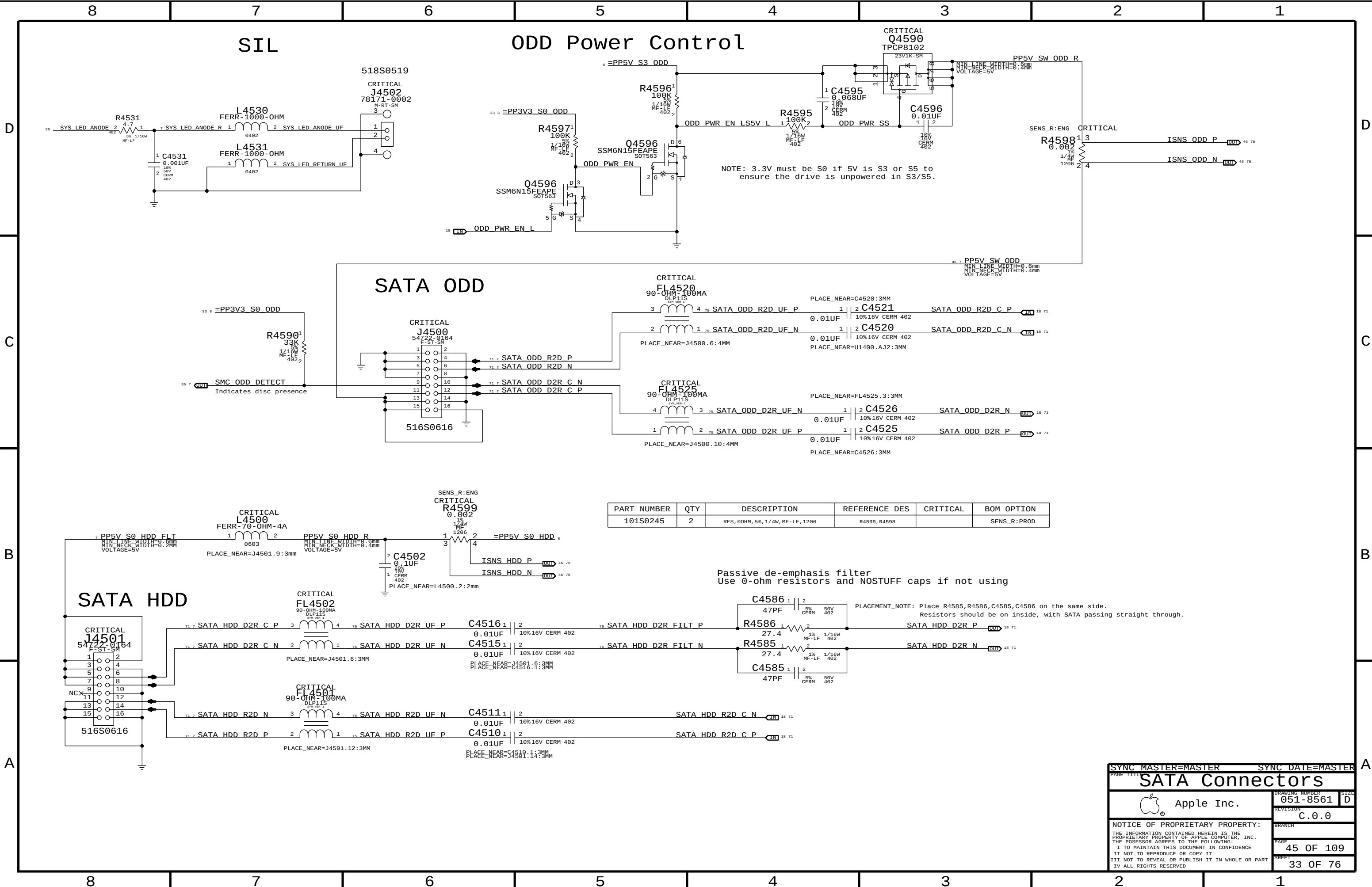
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The image shows a technical drawing of an Ethernet connector, which appears to be a D-subminiature connector. The drawing is oriented vertically, showing the pins and the housing. To the right of the drawing is a document header or title block. It contains the following information:

- Top left: SYNC MASTER=MASTER
- Top right: SYNC DATE=MASTER
- Below those: PAGE 1 OF 109
- Large bold text: ETHERNET CONNECTOR
- Below that: DRAWING NUMBER 051-8561, SIZE D
- REVISION C.0.0
- BRANCH
- PAGE 39 OF 109
- SHEET 32 OF 76

Below the drawing, there is a note:

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
101S0245	2	RES, 00HM, 5%, 1/4W, MF-LF, 1206	R4599, R4598		SENS_R:PROD

Passive de-emphasis filter
Use 0-ohm resistors and N0STUFF caps if not using

SYNC MASTER=MASTER

SYNC DATE=MASTER

SATA Connectors

Apple Inc.

051-8561

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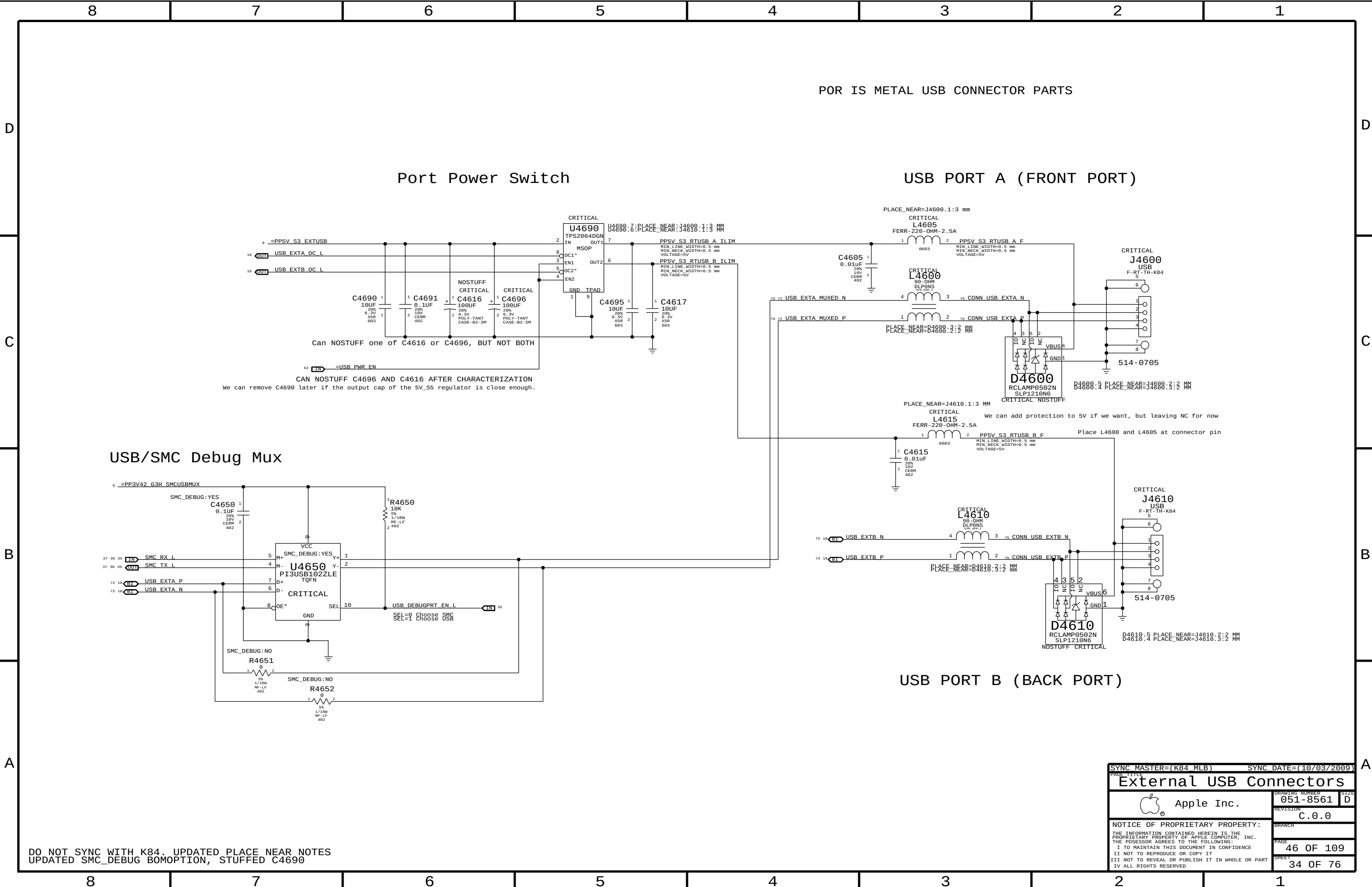
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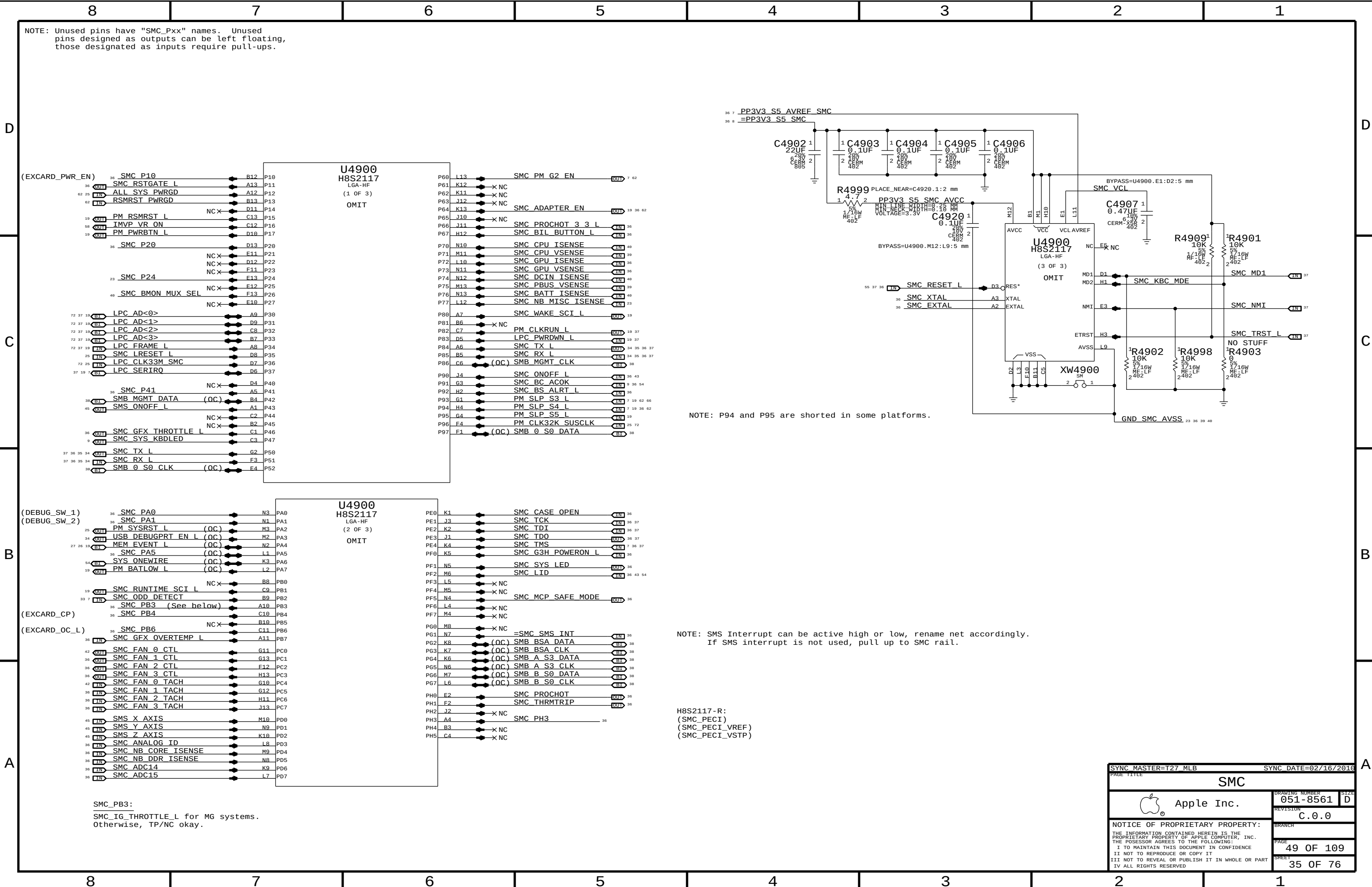
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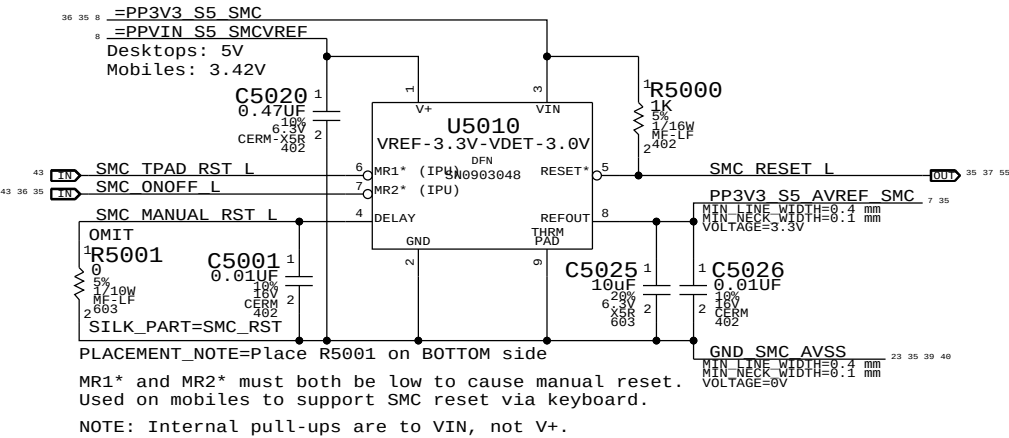


DO NOT SYNC WITH K84. UPDATED PLACE NEAR NOTES
UPDATED SMC_DEBUG_BOMOPTION, STUFFED C4690

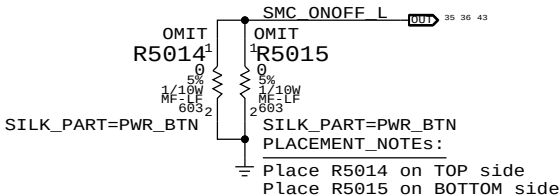
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External USB Connectors			
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		SIZE	D
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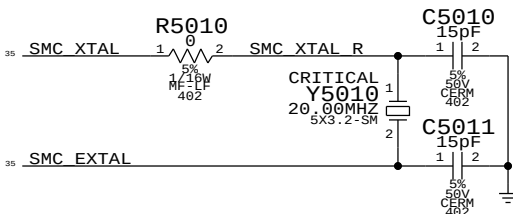
SMC Reset "Button", Supervisor & AVREF Supply



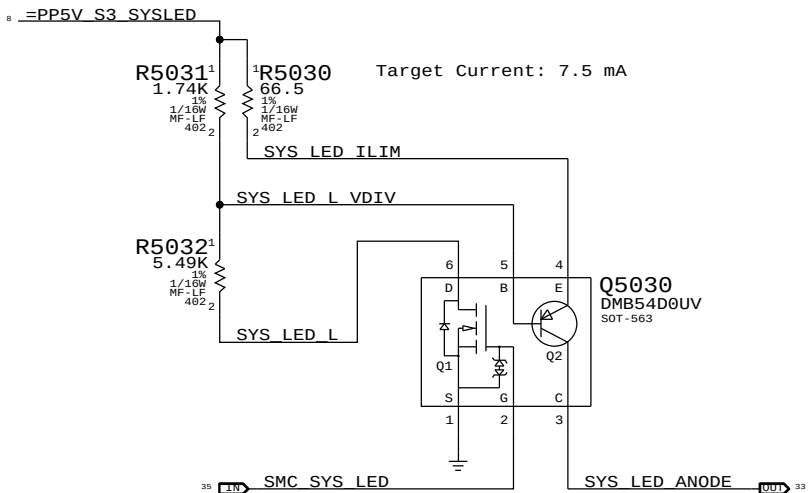
Debug Power "Buttons"



SMC Crystal Circuit

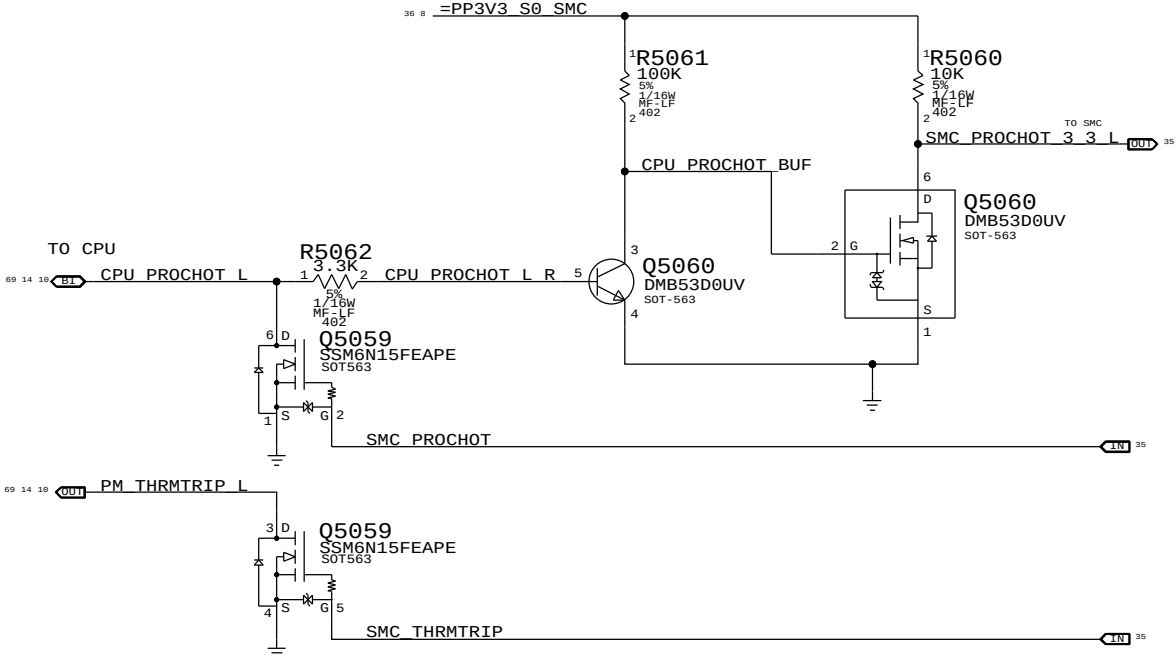


System (Sleep) LED Circuit

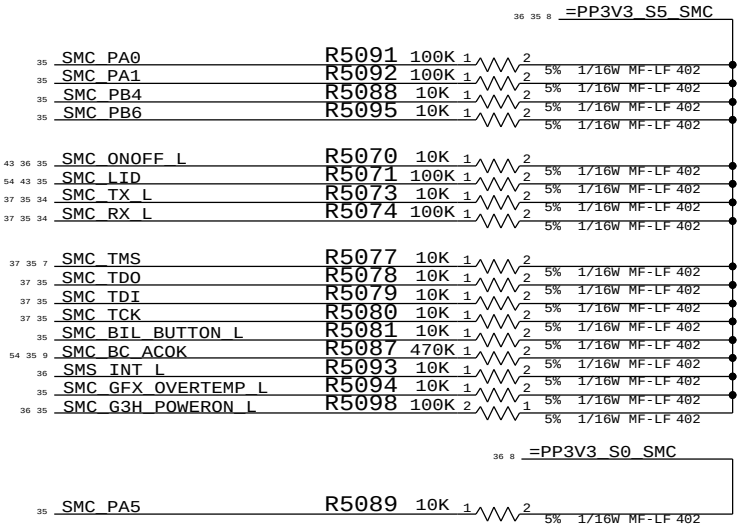


R5030,R5031,R5032 CHANGED FOR DIMMER LED

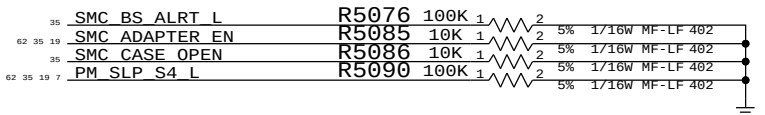
SMC FSB to 3.3V Level Shifting



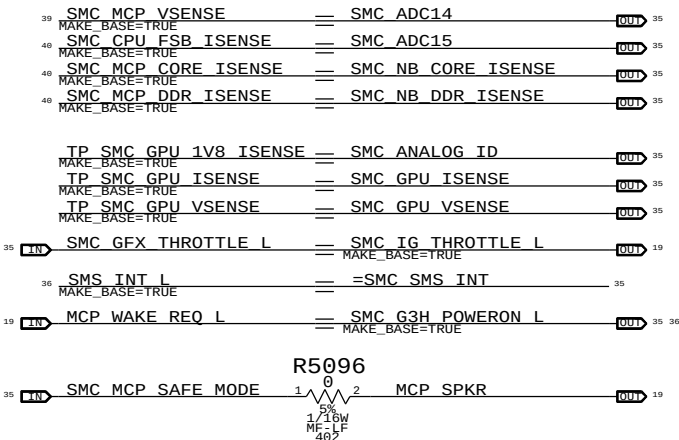
SMC Pull-ups



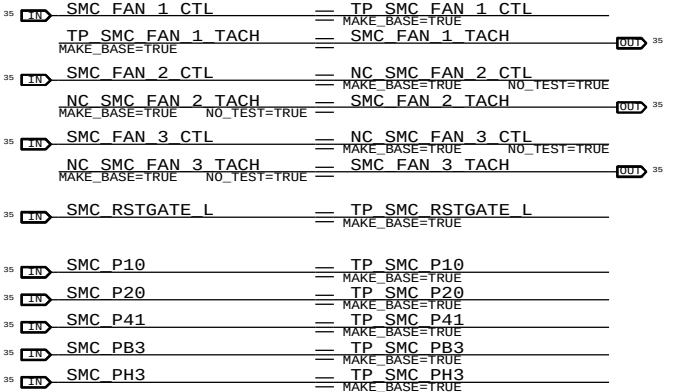
SMC Pull-downs



SMC Aliases



Unused Pins



SYNC_MASTER=(T27_MLB)

SYNC_DATE=(10/27/2009)

SMC Support

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051-8561

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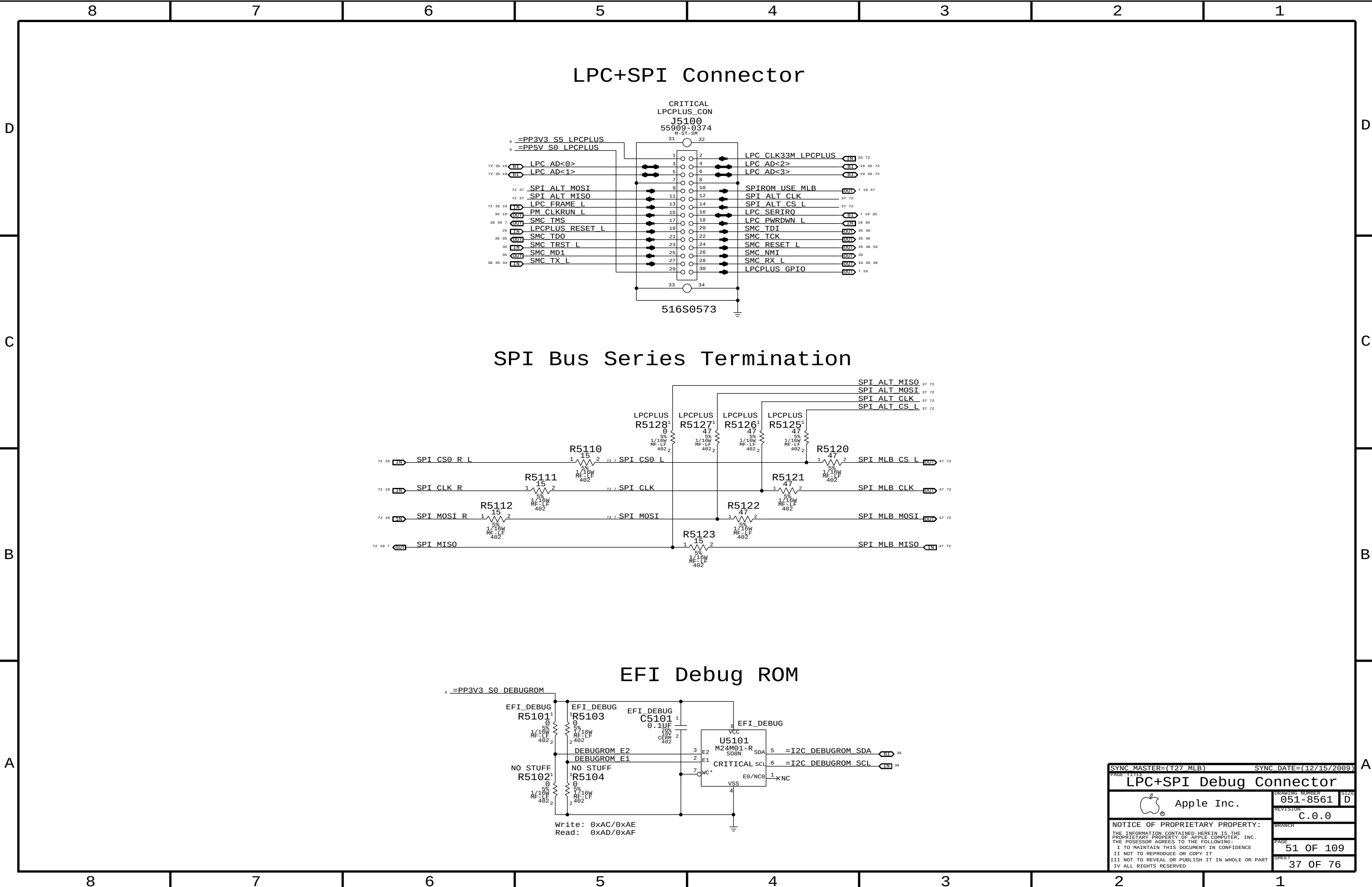
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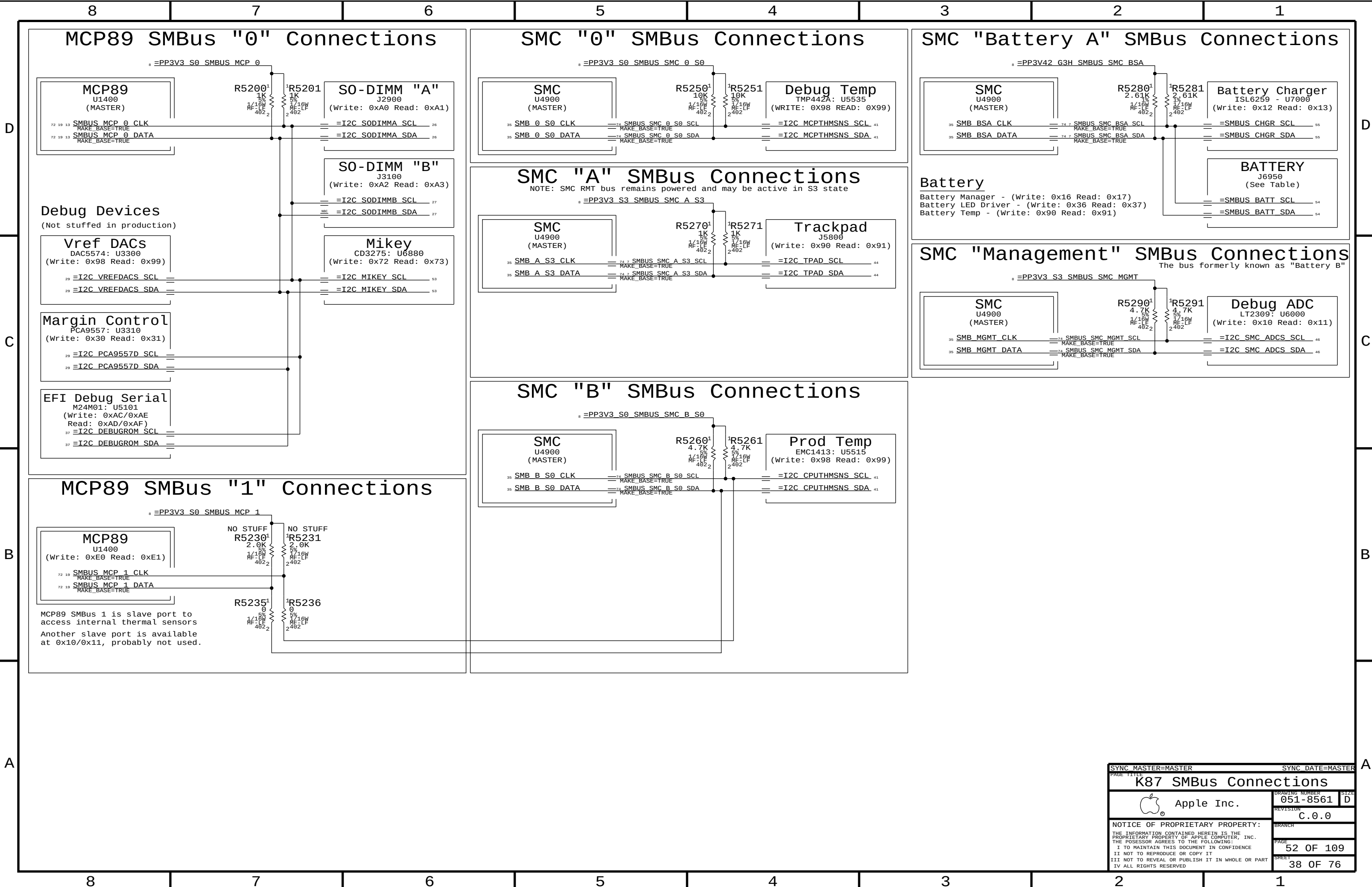
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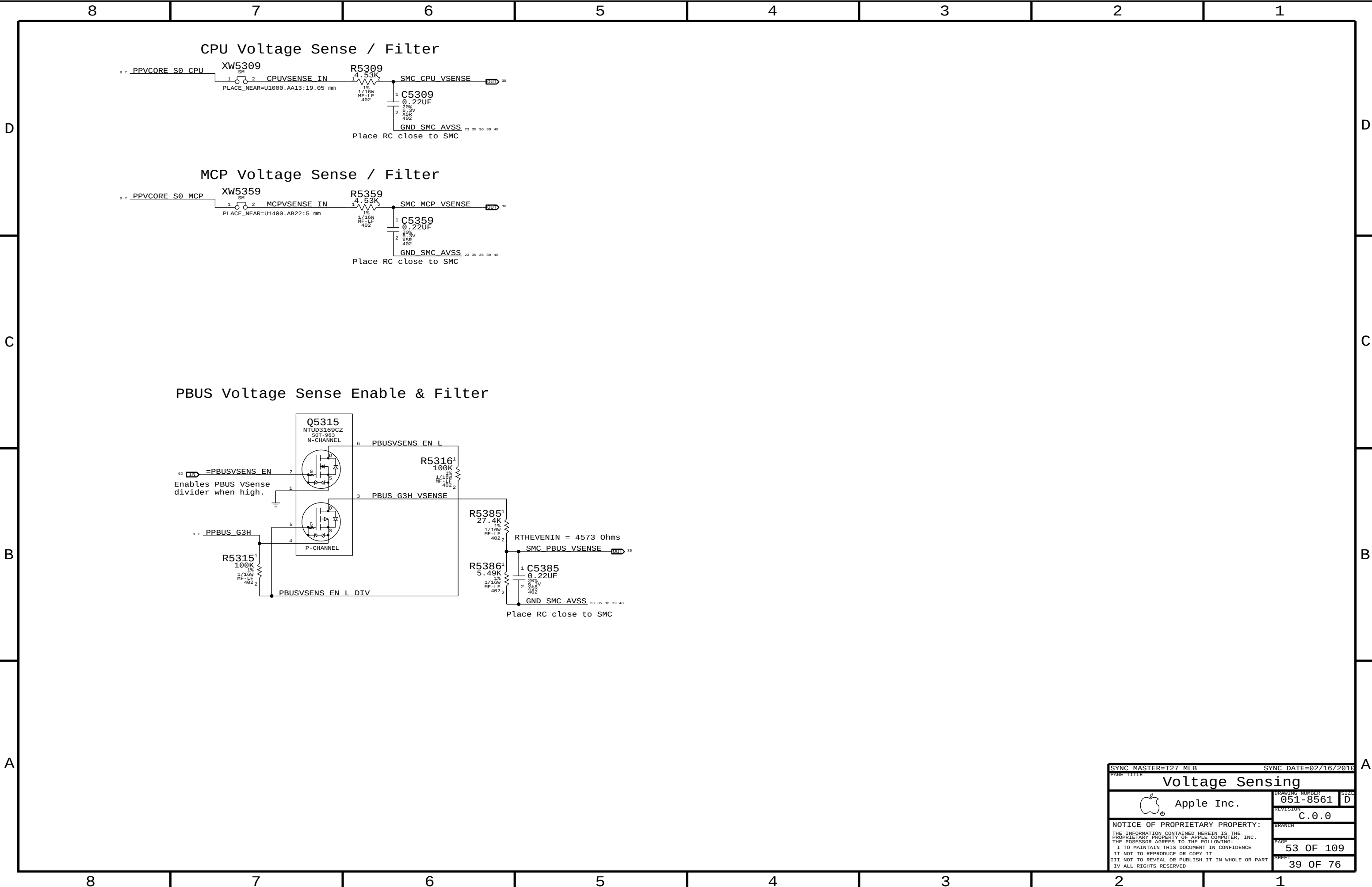


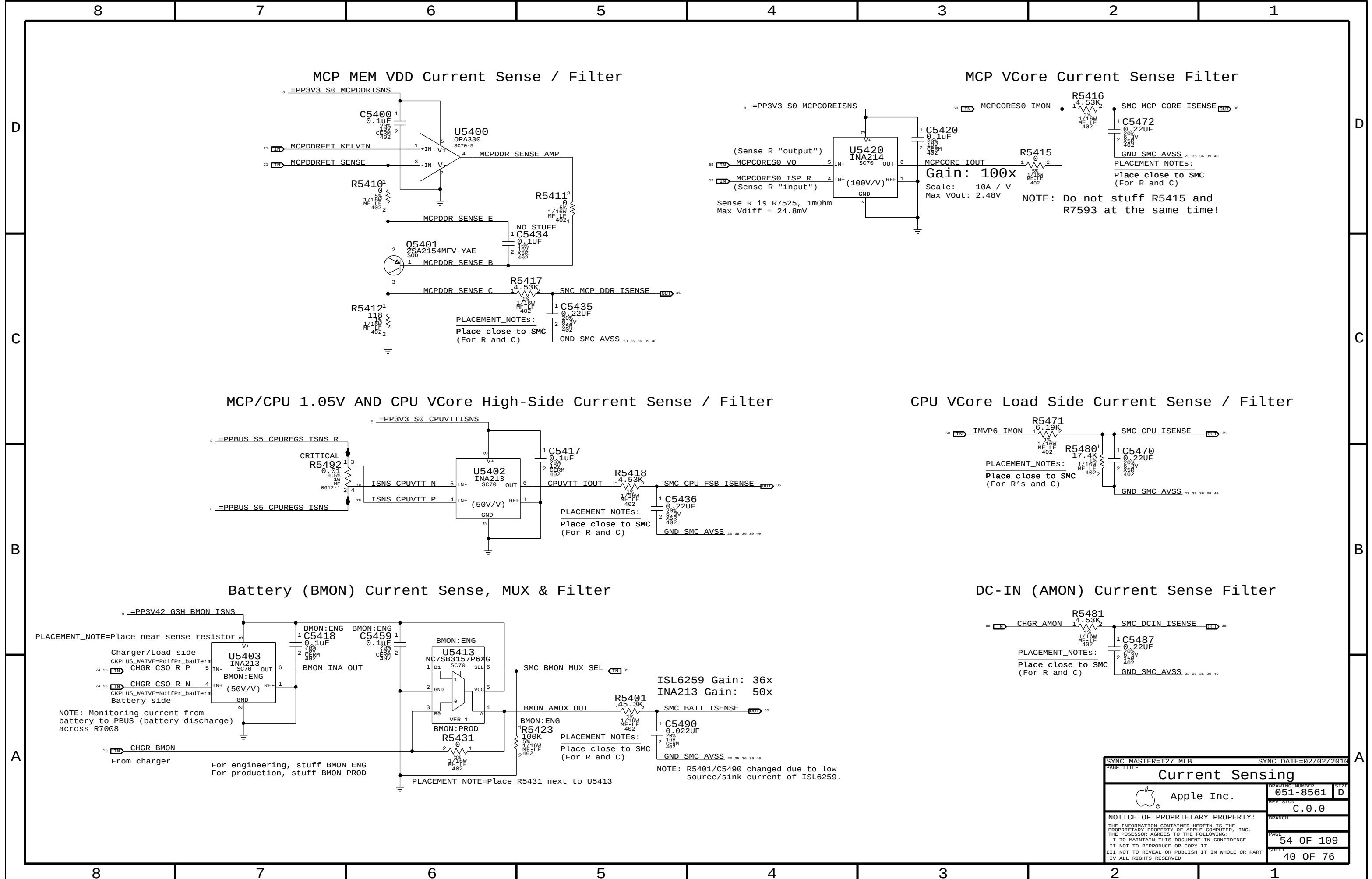
SYNC MASTER=(T27_MLB)

SYNC DATE=(12/15/2009)

PAGE TITLE		LPC+SPI Debug Connector	
Apple Inc.		DRAWING NUMBER	051-8561
		REVISION	C.0.0
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SYNC MASTER=T27 MLB		SYNC DATE=02/02/2016	
PAGE TITLE		Current Sensing	
Apple Inc.		DRAWING NUMBER	051-8561
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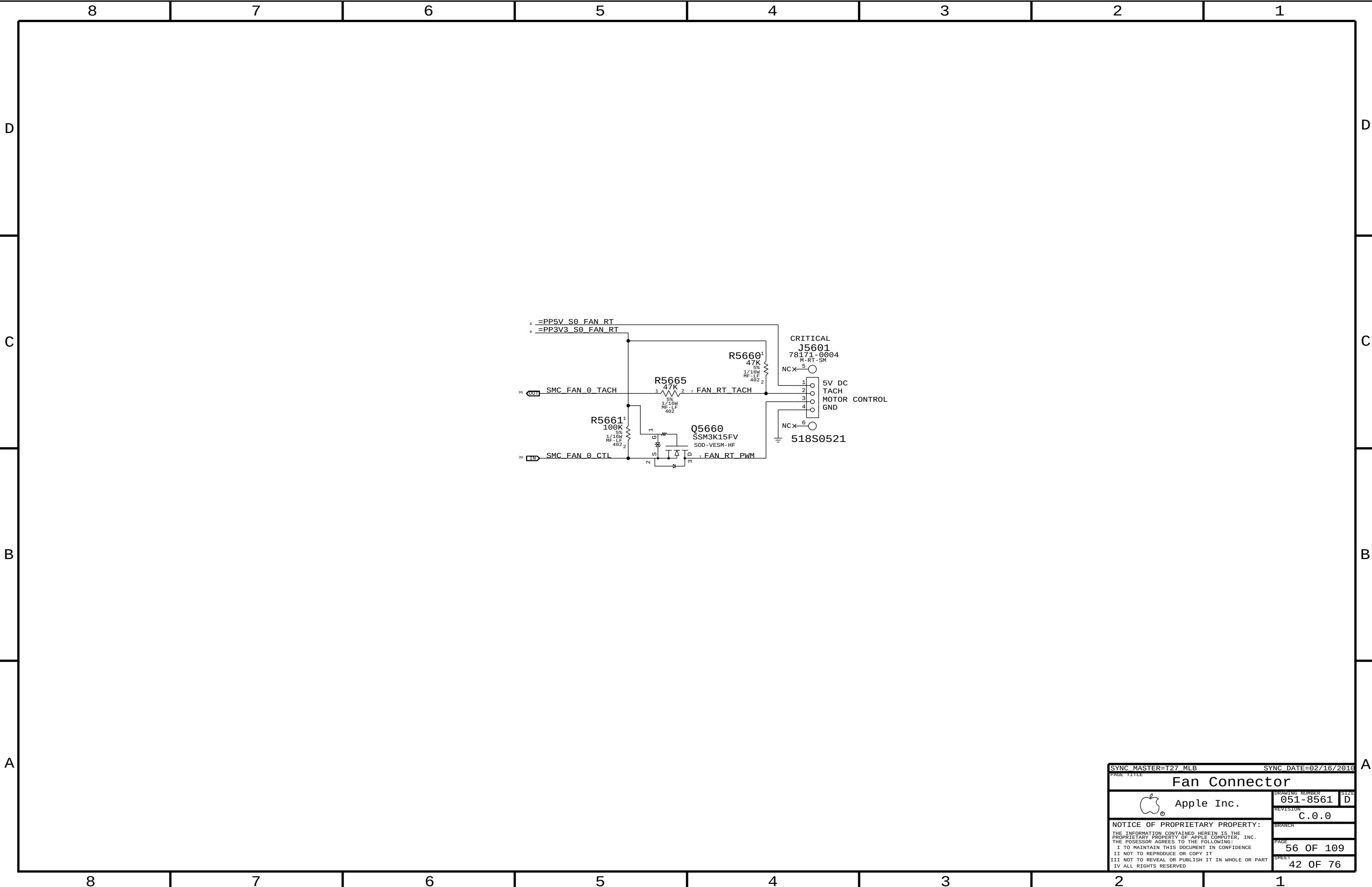
8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---



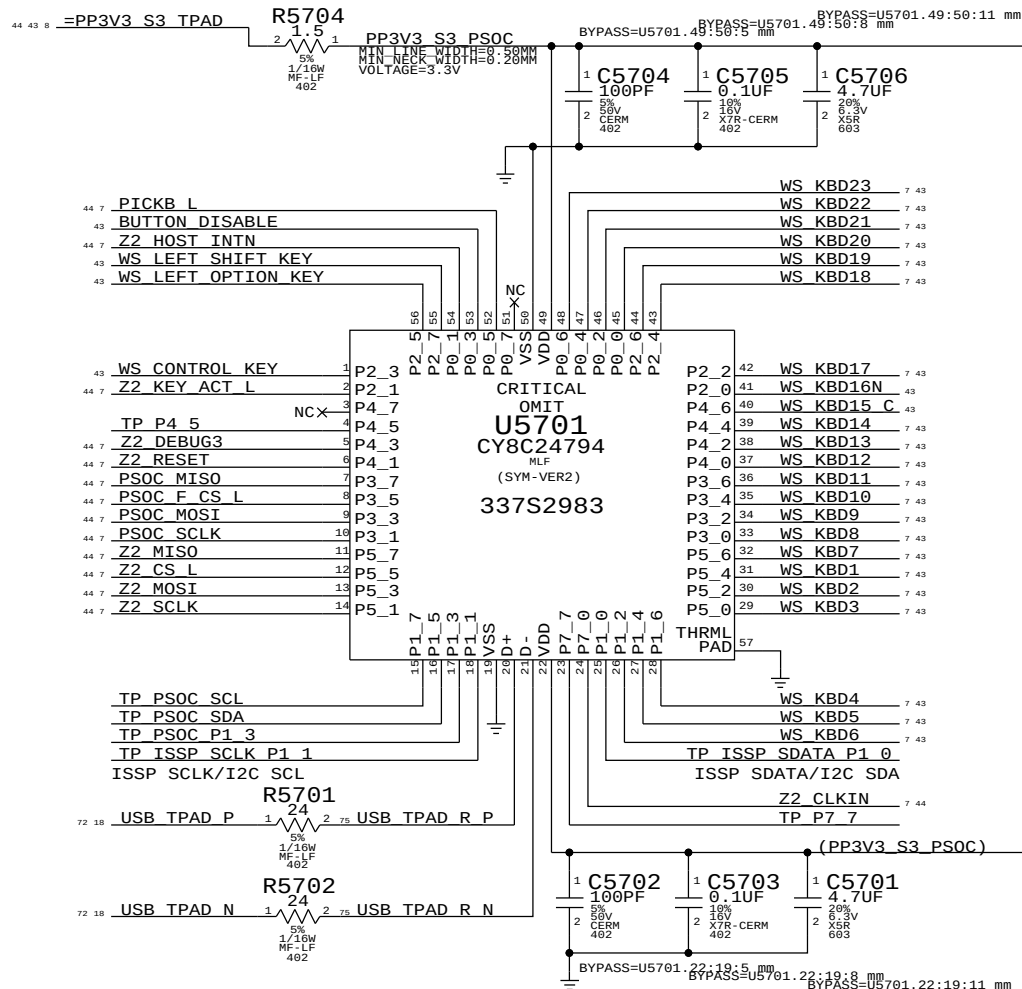
C



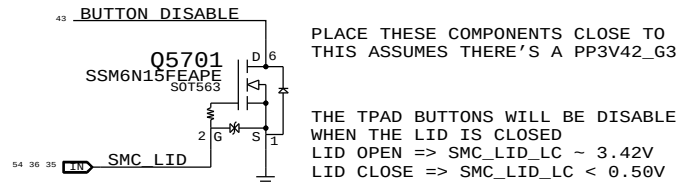
A



- USB INTERFACES TO MLB
- SPI HOST TO Z2
- TRACKPAD PICK BUTTONS
- KEYBOARD SCANNER



PLACE THESE COMPONENTS CLOSE TO J5800
THIS ASSUMES THERE'S A PP3V42_G3H PULL UP ON MLB



IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	100A	2.55 KOHM	0.0255 V	0.255E-6 W
		800A		0.204 V	16.32E-6 W
3V3 LDO	VDD	60mA (MAX)	10 OHM	0.6 V	36E-3 W
	VOUT	60mA (MAX)	0.2 OHM	0.012 V	0.72E-3 W
PS0C	VDD	8mA (TYP)	1.5 OHM	0.012 V	96E-6 W
		14mA (MAX)		0.021 V	294E-6 W
18V BOOSTER	VIN	4mA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

43 8 =PP3V3 S3 TPAD
43 8 =PP3V42 G3H TPAD

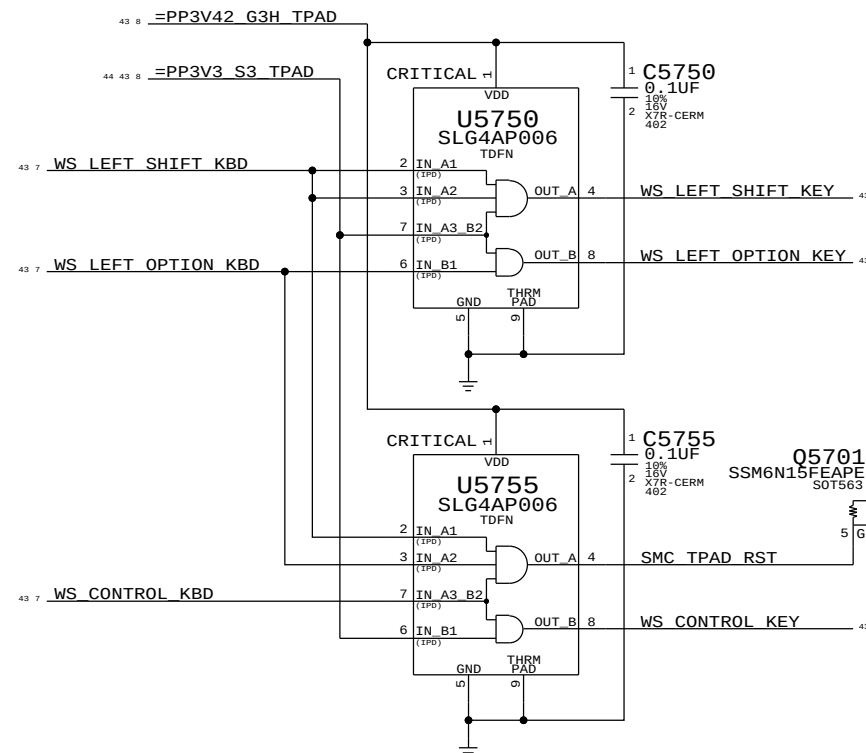
43 7 WS_KBD1
43 7 WS_KBD2
43 7 WS_KBD3
43 7 WS_KBD4
43 7 WS_KBD5
43 7 WS_KBD6
43 7 WS_KBD7
43 7 WS_KBD8
43 7 WS_KBD9
43 7 WS_KBD10
43 7 WS_KBD11
43 7 WS_KBD12
43 7 WS_KBD13
43 7 WS_KBD14
7 WS_KBD15 CAP
7 WS_KBD16 NUM
43 7 WS_KBD17
43 7 WS_KBD18
43 7 WS_KBD19
43 7 WS_KBD20
43 7 WS_KBD21
43 7 WS_KBD22
43 7 WS_KBD23
7 WS_KBD_ONOFF L

43 7 WS_LEFT_SHIFT_KBD
43 7 WS_LEFT_OPTION_KBD
43 7 WS_CONTROL_KBD

NCX 31

F-RT-SM
FF14-30A-R11B-B-3H
J5713
CRITICAL
518S0637

Left shift, option & control keys combined with power button cause SMC RESET# assertion.
Keys ANDed with PSOC power to isolate when PSOC is not powered.



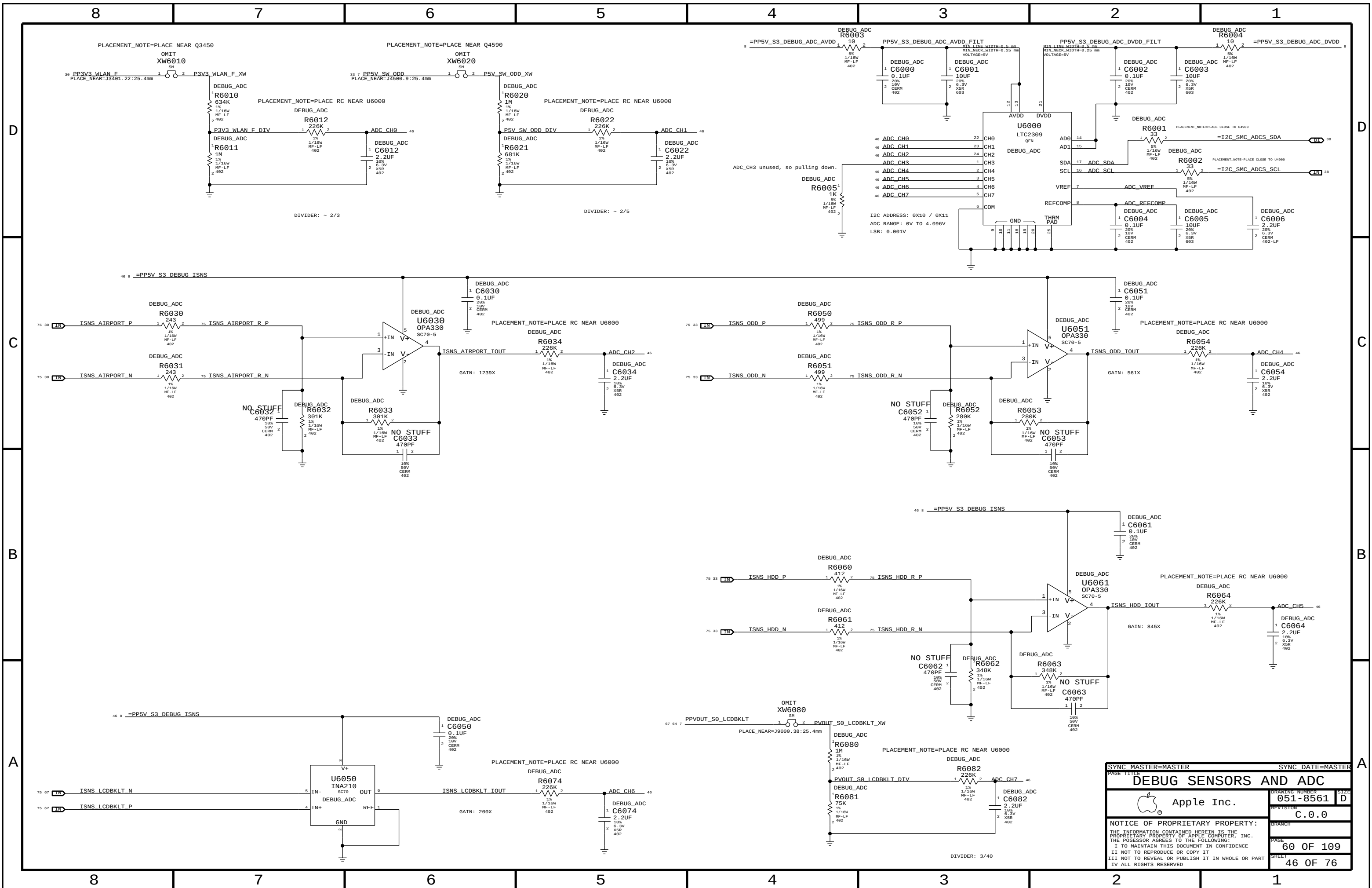
SYNC MASTER=T27 MLB		SYNC DATE=02/16/2010	
PAGE TITLE			
WELLSPRING 1			
	Apple Inc.	DRAWING NUMBER	051-8561
		SIZE	D
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		BRANCH	
		PAGE	57 OF 109
		SHEET	43 OF 76

BOOSTER DESIGN CONSIDERATION:

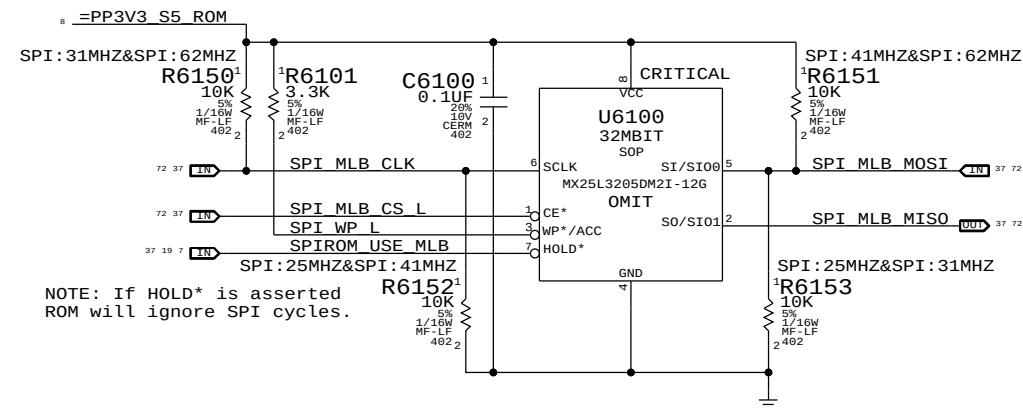
- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812,R5813,C5818 MODIFIED



SYNC MASTER=MASTER		SYNC DATE=MASTER	
PAGE TITLE		WELLSPRING 2	
 Apple Inc.	DRAWING NUMBER	051-8561	SIZE
	REVISION	C.0.0	
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		58 OF 109	
		SHEET	
		44 OF 76	

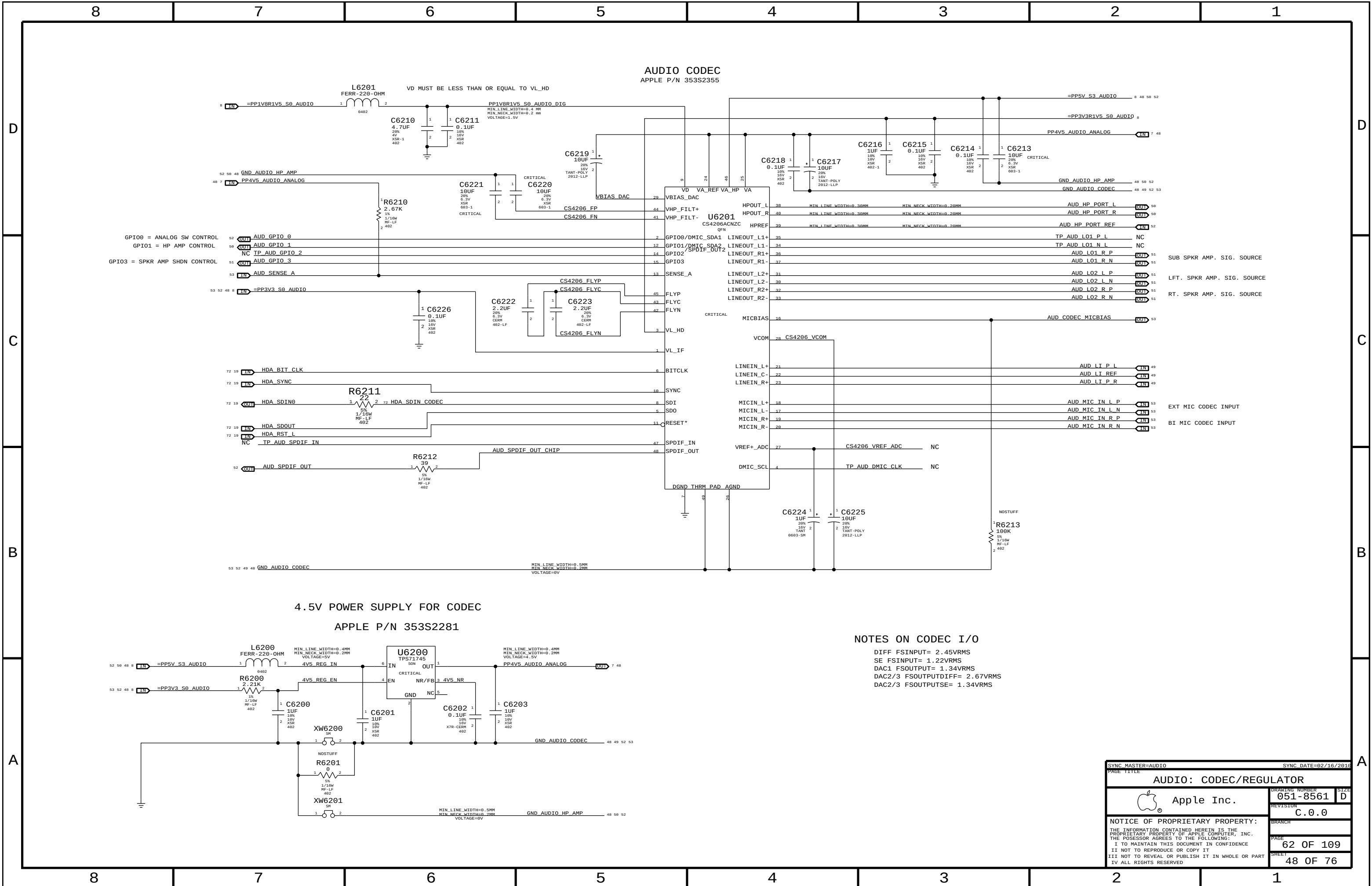


PAGE TITLE		SYNC MASTER=MASTER		SYNC DATE=MASTER	
DEBUG SENSORS AND ADC				DRAWING NUMBER	SIZE
Apple Inc.				051-8561	D
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				PAGE	60 OF 109
				SHEET	46 OF 76



Frequency	SPI_MOSI	SPI_CLK
25.0 MHz	0	0
31.2 MHz	0	1
41.7 MHz	1	0
62.5 MHz	1	1

NOTE: 42 & 62 MHz use FAST_READ command.



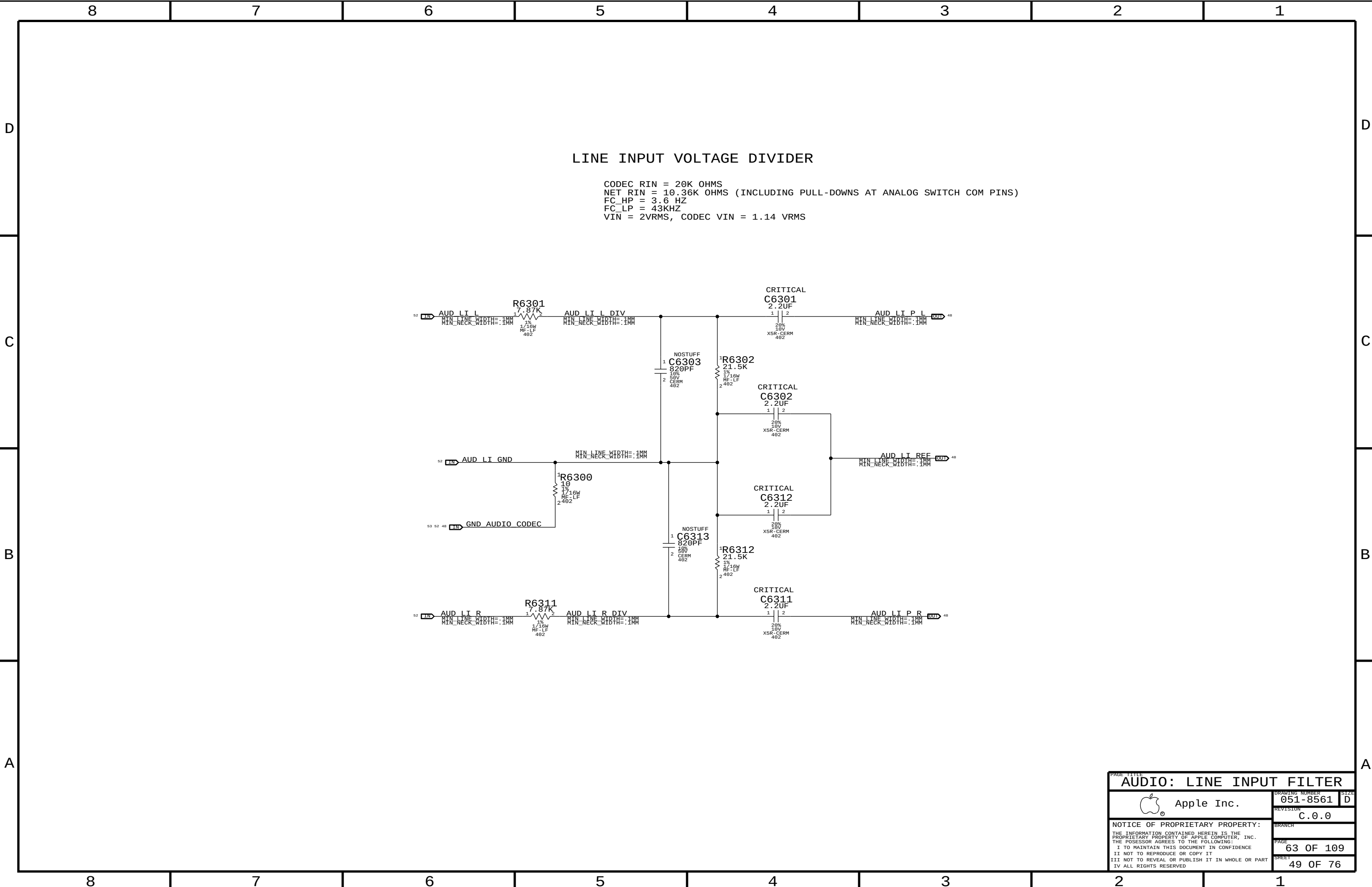
AUDIO CODEC
APPLE P/N 353S2355

4.5V POWER SUPPLY FOR CODEC
APPLE P/N 353S2281

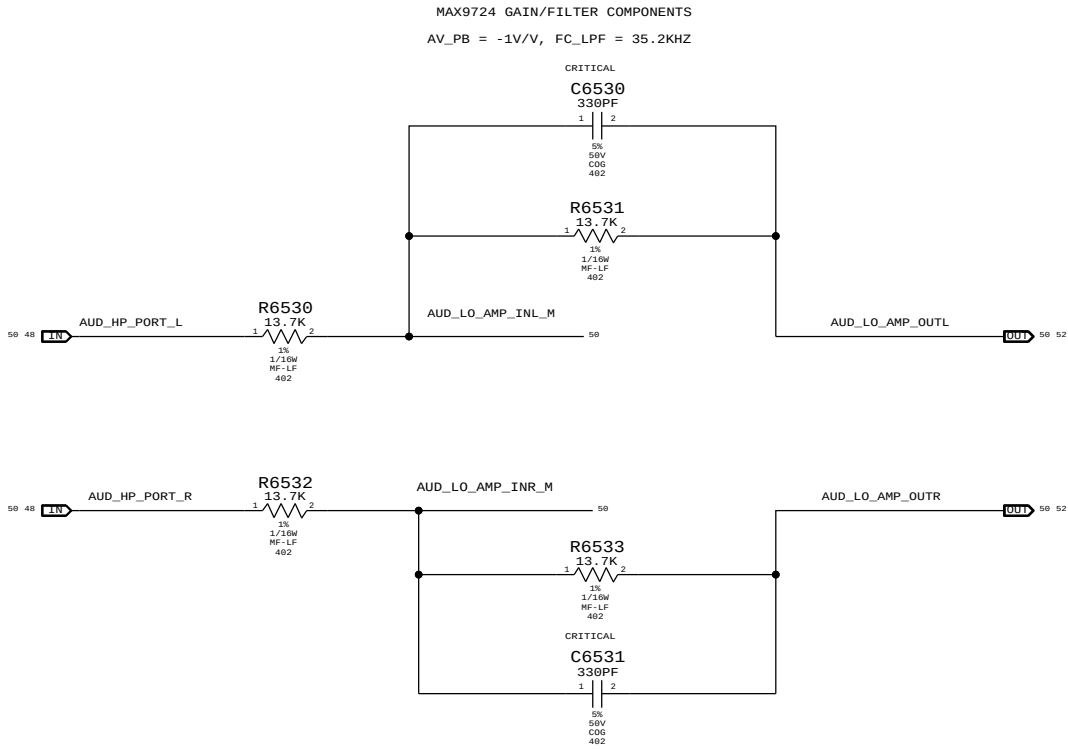
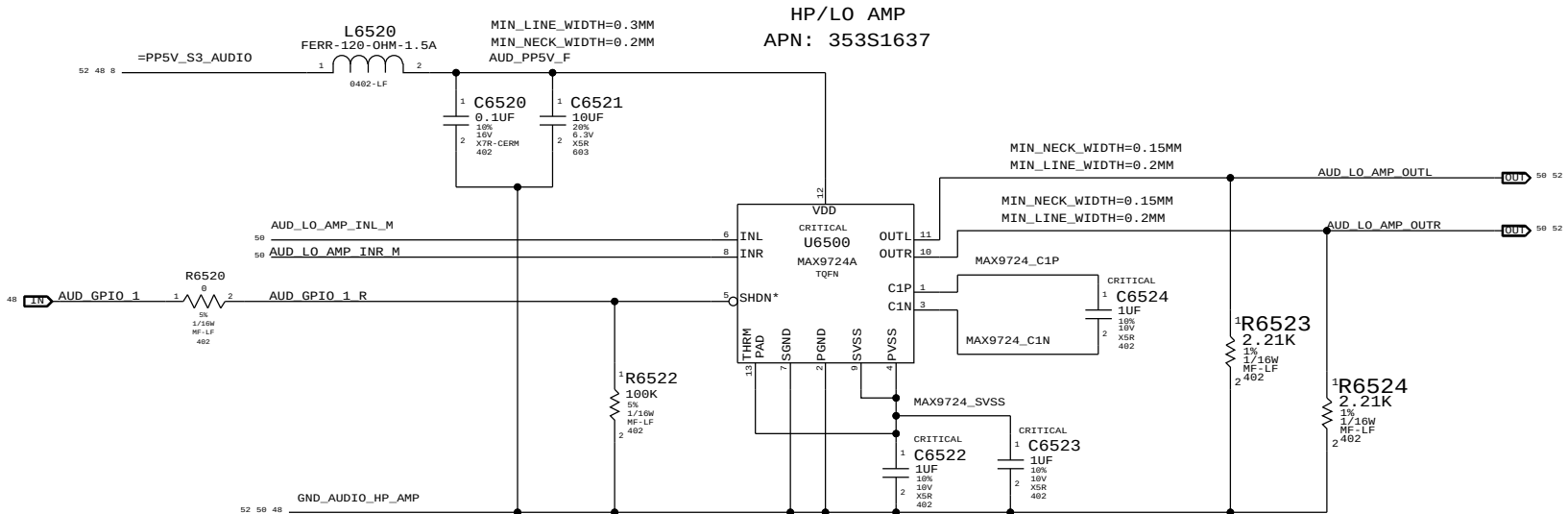
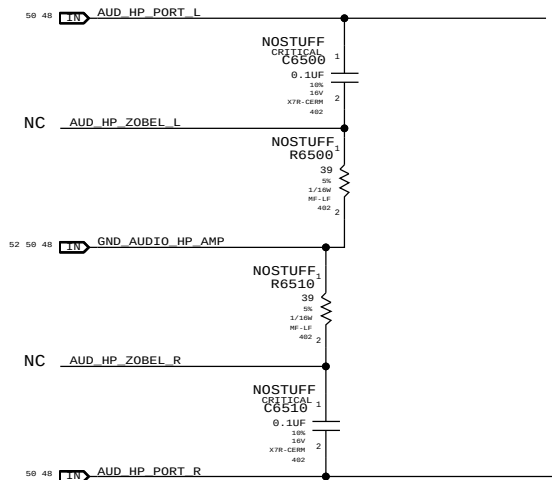
NOTES ON CODEC I/O

DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

SYNC MASTER=AUDIO		SYNC DATE=02/16/2010	
PAGE TITLE		AUDIO: CODEC/REGULATOR	
Apple Inc.		DRAWING NUMBER	051-8561
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		PAGE	62 OF 109
		SHEET	48 OF 76



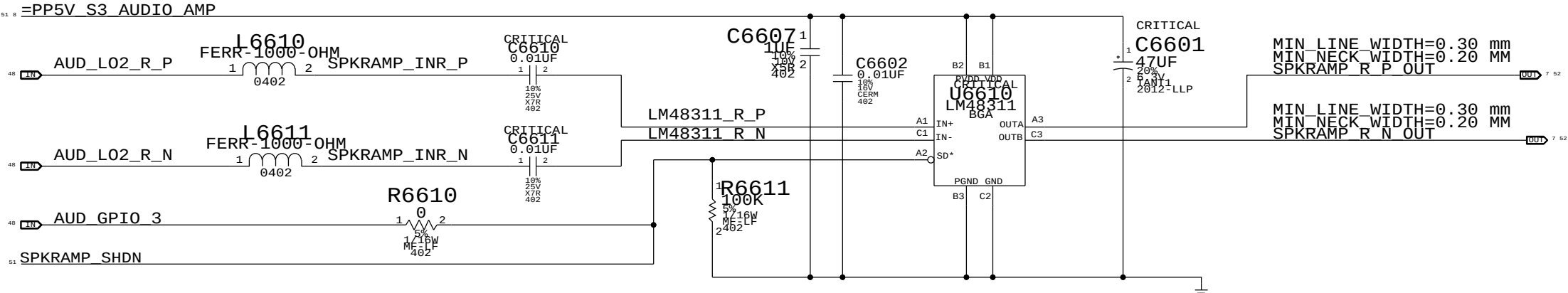
CS4206 HP OUTPUT ZOBEL NETWORK



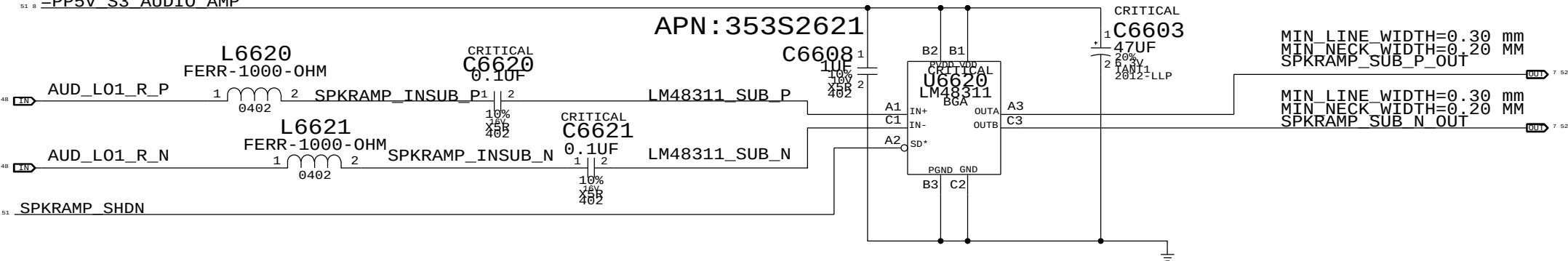
SYNC MASTER=AUDIO		SYNC DATE=02/16/2010	
PAGE TITLE		AUDIO: HEADPHONE FILTER	
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		PAGE	65 OF 109
		SHEET	50 OF 76

SATELLITE 796Hz < HPF FC < 936Hz
SUB 80 Hz < HPF FC < 94 Hz
GAIN 6DB (2V/V)
SPRK AMP. INPUT REFERRED CLIP POINT = ~-6dBFS

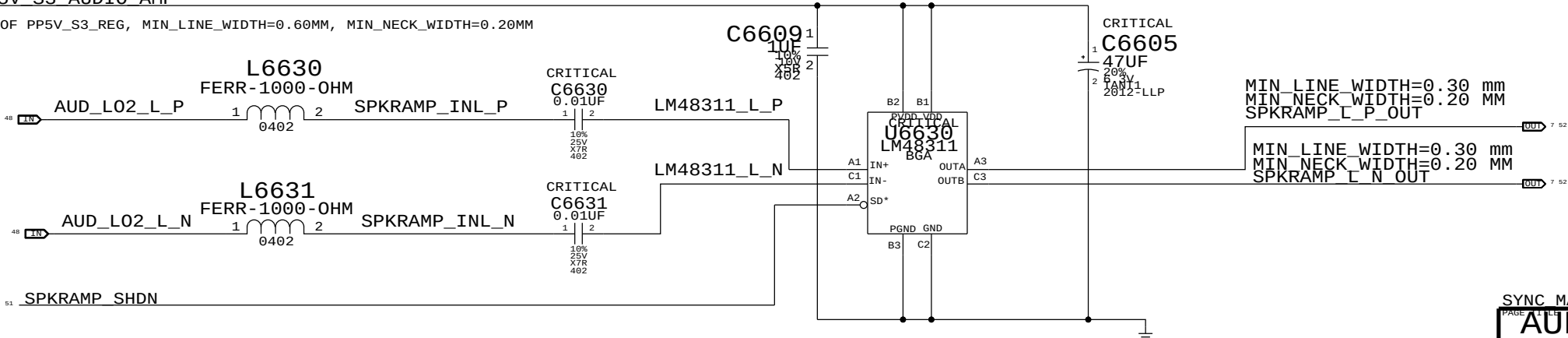
ALIAS OF PP5V_S3_REG, MIN_LINE_WIDTH=0.60MM, MIN_NECK_WIDTH=0.20MM



ALIAS OF PP5V_S3_REG, MIN_LINE_WIDTH=0.60MM, MIN_NECK_WIDTH=0.20MM

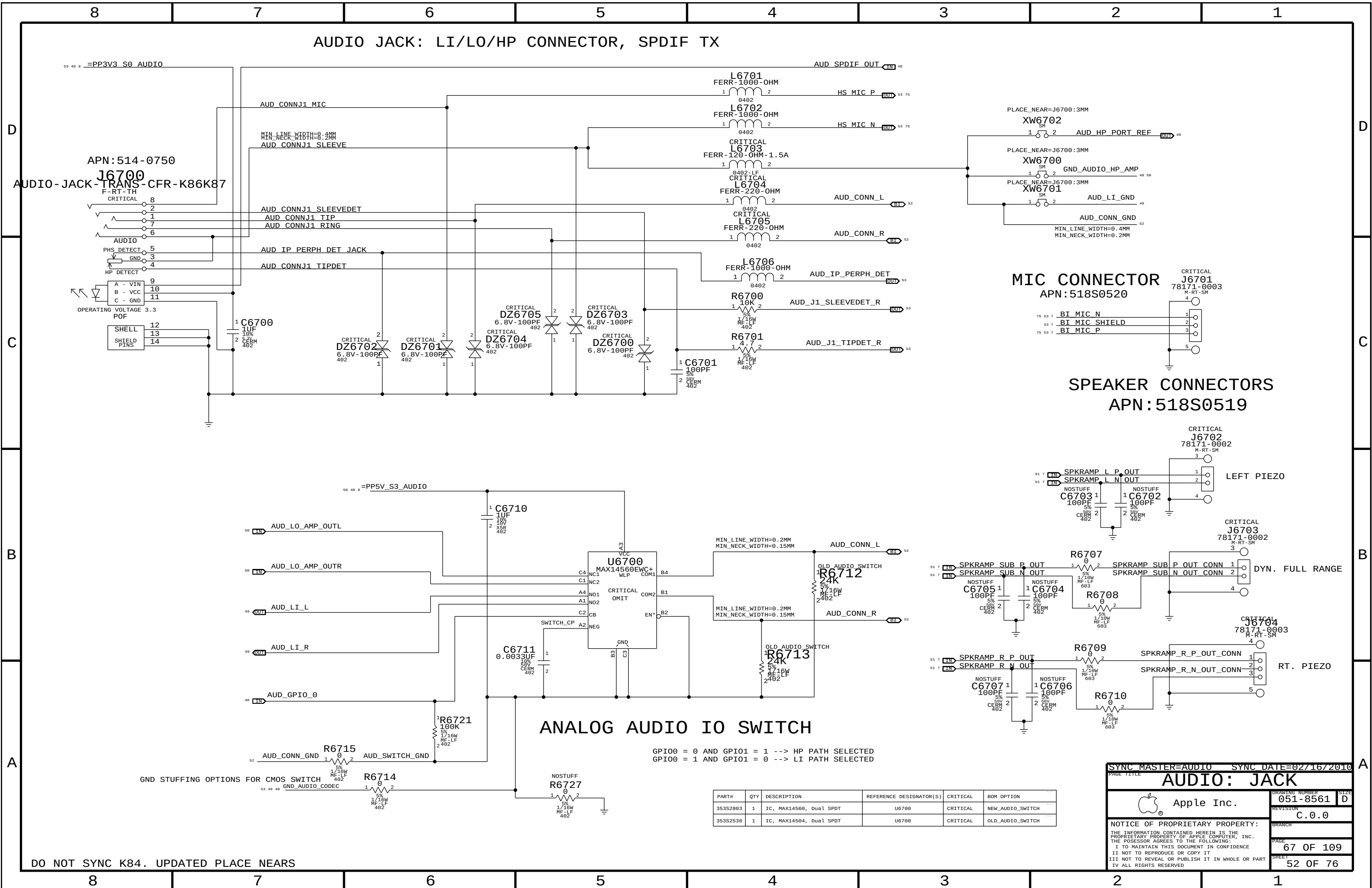


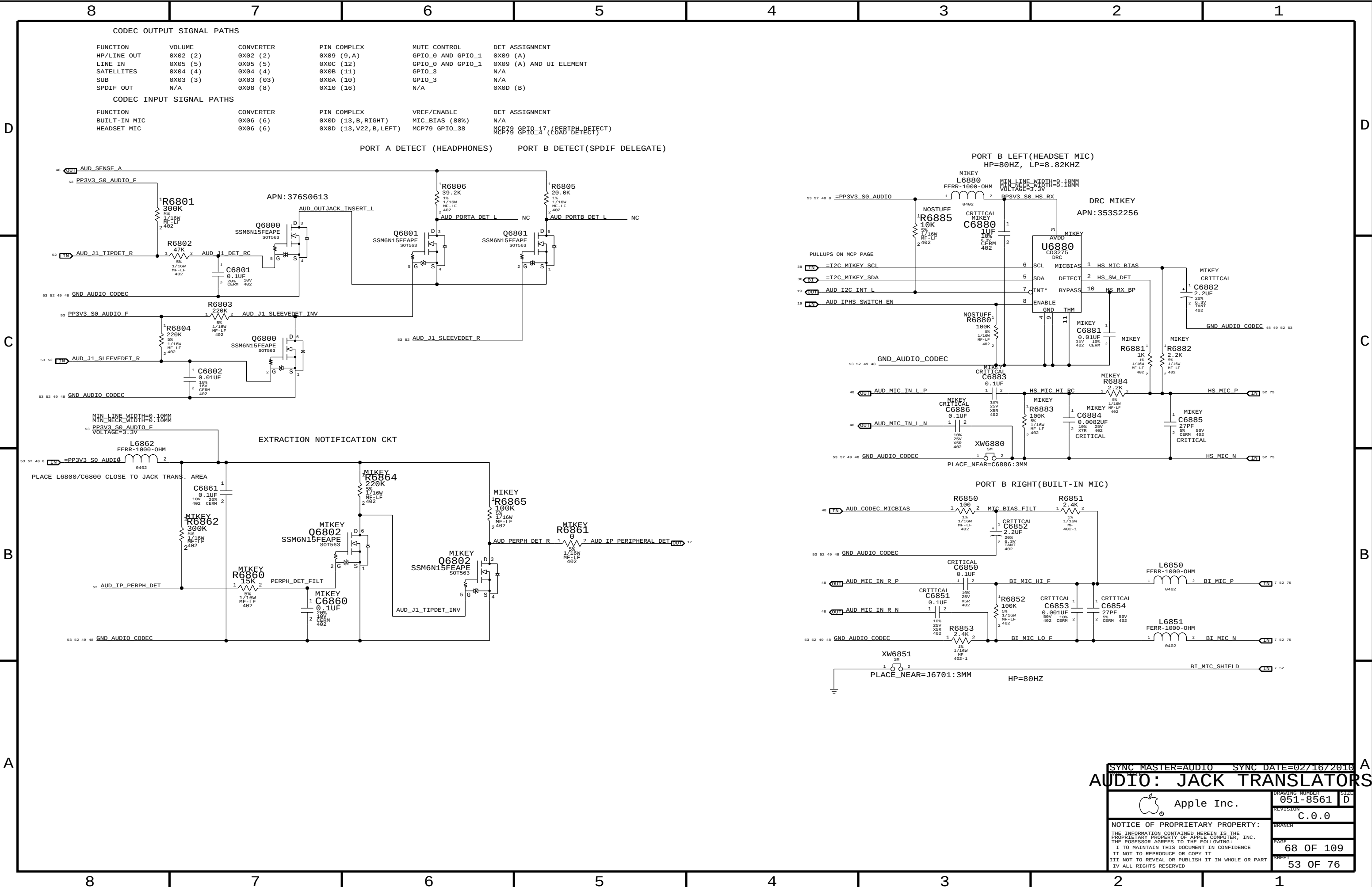
51 8 =PP5V_S3_AUDIO_AMP
ALIAS OF PP5V_S3_REG, MIN_LINE_WIDTH=0.60MM, MIN_NECK_WIDTH=0.20MM



PAGE 1 OF 1
SYNC MASTER=AUDIO SYNC DATE=02/16/2010

AUDIO: SPEAKER AMP		
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SYNC MASTER=AUDIO

SYNC DATE=02/16/2010

AUDIO: JACK TRANSLATORS

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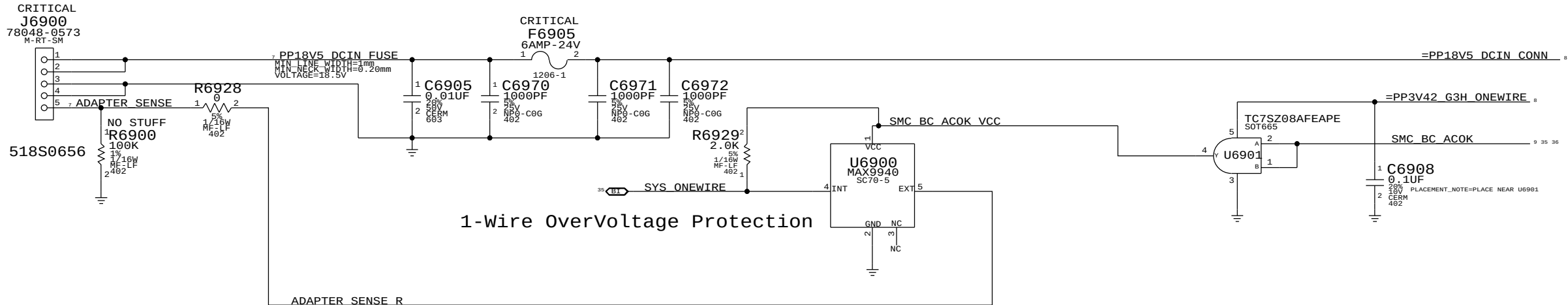
PAGE

68 OF 109

SHEET

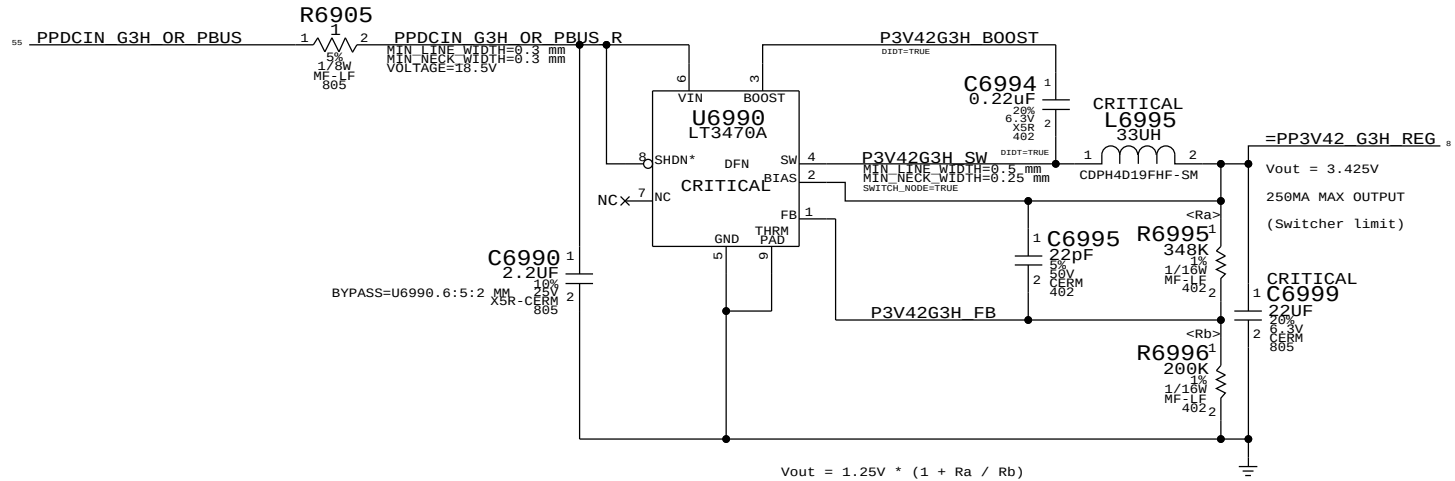
53 OF 76

MagSafe DC Power Jack



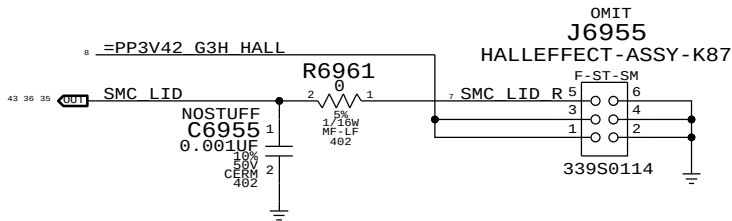
3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator



HALL EFFECT ASSEMBLY

Assembly APN: 339S0114
- BOM: 639-0680
- PCBF: 820-2801
- MCO: 056-3515
- Conn APN:518S0788

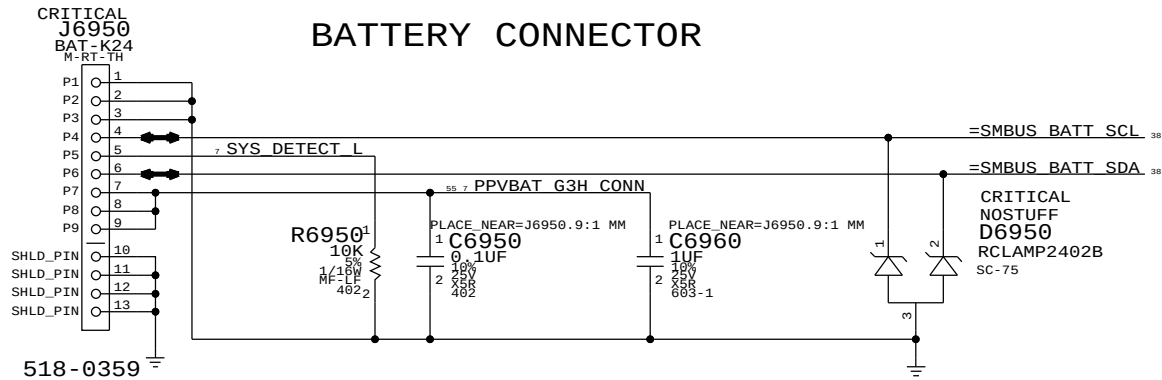


PROTO 0: STUFFING K84 CONNECTOR ONTO MODIFIED K84 PADS
PROTO 1: STUFFING K87 HALL EFFECT ASSEMBLY ONTO K87 PADS

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
607-6831	1	SUB ASSY - HALL EFFECT, K86 K87	J6955	CRITICAL	

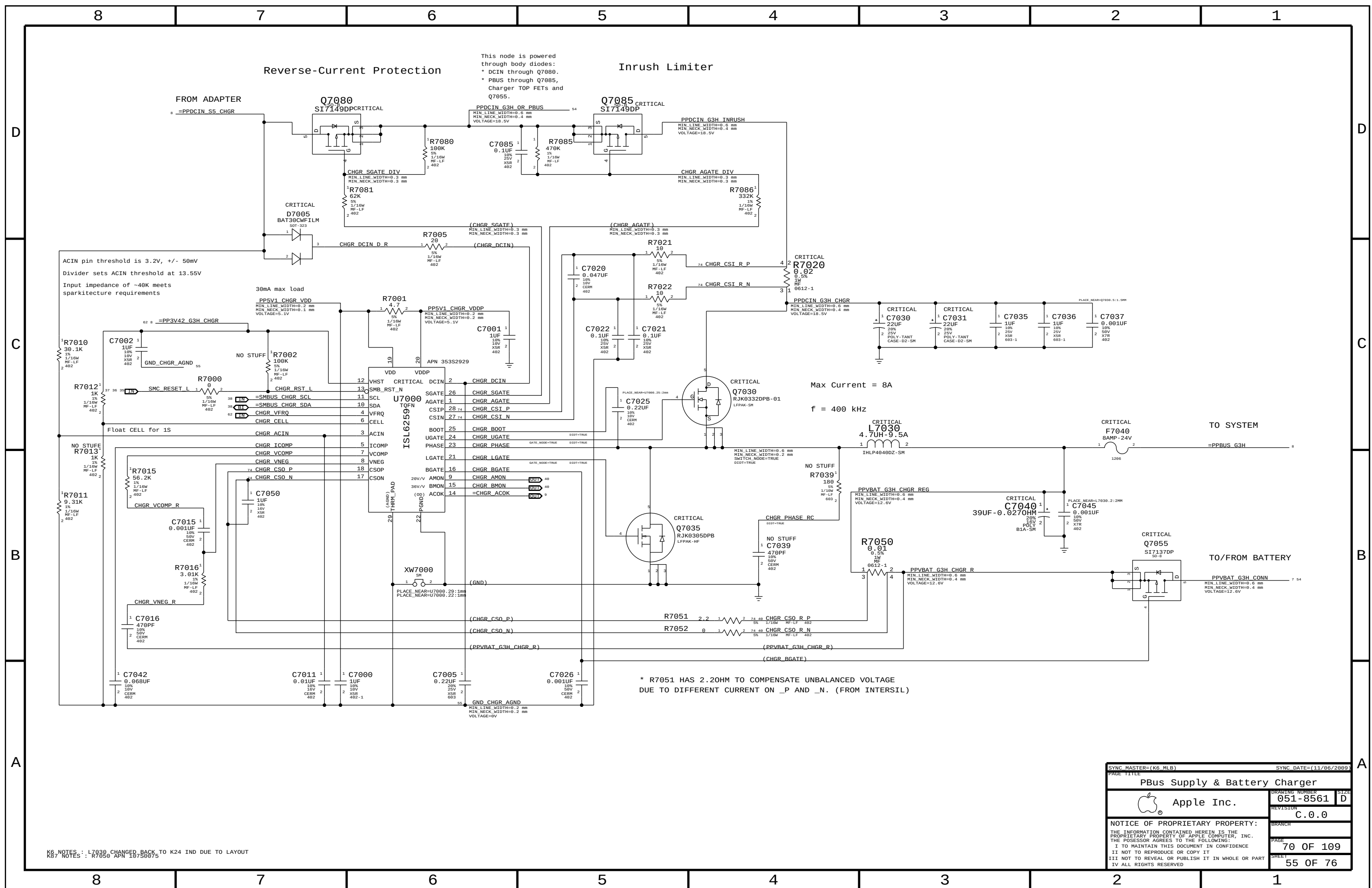
PN: 607-6831 for WCPM. PN: 339S0114 for schematic/board layout

BATTERY CONNECTOR



DO NOT SYNC WITH K84. R6900,C6960,SIGNAL NAMES CHANGED. HALL EFFECT CONNECTOR CHANGED.

PAGE TITLE		SYNC DATE=MASTER	
DC-In & Battery Connectors		051-8561	
Apple Inc.		C.0.0	
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5V_S3/3.3V_S5 POWER SUPPLY

$$V_{OUT} = (2 * R_A / R_B) + 2$$

$$V_{OUT} = (2 * R_C / R_D) + 2$$

ROUTING NOTE:
Place XW7203 by Pin1 OF L7260.

ROUTING NOTE:
Place XW7202 by C7260.

ROUTING NOTE:
Place XW7204 by Pin 2 of L7220.

ROUTING NOTE:
Place XW7205 by C7252.

ROUTING NOTE:
Place XW7201 between Pin 15 and Pin 25 of U7200.

MAX CURRENT = 13.3A
PWM FREQ. = 300 KHZ

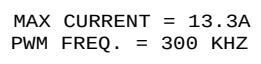
MAX CURRENT = 9.1A
PWM FREQ. = 375 KHZ

SEPERATED MASTER PG000 FOR BOTH 5V AND 3V3.

NOTE: DONT SYNC THIS PAGE FROM T27

SYNC MASTER=(K6 MLB)		SYNC DATE=(10/27/2009)			
PAGE TITLE					
5V/3.3V SUPPLY					
DRAWING NUMBER		051-8561			
REVISION		C.0.0			
BRANCH					
PAGE		72 OF 109			
SHEET		56 OF 76			
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$$V_{OUT} = (2 * RC / RD) + 2$$

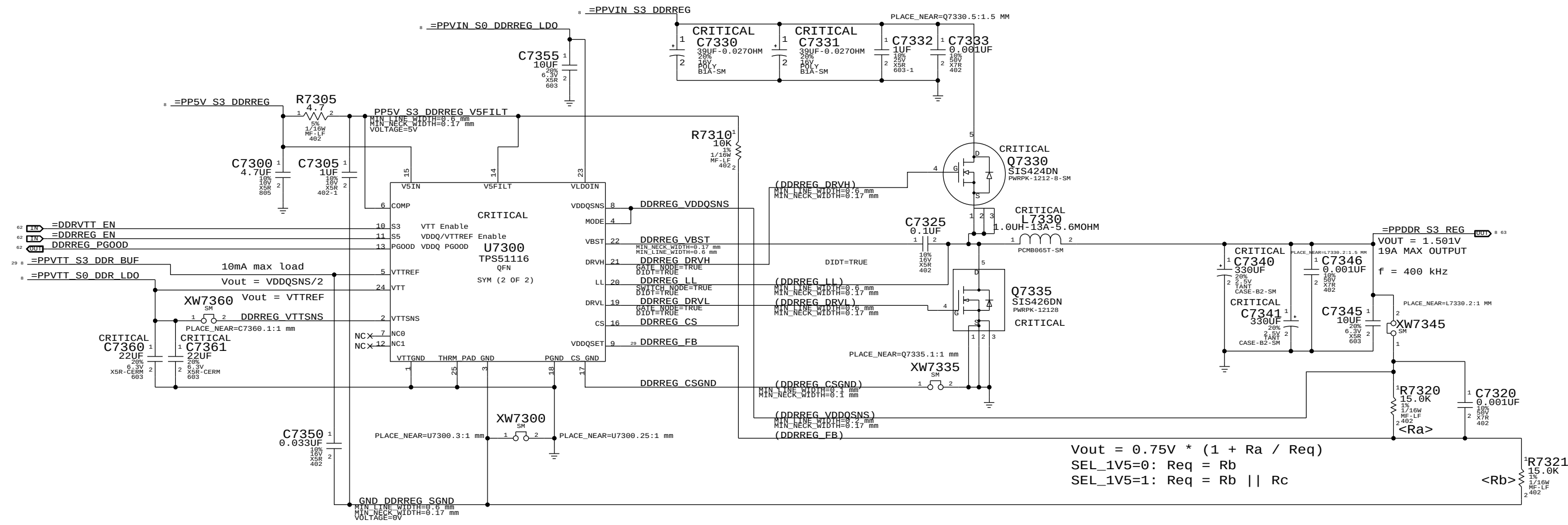


PWM FREQ. = 375 KHZ
MAX CURRENT = 9.1A

NOTE: DONT SYNC THIS PAGE FROM T27

SYNC MASTER=(K6 MLB)		SYNC DATE=(10/27/2009)	
PAGE TITLE			
5V/3.3V SUPPLY			
 Apple Inc.		DRAWING NUMBER 051-8561	SIZE D
		REVISION C.0.0	
BRANCH			
PAGE			
72 OF 109			
SHEET			
56 OF 76			

1.5V/0.75 DDR3 POWER SUPPLY

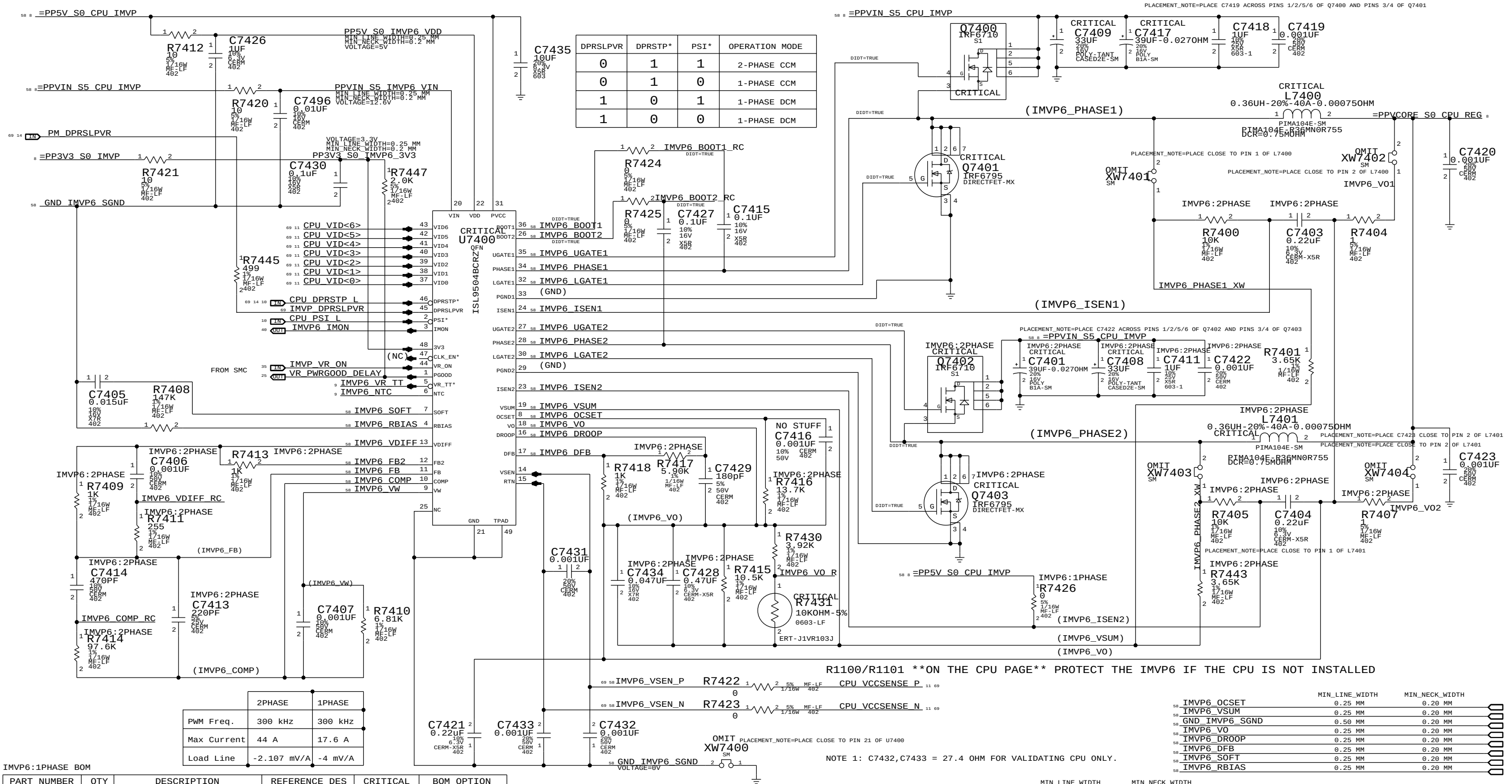


$$V_{out} = 0.75V * (1 + R_a / R_{eq})$$
$$SEL_1V5=0: R_{eq} = R_b$$
$$SEL_1V5=1: R_{eq} = R_b || R_c$$

NOTE: DONT SYNC THIS PAGE FROM T27. C7330 AND C7331 IS CHANGED TO OSCON CAPS
NOTE: DONT SYNC THIS PAGE FROM K6 REMOVED R7380

SYNC MASTER=(K6 MLB)		SYNC DATE=(11/06/2009)	
PAGE TITLE			
1.5V/0.75V		DDR3	SUPPLY
		DRAWING NUMBER	SIZE
Apple Inc.		051-8561	D
		REVISION	
		C.0.0	
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IMVP6 CPU VCORE REGULATOR



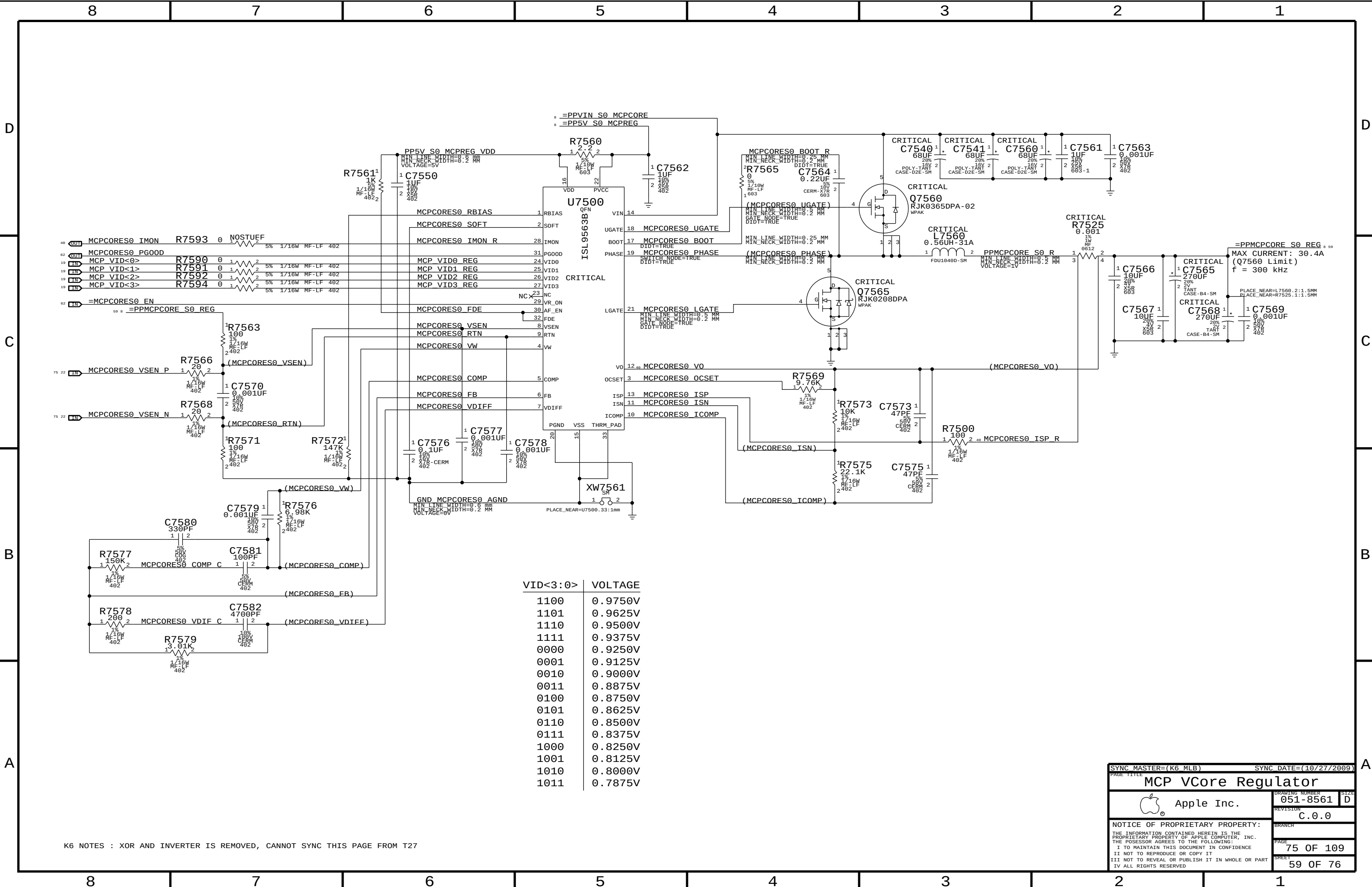
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
11450307	1	RES,MTL FILM,1/16W,8.25K,1,0402,SMD,LF	R7417		IMVP6:1PHASE
11450336	1	RES,MTL FILM,1/16W, 10.9K, 1,0402,SM,LF	R7416		IMVP6:1PHASE
132S0080	1	CAP,CER.,22UF,20,6.3V,X5R,0402	C7428		IMVP6:1PHASE
11450236	1	RES,MTL FILM,1/16W,1.58K,1,0402,SMD,LF	R7409		IMVP6:1PHASE
11450160	1	RES,MTL FILM,1/16W,255 OHM,1,0402,SMD,LF	R7411		IMVP6:1PHASE
132S4720	1	CAP CER 470PF,+/-10%,50V,0402,SMD	C7406		IMVP6:1PHASE
11450410	1	RES,MTL FILM,1/16W,97.6K,1,0402,SMD,LF	R7414		IMVP6:1PHASE
132S0045	1	CAP,CER,1000PF,50V,10%,X7R,0402,SMD	C7414		IMVP6:1PHASE
131S1027	1	CAP,CER,100PF,5V,50V,CC0402	C7413		IMVP6:1PHASE

		MIN. LINE WIDTH	MIN. NECK WIDTH
58	IMVP6_VDIFF	0.25 MM	0.20 MM
58	IMVP6_FB2	0.25 MM	0.20 MM
58	IMVP6_FB	0.25 MM	0.20 MM
58	IMVP6_COMP	0.25 MM	0.20 MM
58	IMVP6_VW	0.25 MM	0.25 MM
69	IMVP6_VSEN_P	0.25 MM	0.25 MM
69	IMVP6_VSEN_N	0.25 MM	0.25 MM

	MIN_LINE_WIDTH	MIN_NECK_WIDTH	
58 IMVP6_PHASE1	1.5 MM	0.25 MM	
58 IMVP6_B00T1	0.25 MM	0.25 MM	
58 IMVP6_UGATE1	1.5 MM	0.25 MM	
58 IMVP6_LGATE1	1.5 MM	0.25 MM	
58 IMVP6_ISEN1	0.25 MM	0.25 MM	
	MIN_LINE_WIDTH	MIN_NECK_WIDTH	
58 IMVP6_PHASE2	1.5 MM	0.25 MM	
58 IMVP6_B00T2	0.25 MM	0.25 MM	
58 IMVP6_UGATE2	0.25 MM	0.25 MM	
58 IMVP6_LGATE2	0.25 MM	0.25 MM	
58 IMVP6_ISEN2	0.25 MM	0.25 MM	

	MIN_LINE_WIDTH	MIN_NECK_WIDTH	
SET	0.25 MM	0.20 MM	
UM	0.25 MM	0.20 MM	
6_SGND	0.50 MM	0.20 MM	
	0.25 MM	0.20 MM	
00P	0.25 MM	0.20 MM	
B	0.25 MM	0.20 MM	
FT	0.25 MM	0.20 MM	
IAS	0.25 MM	0.20 MM	

SYNC MASTER=(K84 MLB)		SYNC DATE=(11/18/2009)	
PAGE TITLE			
IMVP6 CPU VCore Regulator			
 Apple Inc.		DRAWING NUMBER	SIZ
		051-8561	D
		REVISION	
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VID<3:0>	VOLTAGE
1100	0.9750V
1101	0.9625V
1110	0.9500V
1111	0.9375V
0000	0.9250V
0001	0.9125V
0010	0.9000V
0011	0.8875V
0100	0.8750V
0101	0.8625V
0110	0.8500V
0111	0.8375V
1000	0.8250V
1001	0.8125V
1010	0.8000V
1011	0.7875V

K6 NOTES : XOR AND INVERTER IS REMOVED, CANNOT SYNC THIS PAGE FROM T27

SYNC_MASTER=(K6_MLB)

SYNC_DATE=(10/27/2009)

MCP VCore Regulator

Apple Inc.

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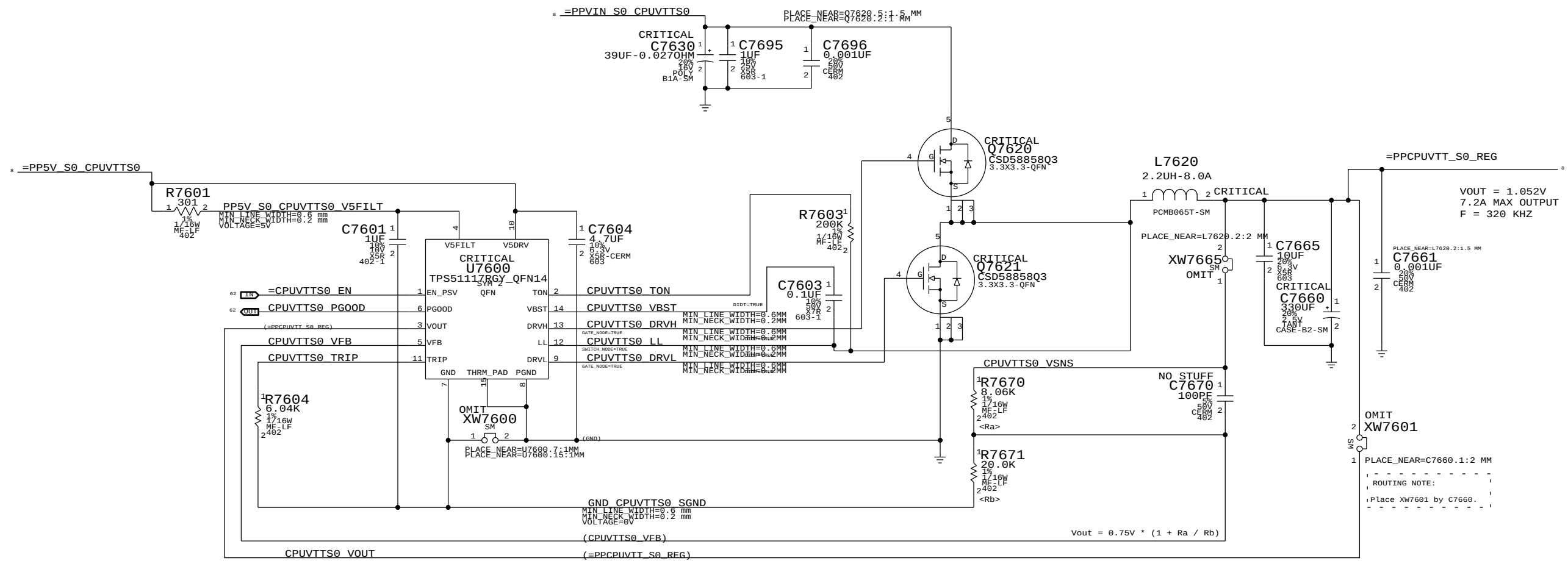
PAGE

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CPUVTT POWER SUPPLY



SYNC MASTER=(K84_MLB)		SYNC DATE=(02/04/2009)	
PAGE TITLE		DRAWING NUMBER	
CPU VTT(1.05V)		051-8561	
SUPPLY		D	
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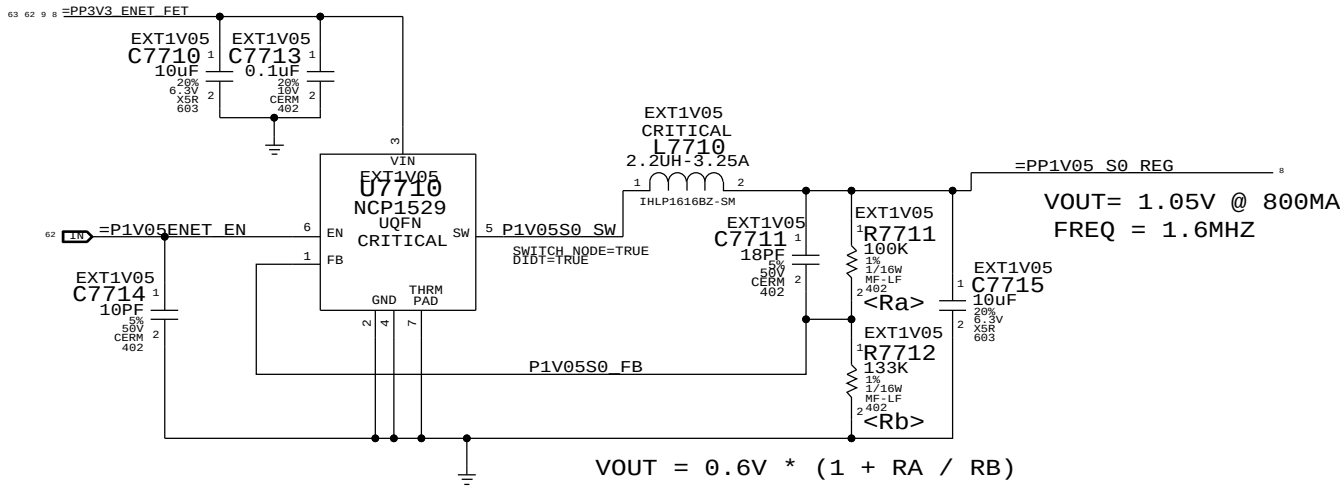
D

C

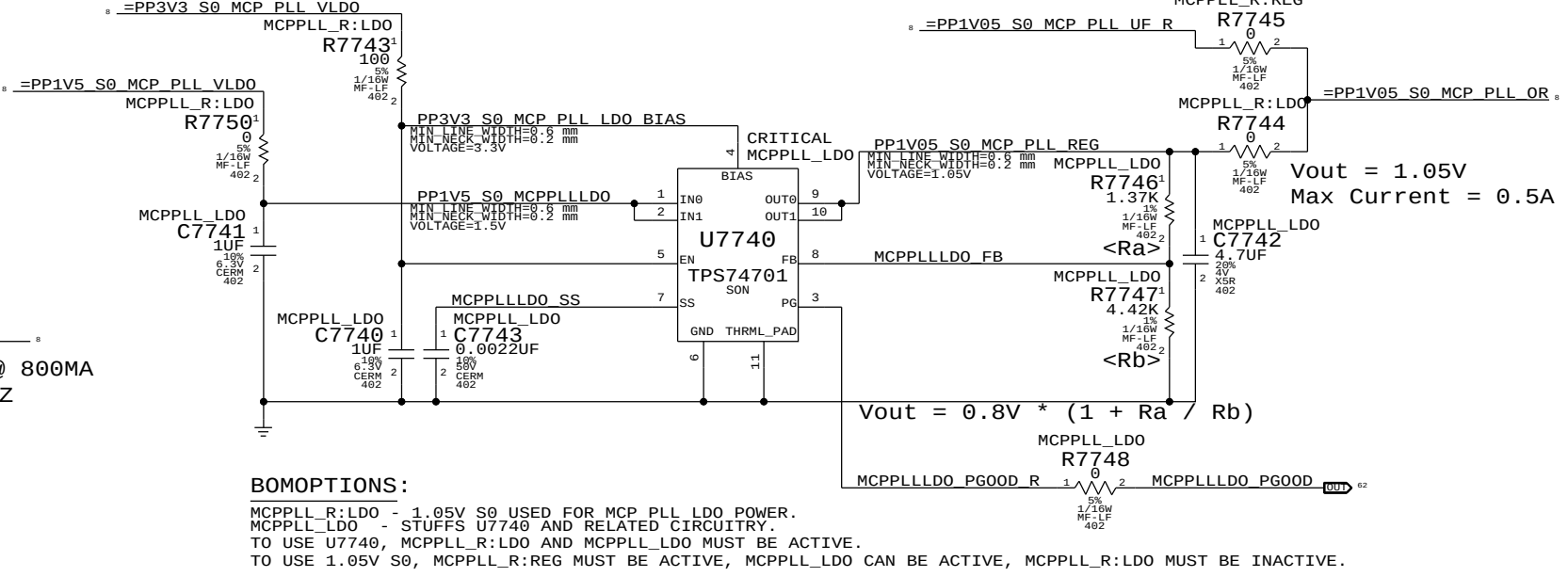
B

A

1.05V ENET Switcher

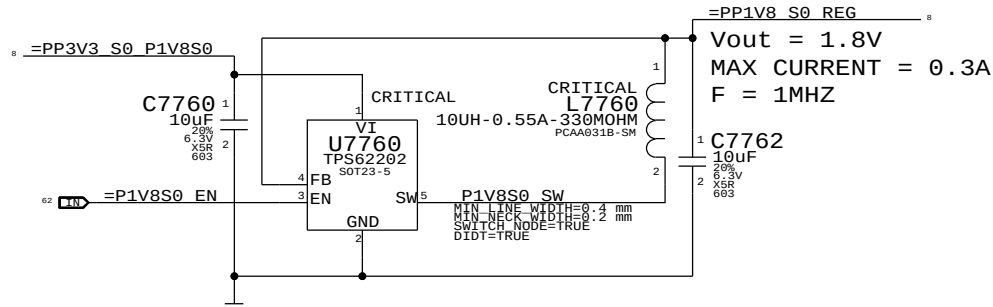


1.05V S0 MCP PLL LDO

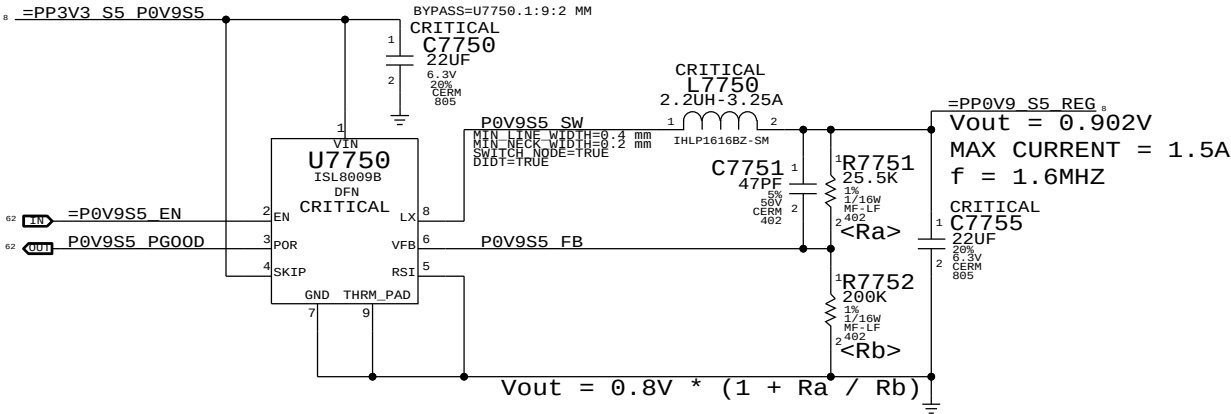


BOMOPTIONS:
MCPPLL_R:LDO - 1.05V S0 USED FOR MCP PLL LDO POWER.
MCPPLL_LDO - STUFFS U7740 AND RELATED CIRCUITRY.
TO USE U7740, MCPPLL_R:LDO AND MCPPLL_LDO MUST BE ACTIVE.
TO USE 1.05V S0, MCPPLL_R:REG MUST BE ACTIVE, MCPPLL_LDO CAN BE ACTIVE, MCPPLL_R:LDO MUST BE INACTIVE.

1.8V S0 Switcher



MCP 0.9V S5 (AUXC) Switcher



K6 NOTES : C7710 AND C7750 HAS BYPASS PROPERTY, SHOULD BE ADDED INCASE THIS PAGE IS SYNC'ED FROM T27

SYNC MASTER=MASTER		SYNC DATE=MASTER	
PAGE TITLE		SIZE	
Misc Power Supplies		051-8561 D	
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8	7	6	5	4	3	2	1
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State	SMC_PM_G2_ENABLE	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1
Sleep (S3)	1	1	0
Soft-Off (S5)	1	0	0
Battery Off (G3Hot)	0	0	0

[illegible]

C



PM WLAN EN L

Pull-up is with power FET.

30

7890

SSM6N15FEAPE SOT563

AP_PWR_EN

30 19

AC_OR_S0_L

7890

SSM6N15FEAPE SOT563

SMC_ADAPTER_EN

36 35 19

7891

SSM6N15FEAPE SOT563

PM_SLP_S3_L

66 62 35 19 7

[illegible]

Worst-Case Thresholds:
Q2: 0.XXXV
Q3: 0.640V
Q4: 0.660V
3.3V w/Divider: 2.345V

B



VTT rail must ramp up in about the same time as MEMVDD rail (Q2300).

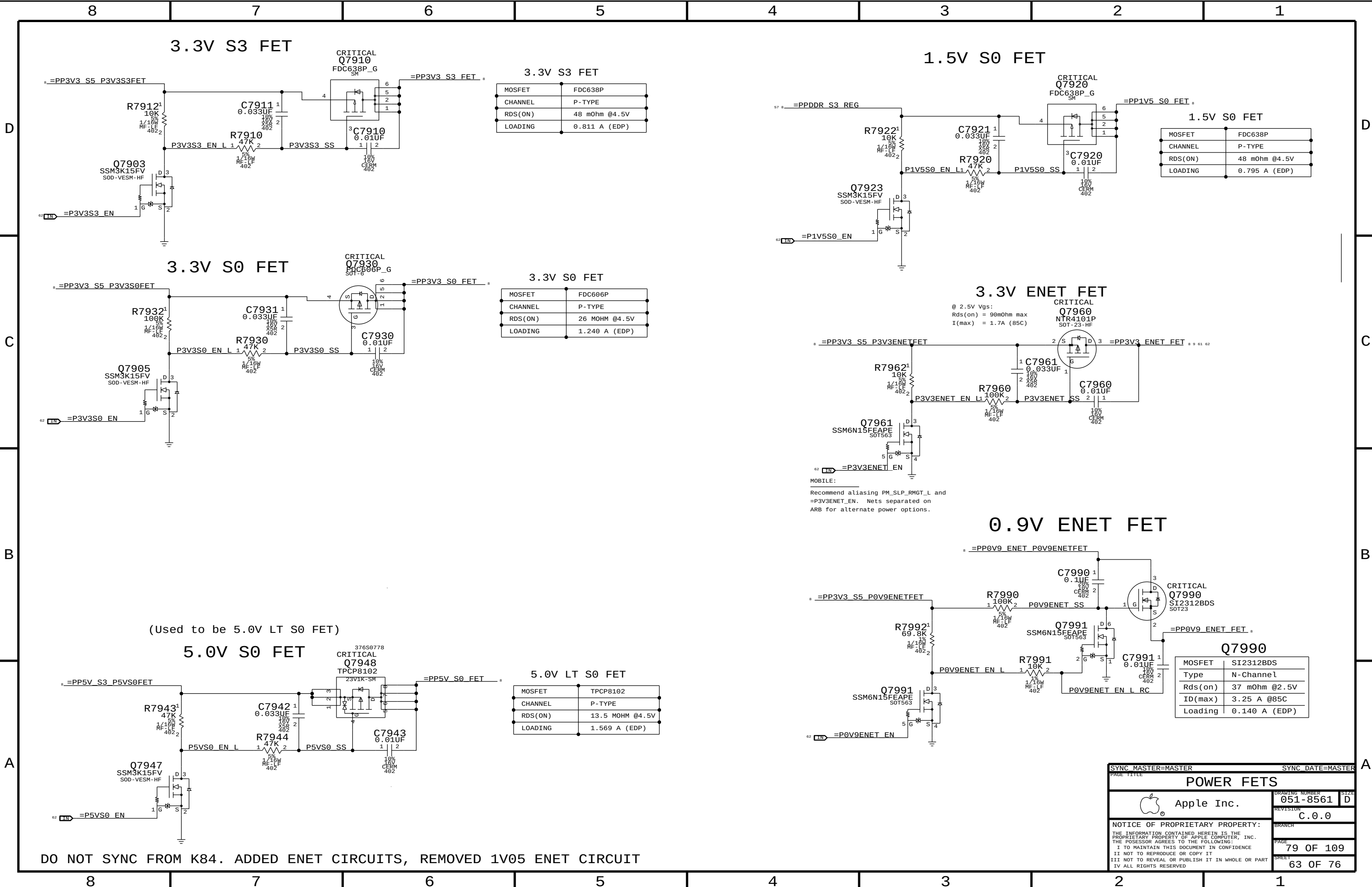
VTT rail must ramp up in about the same time as MEMVDD rail (Q2300).

```
57  IN  DDRREG PG00D  =  TP  DDRREG PG00D
      =  MAKE BASE=TRUE
```

SYMC MASTER=(T27 MLB)		SYMC DATE=(10/27/2009)	
PAGE TITLE			
Power Sequencing			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-8561	D
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		78	OF 109
		SHEET	
		62	OF 76

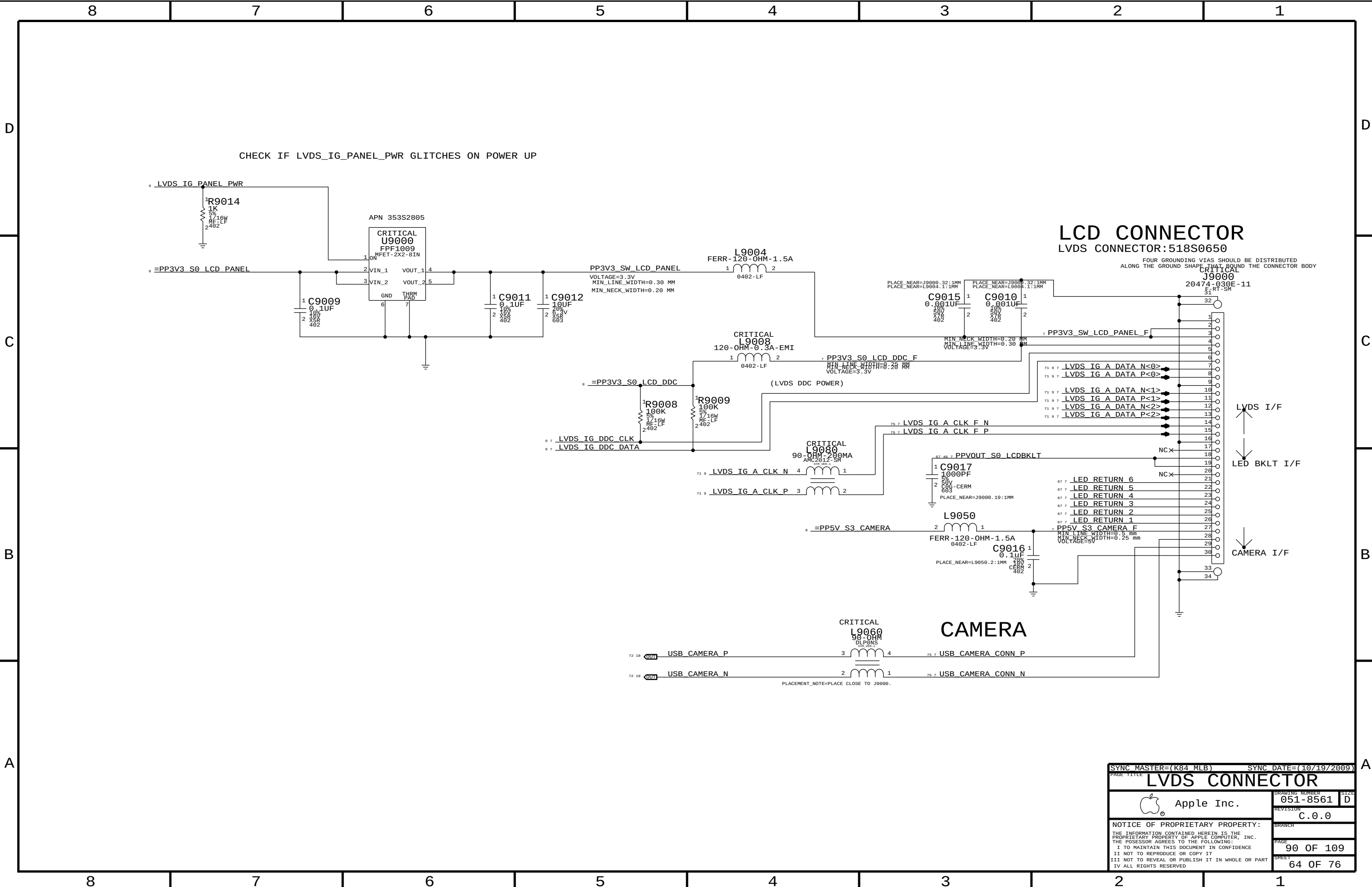
8	7	6	5	4	3	2	1
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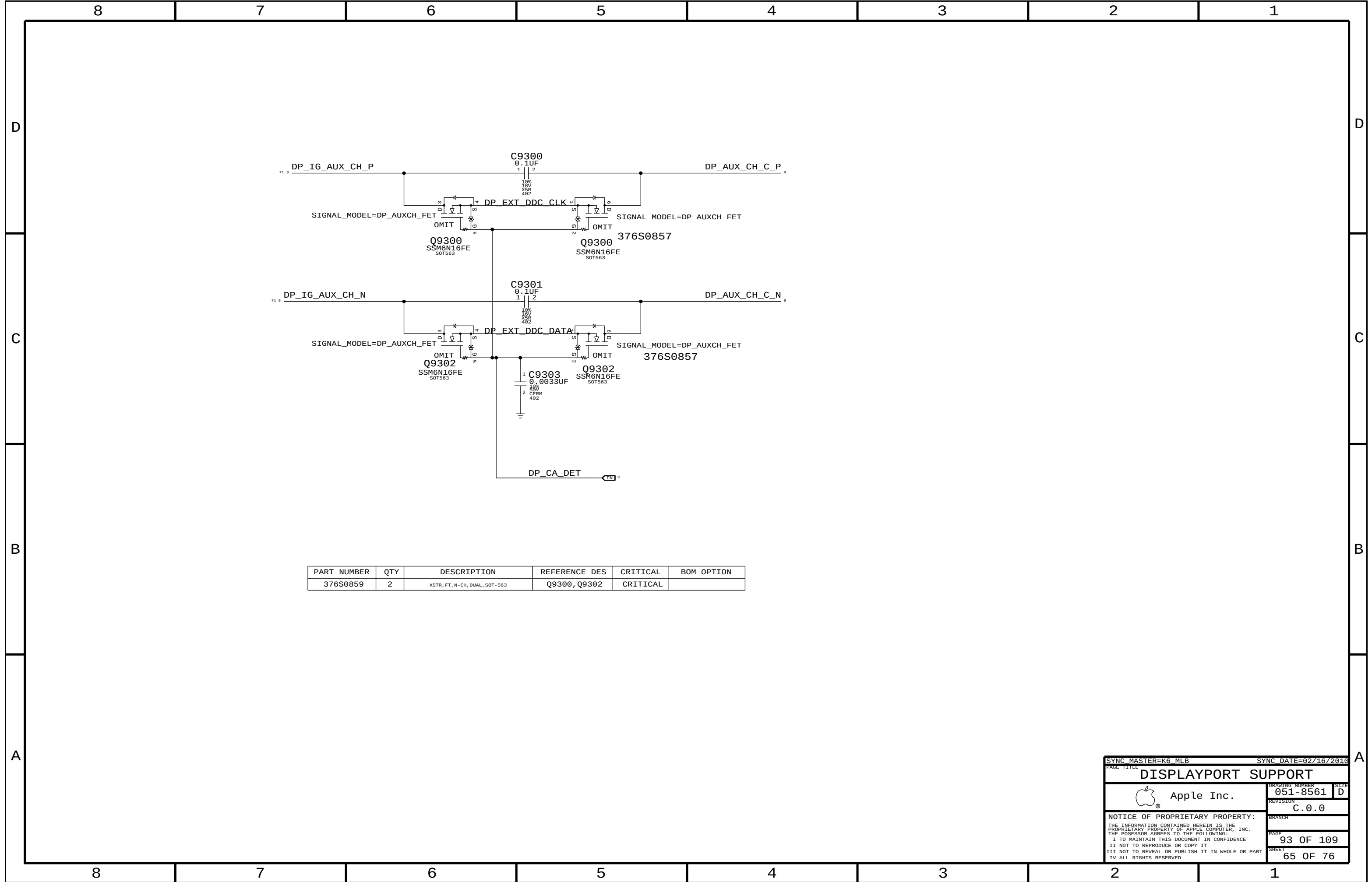
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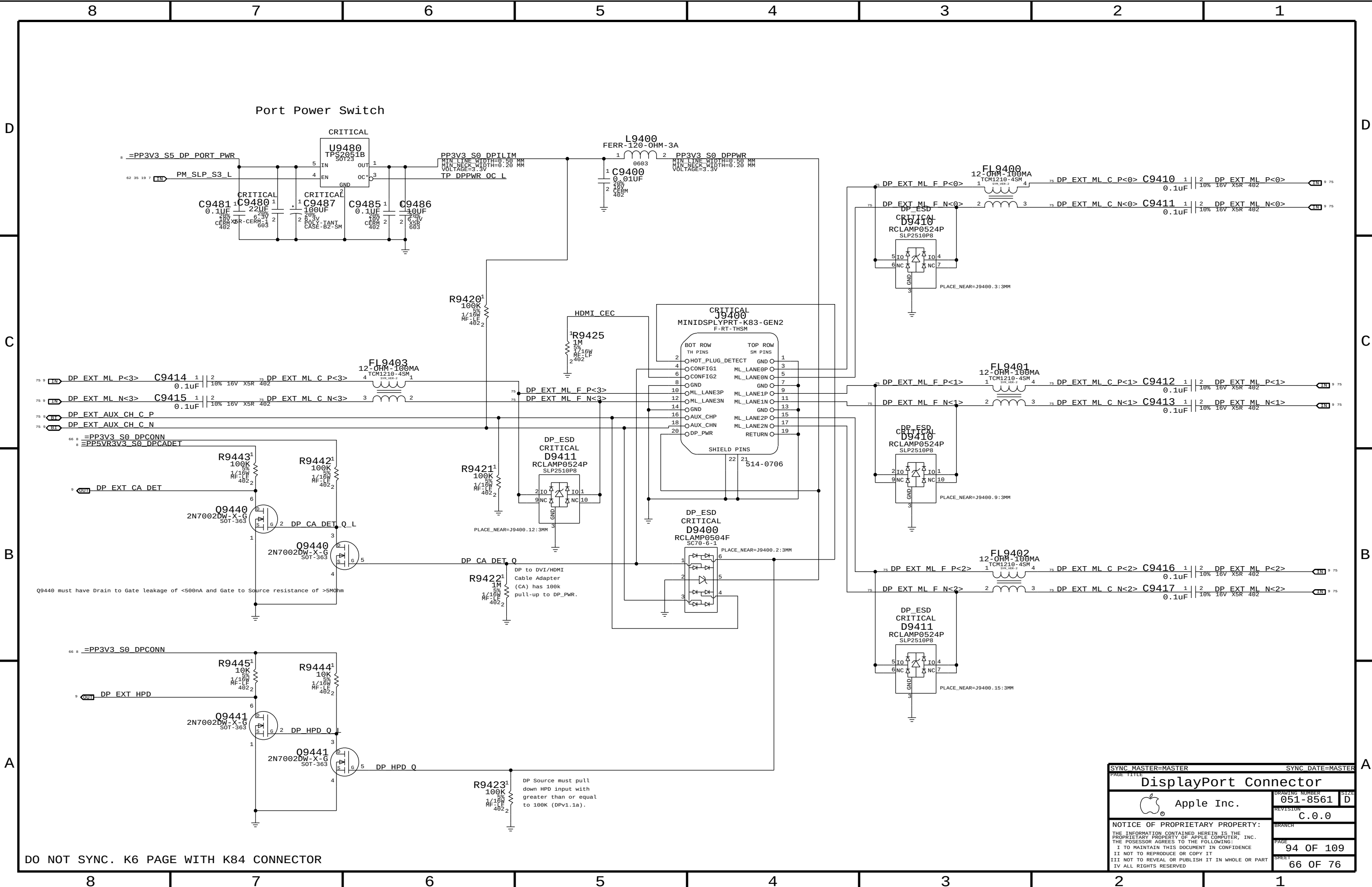


DO NOT SYNC FROM K84. ADDED ENET CIRCUITS, REMOVED 1V05 ENET CIRCUIT

SYNC MASTER=MASTER		SYNC DATE=MASTER	
PAGE TITLE		POWER FETS	
 Apple Inc.		DRAWING NUMBER	051-8561
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		SHEET	63 OF 76

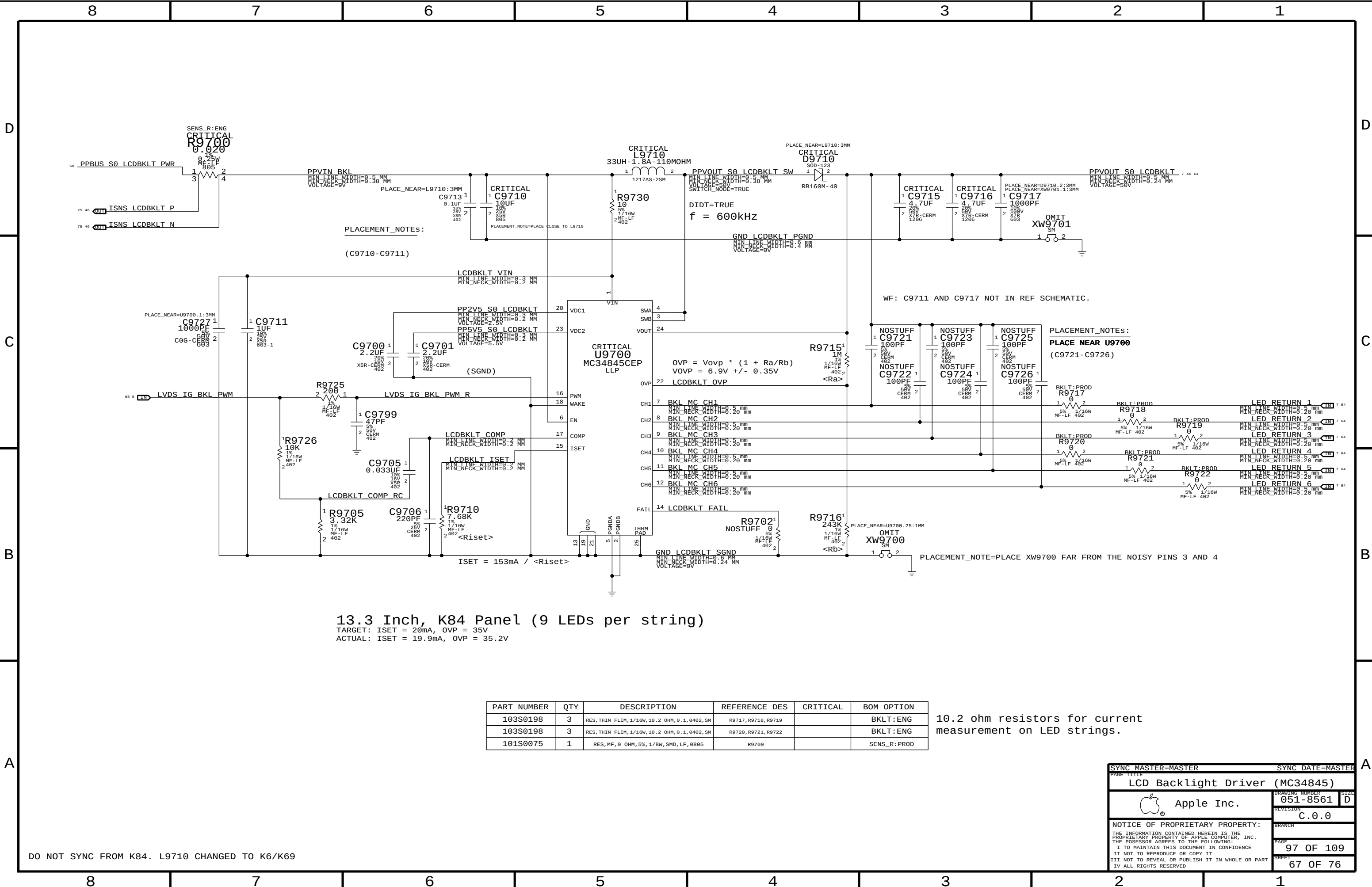






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PAGE TITLE		SYNC DATE=MASTER	
DisplayPort Connector		DRAWING NUMBER	
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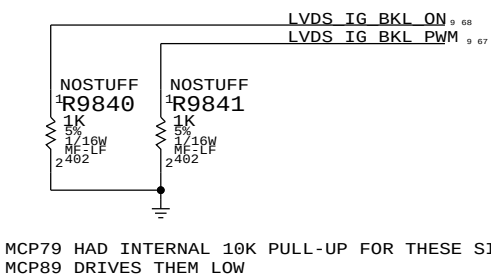
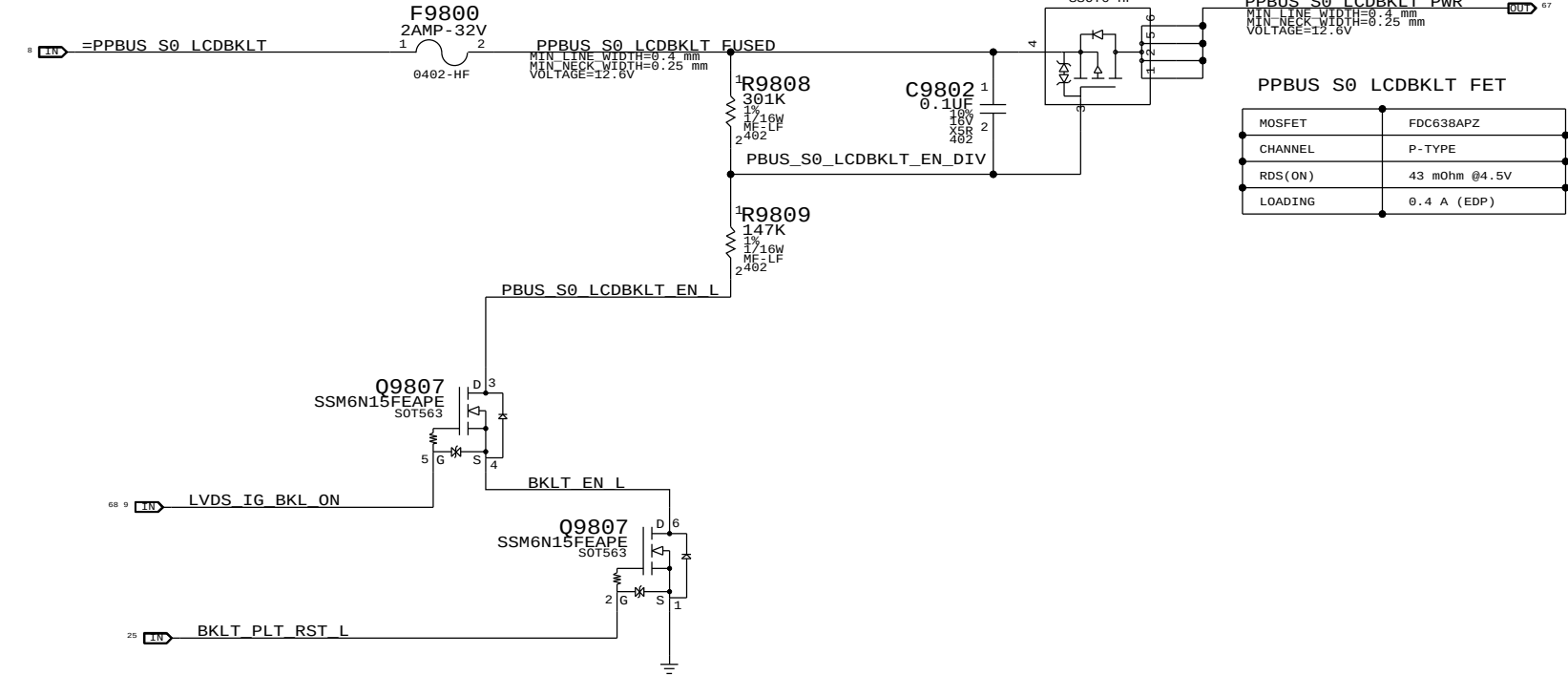
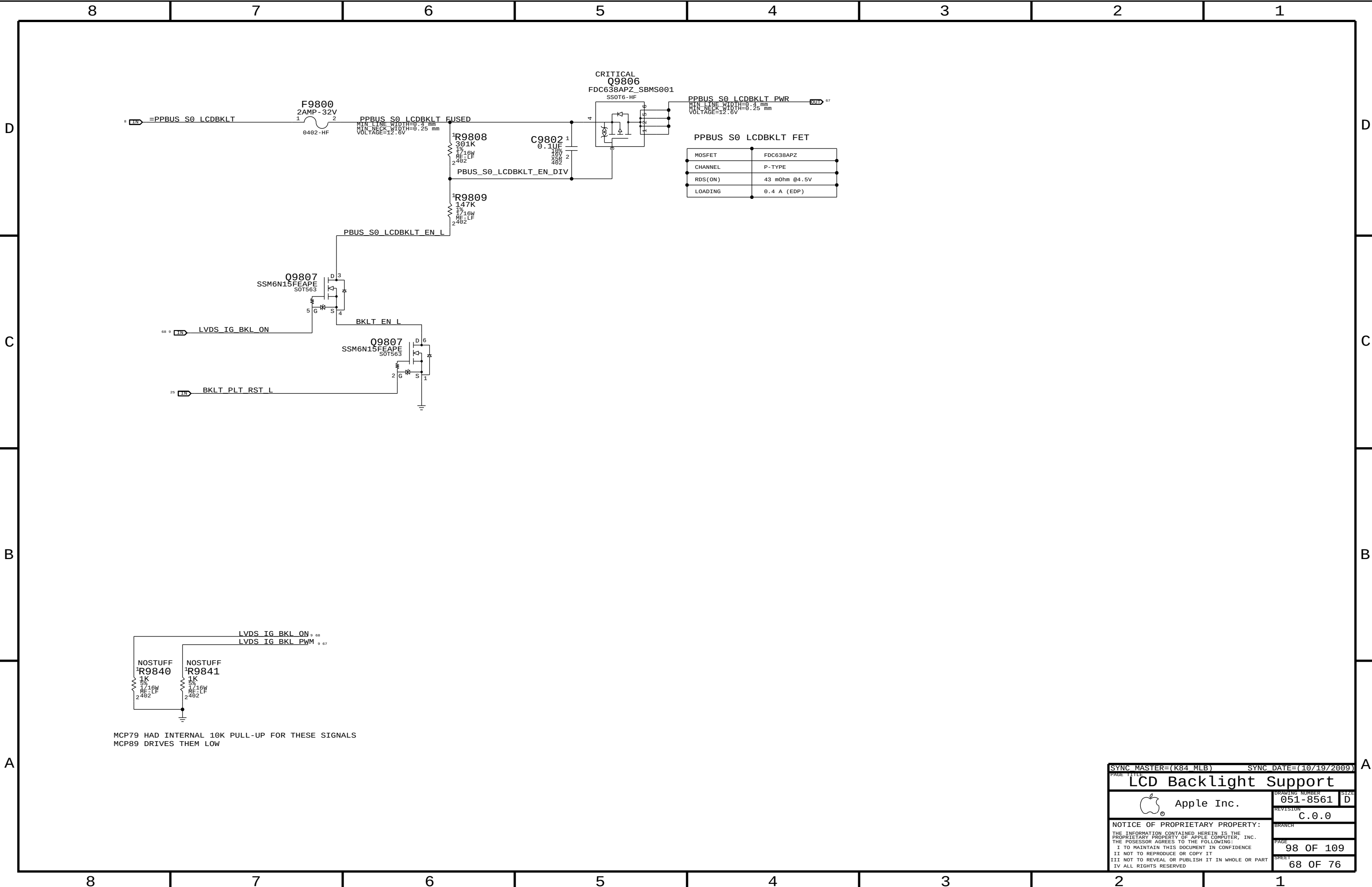
13.3 Inch, K84 Panel (9 LEDs per string)
TARGET: ISET = 20mA, OVP = 35V
ACTUAL: ISET = 19.9mA, OVP = 35.2V

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9717, R9718, R9719		BKLT:ENG
103S0198	3	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9720, R9721, R9722		BKLT:ENG
101S0075	1	RES, MF, 0 OHM, 5%, 1/8W, SMD, LF, 0805	R9700		SENS_R: PROD

10.2 ohm resistors for current measurement on LED strings.

DO NOT SYNC FROM K84. L9710 CHANGED TO K6/K69

SYNC MASTER=MASTER		SYNC DATE=MASTER	
PAGE TITLE			
LCD Backlight Driver		(MC34845)	
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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_20THER	*	25 MIL	?

NV DG says 3x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 4x inner, 5x outer

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM	MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM	MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM	MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM	MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM	MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

DDR3: Need to support MEM_*-style wildcards!
DQS signals should be matched within 5 ps of associated DQS pair.
DQS intra-pair matching should be within 1 ps, inter-pair matching should be within 360 ps.
No DQS to clock matching requirement.
CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 2 ps.
CMD/CTRL signals should be matched within 150 ps.
All memory signals maximum length is 1.030 ps.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.2.3
SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Section 6.2

MCP MEM COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MEM_COMP	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_MEM_COMP	*	=2x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.2.2

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	MEM_A_CLK	MEM_70D	MEM_CLK	MEM A CLK P<5..0>	15 26
	MEM_A_CLK	MEM_70D	MEM_CLK	MEM A CLK N<5..0>	15 26
	MEM_A_CKE	MEM_40S	MEM_CTRL	MEM A CKE<3..0>	15 21 26
	MEM_A_CNTRL	MEM_40S	MEM_CTRL	MEM A CS L<3..0>	15 26
	MEM_A_CNTRL	MEM_40S	MEM_CTRL	MEM A ODT<3..0>	15 26
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM A A<15..0>	15 26
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM A BA<2..0>	15 26
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM A RAS L	15 26
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM A CAS L	15 26
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM A WE L	15 26
	MEM_A_DQ_BYTE0	MEM_40S	MEM_DATA	MEM A DQ<7..0>	15 28
	MEM_A_DQ_BYTE1	MEM_40S	MEM_DATA	MEM A DQ<15..8>	15 28
	MEM_A_DQ_BYTE2	MEM_40S	MEM_DATA	MEM A DQ<23..16>	15 28
	MEM_A_DQ_BYTE3	MEM_40S	MEM_DATA	MEM A DQ<31..24>	15 28
	MEM_A_DQ_BYTE4	MEM_40S	MEM_DATA	MEM A DQ<39..32>	15 28
	MEM_A_DQ_BYTE5	MEM_40S	MEM_DATA	MEM A DQ<47..40>	15 28
	MEM_A_DQ_BYTE6	MEM_40S	MEM_DATA	MEM A DQ<55..48>	15 28
	MEM_A_DQ_BYTE7	MEM_40S	MEM_DATA	MEM A DQ<63..56>	15 28
	MEM_A_DQ_BYTE0	MEM_40S	MEM_DATA	MEM A DM<0>	15 28
	MEM_A_DQ_BYTE1	MEM_40S	MEM_DATA	MEM A DM<1>	15 28
	MEM_A_DQ_BYTE2	MEM_40S	MEM_DATA	MEM A DM<2>	15 28
	MEM_A_DQ_BYTE3	MEM_40S	MEM_DATA	MEM A DM<3>	15 28
	MEM_A_DQ_BYTE4	MEM_40S	MEM_DATA	MEM A DM<4>	15 28
	MEM_A_DQ_BYTE5	MEM_40S	MEM_DATA	MEM A DM<5>	15 28
	MEM_A_DQ_BYTE6	MEM_40S	MEM_DATA	MEM A DM<6>	15 28
	MEM_A_DQ_BYTE7	MEM_40S	MEM_DATA	MEM A DM<7>	15 28
	MEM_A_DQS0	MEM_70D	MEM_DQS	MEM A DQS P<0>	15 28
	MEM_A_DQS0	MEM_70D	MEM_DQS	MEM A DQS N<0>	15 28
	MEM_A_DQS1	MEM_70D	MEM_DQS	MEM A DQS P<1>	15 28
	MEM_A_DQS1	MEM_70D	MEM_DQS	MEM A DQS N<1>	15 28
	MEM_A_DQS2	MEM_70D	MEM_DQS	MEM A DQS P<2>	15 28
	MEM_A_DQS2	MEM_70D	MEM_DQS	MEM A DQS N<2>	15 28
	MEM_A_DQS3	MEM_70D	MEM_DQS	MEM A DQS P<3>	15 28
	MEM_A_DQS3	MEM_70D	MEM_DQS	MEM A DQS N<3>	15 28
	MEM_A_DQS4	MEM_70D	MEM_DQS	MEM A DQS P<4>	15 28
	MEM_A_DQS4	MEM_70D	MEM_DQS	MEM A DQS N<4>	15 28
	MEM_A_DQS5	MEM_70D	MEM_DQS	MEM A DQS P<5>	15 28
	MEM_A_DQS5	MEM_70D	MEM_DQS	MEM A DQS N<5>	15 28
	MEM_A_DQS6	MEM_70D	MEM_DQS	MEM A DQS P<6>	15 28
	MEM_A_DQS6	MEM_70D	MEM_DQS	MEM A DQS N<6>	15 28
	MEM_A_DQS7	MEM_70D	MEM_DQS	MEM A DQS P<7>	15 28
	MEM_A_DQS7	MEM_70D	MEM_DQS	MEM A DQS N<7>	15 28
	MEM_B_CLK	MEM_70D	MEM_CLK	MEM B CLK P<5..0>	15 27
	MEM_B_CLK	MEM_70D	MEM_CLK	MEM B CLK N<5..0>	15 27
	MEM_B_CKE	MEM_40S	MEM_CTRL	MEM B CKE<3..0>	15 21 27
	MEM_B_CNTRL	MEM_40S	MEM_CTRL	MEM B CS L<3..0>	15 27
	MEM_B_CNTRL	MEM_40S	MEM_CTRL	MEM B ODT<3..0>	15 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<15..0>	15 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM B BA<2..0>	15 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM B RAS L	15 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM B CAS L	15 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM B WE L	15 27
	MEM_B_DQ_BYTE0	MEM_40S	MEM_DATA	MEM B DQ<7..0>	15 28
	MEM_B_DQ_BYTE1	MEM_40S	MEM_DATA	MEM B DQ<15..8>	15 28
	MEM_B_DQ_BYTE2	MEM_40S	MEM_DATA	MEM B DQ<23..16>	15 28
	MEM_B_DQ_BYTE3	MEM_40S	MEM_DATA	MEM B DQ<31..24>	15 28
	MEM_B_DQ_BYTE4	MEM_40S	MEM_DATA	MEM B DQ<39..32>	15 28
	MEM_B_DQ_BYTE5	MEM_40S	MEM_DATA	MEM B DQ<47..40>	15 28
	MEM_B_DQ_BYTE6	MEM_40S	MEM_DATA	MEM B DQ<55..48>	15 28
	MEM_B_DQ_BYTE7	MEM_40S	MEM_DATA	MEM B DQ<63..56>	15 28
	MEM_B_DQ_BYTE0	MEM_40S	MEM_DATA	MEM B DM<0>	15 28
	MEM_B_DQ_BYTE1	MEM_40S	MEM_DATA	MEM B DM<1>	15 28
	MEM_B_DQ_BYTE2	MEM_40S	MEM_DATA	MEM B DM<2>	15 28
	MEM_B_DQ_BYTE3	MEM_40S	MEM_DATA	MEM B DM<3>	15 28
	MEM_B_DQ_BYTE4	MEM_40S	MEM_DATA	MEM B DM<4>	15 28
	MEM_B_DQ_BYTE5	MEM_40S	MEM_DATA	MEM B DM<5>	15 28
	MEM_B_DQ_BYTE6	MEM_40S	MEM_DATA	MEM B DM<6>	15 28
	MEM_B_DQ_BYTE7	MEM_40S	MEM_DATA	MEM B DM<7>	15 28
	MEM_B_DQS0	MEM_70D	MEM_DQS	MEM B DQS P<0>	15 28
	MEM_B_DQS0	MEM_70D	MEM_DQS	MEM B DQS N<0>	15 28
	MEM_B_DQS1	MEM_70D	MEM_DQS	MEM B DQS P<1>	15 28
	MEM_B_DQS1	MEM_70D	MEM_DQS	MEM B DQS N<1>	15 28
	MEM_B_DQS2	MEM_70D	MEM_DQS	MEM B DQS P<2>	15 28
	MEM_B_DQS2	MEM_70D	MEM_DQS	MEM B DQS N<2>	15 28
	MEM_B_DQS3	MEM_70D	MEM_DQS	MEM B DQS P<3>	15 28
	MEM_B_DQS3	MEM_70D	MEM_DQS	MEM B DQS N<3>	15 28
	MEM_B_DQS4	MEM_70D	MEM_DQS	MEM B DQS P<4>	15 28
	MEM_B_DQS4	MEM_70D	MEM_DQS	MEM B DQS N<4>	15 28
	MEM_B_DQS5	MEM_70D	MEM_DQS	MEM B DQS P<5>	15 28
	MEM_B_DQS5	MEM_70D	MEM_DQS	MEM B DQS N<5>	15 28
	MEM_B_DQS6	MEM_70D	MEM_DQS	MEM B DQS P<6>	15 28
	MEM_B_DQS6	MEM_70D	MEM_DQS	MEM B DQS N<6>	15 28
	MEM_B_DQS7	MEM_70D	MEM_DQS	MEM B DQS P<7>	15 28
	MEM_B_DQS7	MEM_70D	MEM_DQS	MEM B DQS N<7>	15 28
	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP VDD	15
	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP GND	15

SYNC MASTER-T27 MLB		SYNC DATE=02/16/2016	
PAGE TITLE			
Memory Constraints			
 Apple Inc.	DRAWING NUMBER	051-8561	
	SIZE	D	
	REVISION	C.0.0	
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		SHEET 70 OF 76	

87654321

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
CLK_PCIE_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20_MIL	?
MCP_PEX_COMP	*	8_MIL	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.3

NEED PCIe Gen1/Gen2 notes!

Analog Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CRT_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CRT	*	20_MIL	?
CRT_2CRT	*	15_MIL	?
CRT_2CLK	*	50_MIL	?
CRT_2SWITCHER	*	250_MIL	?
CRT_SYNC	*	=4X_DIELECTRIC	?
MCP_DAC_COMP	*	=2X_DIELECTRIC	?

CRT signal single-ended impedance varies by location:

- 37.5-ohm from MCP to first termination resistor.
- 50-ohm from first to second termination resistor.
- 75-ohm from output of three-pole filter to connector (if possible).

R/G/B signals should be matched as close as possible and < 10 inches.
SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.4.1.

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
DP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
LVDS_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
MCP_DV_COMP	*	Y	20_MIL	20_MIL	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	*	=3X_DIELECTRIC	?
LVDS	*	=3X_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be matched within 100 mils.
NOTE: NV DG recommends 90 ohm differential for LVDS, but cable/display assume 100 ohm.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 100 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max trace length: LVDS 10 inches, DP 8.5 inches.
SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.4.2

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA	*	=3X_DIELECTRIC	?
SATA_TERM	*	8_MIL	?

SATA intra-pair matching should be 1 ps.
Max trace length: 12 inches for SATA Gen1/Gen2, TBD for SATA Gen3.
SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.6

MCP89 Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
PEG_R2D	PCIE_90D	PCIE	PEG_R2D_P<15..0>
PEG_R2D	PCIE_90D	PCIE	PEG_R2D_N<15..0>
PEG_R2D	PCIE_90D	PCIE	PEG_R2D_C_P<15..0>
PEG_R2D	PCIE_90D	PCIE	PEG_R2D_C_N<15..0>
PEG_D2R	PCIE_90D	PCIE	PEG_D2R_P<15..0>
PEG_D2R	PCIE_90D	PCIE	PEG_D2R_N<15..0>
PEG_D2R	PCIE_90D	PCIE	PEG_D2R_C_P<15..0>
PEG_D2R	PCIE_90D	PCIE	PEG_D2R_C_N<15..0>
PCIE_AP_R2D_P	PCIE_90D	PCIE	PCIE_AP_R2D_P
PCIE_AP_R2D_N	PCIE_90D	PCIE	PCIE_AP_R2D_N
PCIE_AP_R2D_C_P	PCIE_90D	PCIE	PCIE_AP_R2D_C_P
PCIE_AP_R2D_C_N	PCIE_90D	PCIE	PCIE_AP_R2D_C_N
PCIE_AP_D2R_P	PCIE_90D	PCIE	PCIE_AP_D2R_P
PCIE_AP_D2R_N	PCIE_90D	PCIE	PCIE_AP_D2R_N
PCIE_ENET_R2D_P	PCIE_90D	PCIE	PCIE_ENET_R2D_P
PCIE_ENET_R2D_N	PCIE_90D	PCIE	PCIE_ENET_R2D_N
PCIE_ENET_R2D_C_P	PCIE_90D	PCIE	PCIE_ENET_R2D_C_P
PCIE_ENET_R2D_C_N	PCIE_90D	PCIE	PCIE_ENET_R2D_C_N
PCIE_ENET_D2R_P	PCIE_90D	PCIE	PCIE_ENET_D2R_P
PCIE_ENET_D2R_N	PCIE_90D	PCIE	PCIE_ENET_D2R_N
PCIE_ENET_D2R_C_P	PCIE_90D	PCIE	PCIE_ENET_D2R_C_P
PCIE_ENET_D2R_C_N	PCIE_90D	PCIE	PCIE_ENET_D2R_C_N
PCIE_FW_R2D_P	PCIE_90D	PCIE	PCIE_FW_R2D_P
PCIE_FW_R2D_N	PCIE_90D	PCIE	PCIE_FW_R2D_N
PCIE_FW_R2D_C_P	PCIE_90D	PCIE	PCIE_FW_R2D_C_P
PCIE_FW_R2D_C_N	PCIE_90D	PCIE	PCIE_FW_R2D_C_N
PCIE_FW_D2R_P	PCIE_90D	PCIE	PCIE_FW_D2R_P
PCIE_FW_D2R_N	PCIE_90D	PCIE	PCIE_FW_D2R_N
PCIE_FW_D2R_C_P	PCIE_90D	PCIE	PCIE_FW_D2R_C_P
PCIE_FW_D2R_C_N	PCIE_90D	PCIE	PCIE_FW_D2R_C_N
MCP_PEG_REECLK	CLK_PCIE_100D	CLK_PCIE	PEG_CLK100M_P
MCP_PEG_REECLK	CLK_PCIE_100D	CLK_PCIE	PEG_CLK100M_N
MCP_PE1_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_AP_P
MCP_PE1_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_AP_N
MCP_PE2_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_ENET_P
MCP_PE2_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_ENET_N
MCP_PE3_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_P
MCP_PE3_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_N
MCP_PEX_CLK_COMP		MCP_PEX_COMP	MCP_PEX0_TERM
CRT_RED	CRT_50S	CRT	CRT_IG_R_C_PR
CRT_GREEN	CRT_50S	CRT	CRT_IG_G_Y_Y
CRT_BLUE	CRT_50S	CRT	CRT_IG_B_COMP_PB
CRT_SYNC	CRT_50S	CRT_SYNC	CRT_IG_HSYNC
CRT_SYNC	CRT_50S	CRT_SYNC	CRT_IG_VSYNC
MCP_DAC_RSET		MCP_DAC_COMP	MCP_TV_DAC_RSET
MCP_DAC_VREF		MCP_DAC_COMP	MCP_TV_DAC_VREF
TMDS_IG_TXC	DP_90D	DISPLAYPORT	TMDS_IG_TXC_P
TMDS_IG_TXC	DP_90D	DISPLAYPORT	TMDS_IG_TXC_N
TMDS_IG_TXD	DP_90D	DISPLAYPORT	TMDS_IG_TXD_P<5..0>
TMDS_IG_TXD	DP_90D	DISPLAYPORT	TMDS_IG_TXD_N<5..0>
DP_EXT_ML	DP_90D	DISPLAYPORT	DP_IG_ML_P<3..0>
DP_EXT_ML	DP_90D	DISPLAYPORT	DP_IG_ML_N<3..0>
DP_EXT_AUX_CH	DP_90D	DISPLAYPORT	DP_IG_AUX_CH_P
DP_EXT_AUX_CH	DP_90D	DISPLAYPORT	DP_IG_AUX_CH_N
MCP_TMDS0_RSET	MCP_DV_COMP		MCP_TMDS0_RSET
MCP_TMDS0_VPROBE			MCP_TMDS0_VPROBE
LVDS_IG_A_CLK	LVDS_100D	LVDS	LVDS_IG_A_CLK_P
LVDS_IG_A_CLK	LVDS_100D	LVDS	LVDS_IG_A_CLK_N
LVDS_IG_A_DATA	LVDS_100D	LVDS	LVDS_IG_A_DATA_P<2..0>
LVDS_IG_A_DATA	LVDS_100D	LVDS	LVDS_IG_A_DATA_N<2..0>
LVDS_IG_A_DATA3	LVDS_100D	LVDS	LVDS_IG_A_DATA_P<3>
LVDS_IG_A_DATA3	LVDS_100D	LVDS	LVDS_IG_A_DATA_N<3>
LVDS_IG_B_CLK	LVDS_100D	LVDS	LVDS_IG_B_CLK_P
LVDS_IG_B_CLK	LVDS_100D	LVDS	LVDS_IG_B_CLK_N
LVDS_IG_B_DATA	LVDS_100D	LVDS	LVDS_IG_B_DATA_P<2..0>
LVDS_IG_B_DATA	LVDS_100D	LVDS	LVDS_IG_B_DATA_N<2..0>
LVDS_IG_B_DATA3	LVDS_100D	LVDS	LVDS_IG_B_DATA_P<3>
LVDS_IG_B_DATA3	LVDS_100D	LVDS	LVDS_IG_B_DATA_N<3>
MCP_IFPAB_RSET	MCP_DV_COMP		MCP_IFPAB_RSET
MCP_IFPAB_VPROBE			MCP_IFPAB_VPROBE
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_C_P
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_C_N
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_P
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_N
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_P
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_N
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_C_P
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_C_N
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_C_P
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_C_N
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_P
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_N
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_P
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_N
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_C_P
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_C_N
MCP_SATA_TERM		SATA_TERM	MCP_SATA_TERM

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MCP Constraints 1

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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5x_DIELECTRIC	?
CLK_LPC	*	=2x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.7

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_USB_RBIA5	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP, BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.8

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.9

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?
MCP_HDA_COMP	*	8 MIL	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.10

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=1.5x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.11

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=1.5x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.12

MCP89 Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	I _{PC} AD	I _{PC} 55S	I _{PC}	LPC AD<3..0>
	I _{PC} FRAME I	LPC 55S	LPC	LPC FRAME L
	I _{PC} RESET I	LPC 55S	LPC	LPC RESET L
	MCP I _{PC} CLK0	CLK I _{PC} 55S	CLK I _{PC}	LPC CLK33M SMC R
		CLK I _{PC} 55S	CLK I _{PC}	LPC CLK33M SMC
		CLK I _{PC} 55S	CLK I _{PC}	LPC CLK33M LPCPLUS
	USB_EXT_A	USB_90D	USB	USB_EXT_A P
		USB_90D	USB	USB_EXT_A N
		USB_90D	USB	USB_EXT_A MUXED P
		USB_90D	USB	USB_EXT_A MUXED N
	USB_MINI	USB_90D	USB	USB_MINI P
		USB_90D	USB	USB_MINI N
	USB_EXT_D	USB_90D	USB	USB_EXT_D P
		USB_90D	USB	USB_EXT_D N
	USB_CAMERA	USB_90D	USB	USB_CAMERA P
		USB_90D	USB	USB_CAMERA N
	USB_BT	USB_90D	USB	USB_BT P
		USB_90D	USB	USB_BT N
	USB_TPAD	USB_90D	USB	USB_TPAD P
		USB_90D	USB	USB_TPAD N
	USB_IR	USB_90D	USB	USB_IR P
		USB_90D	USB	USB_IR N
	USB_EXTR	USB_90D	USB	USB_EXTR P
		USB_90D	USB	USB_EXTR N
	USB_T57	USB_90D	USB	USB_T57 P
		USB_90D	USB	USB_T57 N
	USB_EXTC	USB_90D	USB	USB_EXTC P
		USB_90D	USB	USB_EXTC N
	USB_SDCARD	USB_90D	USB	USB_SDCARD P
		USB_90D	USB	USB_SDCARD N
	USB_WM	USB_90D	USB	USB_WM P
		USB_90D	USB	USB_WM N
	MCP_USB_RBIAS	MCP_USB_RBIAS		MCP_USB_RBIAS_GND
	SMBUS_MCP_0_CLK	SMB 55S	SMB	SMBUS_MCP_0_CLK
	SMBUS_MCP_0_DATA	SMB 55S	SMB	SMBUS_MCP_0_DATA
	(SMBUS_SMC_MGMT_SCL)	SMB 55S	SMB	SMBUS_MCP_1_CLK
	(SMBUS_SMC_MGMT_SDA)	SMB 55S	SMB	SMBUS_MCP_1_DATA
	HDA_BIT_CLK	HDA 55S	HDA	HDA_BIT_CLK
		HDA 55S	HDA	HDA_BIT_CLK_R
	HDA_SYNC	HDA 55S	HDA	HDA_SYNC
		HDA 55S	HDA	HDA_SYNC_R
	HDA_RST_L	HDA 55S	HDA	HDA_RST_R_L
		HDA 55S	HDA	HDA_RST_L
	HDA_SDIN0	HDA 55S	HDA	HDA_SDIN0
		HDA 55S	HDA	HDA_SDIN_CODEC
	HDA_SDOUT	HDA 55S	HDA	HDA_SDOUT
		HDA 55S	HDA	HDA_SDOUT_R
	MCP_HDA_PULLDN_COMP		MCP_HDA_COMP	MCP_HDA_PULLDN_COMP
	MCP_SUS_CLK	CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK_R
		CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK
	SPI_CLK	SPT 55S	SPT	SPI_CLK_R
		SPT 55S	SPT	SPI_CLK
	SPT_MOSI	SPT 55S	SPT	SPI_MOSI_R
		SPT 55S	SPT	SPI_MOSI
	SPT_MISO	SPT 55S	SPT	SPI_MISO
	SPT_CS0	SPT 55S	SPT	SPI_CS0_R_L
		SPT 55S	SPT	SPI_CS0_L
		SPT 55S	SPT	SPI_MLB_CLK
		SPT 55S	SPT	SPI_MLB_MOSI
		SPT 55S	SPT	SPI_MLB_MISO
		SPT 55S	SPT	SPI_MLB_CS_L
		SPT 55S	SPT	SPI_ALT_CLK
		SPT 55S	SPT	SPI_ALT_MOSI
		SPT 55S	SPT	SPI_ALT_MISO
		SPT 55S	SPT	SPI_ALT_CS_L

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MCP RGMII (Ethernet) Constraints

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_BUF0_CLK	*	=3:1_SPACING	?
ENET_MII	*	12 MIL	?

88E1116R (Ethernet PHY) Constraints

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	25 MIL	?

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	MCP_MII_COMP_VDD	MCP_MII_COMP		MCP MII COMP VDD 18
	MCP_MII_COMP_GND	MCP_MII_COMP		MCP MII COMP GND 18
	MCP_CLK25M_BUF0	ENET_MII_55S	MCP_BUF0_CLK	MCP CLK25M BUF0 R 9
	MCP_CLK25M_CKXTAL1	ENET_MII_55S	MCP_BUF0_CLK	RTL8211 CLK25M CKXTAL1 31
	ENET_INTR_L	ENET_MII_55S	ENET_MII	ENET INTR L 18
	ENET_MDIO	ENET_MII_55S	ENET_MII	ENET MDIO 18 31
	ENET_MDC	ENET_MII_55S	ENET_MII	ENET MDC 9 31
	ENET_PWRDWN_L	ENET_MII_55S	ENET_MII	ENET PWRDWN L 18
	ENET_CLK125M_RXCLK_R	ENET_MII_55S	ENET_MII	ENET CLK125M RXCLK R 31
	ENET_CLK125M_RXCLK	ENET_MII_55S	ENET_MII	ENET CLK125M RXCLK 31
	ENET_RXD_R<3..0>	ENET_MII_55S	ENET_MII	ENET RXD R<3..0> 31
	ENET_RXD<0>	ENET_MII_55S	ENET_MII	ENET RXD<0> 31
	ENET_RXD<3..1>	ENET_MII_55S	ENET_MII	ENET RXD<3..1> 31
	ENET_RX_CTRL	ENET_MII_55S	ENET_MII	ENET RX CTRL 31
	ENET_RXCTL_R	ENET_MII_55S	ENET_MII	ENET RXCTL R 31
	ENET_CLK125M_TXCLK_R	ENET_MII_55S	ENET_MII	ENET CLK125M TXCLK R 31
	ENET_CLK125M_TXCLK	ENET_MII_55S	ENET_MII	ENET CLK125M TXCLK 9 31
	ENET_TXD<0>	ENET_MII_55S	ENET_MII	ENET TXD<0> 9 31
	ENET_TXD<3..1>	ENET_MII_55S	ENET_MII	ENET TXD<3..1> 9 31
	ENET_TX_CTRL	ENET_MII_55S	ENET_MII	ENET TX CTRL 9 31
	ENET_RESET_L	ENET_MII_55S	ENET_MII	ENET RESET L 9 31
	ENET_MDI_P<3..0>	ENET_MDI_100D	ENET_MDI	ENET MDI P<3..0> 31 32
	ENET_MDI_N<3..0>	ENET_MDI_100D	ENET_MDI	ENET MDI N<3..0> 31 32
	ENET_MDI_TRAN_P<3..0>	ENET_MDI_100D	ENET_MDI	ENET MDI TRAN P<3..0> 31 32
	ENET_MDI_TRAN_N<3..0>	ENET_MDI_100D	ENET_MDI	ENET MDI TRAN N<3..0> 32

A

D

C

B

A

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
MEM_POWER	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DATA	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_POWER	*	PWR_P2MM
MEM_CMD	MEM_POWER	*	PWR_P2MM
MEM_CTRL	MEM_POWER	*	PWR_P2MM
MEM_DATA	MEM_POWER	*	PWR_P2MM
MEM_DQS	MEM_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PwR_P2MM
SATA	SB_POWER	*	PwR_P2MM
USB	SB_POWER	*	PwR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_FSB	GND	*	GND_P2MM
CPU_COMP	GND	*	GND_P2MM
CPU_GTLREF	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LVDS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

MCP Fanout Constraint Relaxations

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	5.8 MM OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_MEM_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_MII_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_USB_RBIAS OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.25 MM OVERRIDE	250 MIL OVERRIDE	OVERRIDE	OVERRIDE

Misc Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	(PCIE_AP)	CLK_PCIE 100D	CLK_PCIE	PCIE CLK100M AP CONN P
		CLK_PCIE 100D	CLK_PCIE	PCIE CLK100M AP CONN N
	(USB_EXT_A)	USB_90D	USB	USB_EXT_A MUXED_P
	(USB_EXT_A)	USB_90D	USB	USB_EXT_A MUXED_N
	(USB_EXT_A)	USB_90D	USB	USB_LT1_P
	(USB_EXT_A)	USB_90D	USB	USB_LT1_N
	(USB_TPAD)	USB_90D	USB	USB_TPAD_R_P
	(USB_TPAD)	USB_90D	USB	USB_TPAD_R_N
	(USB_CAMERA)	USB_90D	USB	USB_CAMERA_CONN_P
	(USB_CAMERA)	USB_90D	USB	USB_CAMERA_CONN_N
		USB_90D	USB	USB_LT2_P
		USB_90D	USB	USB_LT2_N
		ENET_MDT 100D	ENET_CONN	ENETCONN P<3..0>
		ENET_MDT 100D	ENET_CONN	ENETCONN N<3..0>
		SATA_90D	SATA	SATA_ODD_R2D_UF_P
		SATA_90D	SATA	SATA_ODD_R2D_UF_N
		SATA_90D	SATA	SATA_ODD_D2R_UF_P
		SATA_90D	SATA	SATA_ODD_D2R_UF_N
		SATA_90D	SATA	SATA_HDD_D2R_FILT_P
		SATA_90D	SATA	SATA_HDD_D2R_FILT_N
		SATA_90D	SATA	SATA_HDD_D2R_UF_P
		SATA_90D	SATA	SATA_HDD_D2R_UF_N
		SATA_90D	SATA	SATA_HDD_R2D_UF_P
		SATA_90D	SATA	SATA_HDD_R2D_UF_N
		SATA_90D	SATA	SATA_HDD_D2R_RDRV_IN_P
		SATA_90D	SATA	SATA_HDD_D2R_RDRV_IN_N
		SATA_90D	SATA	SATA_HDD_R2D_RDRV_IN_P
		SATA_90D	SATA	SATA_HDD_R2D_RDRV_IN_N
		SATA_90D	SATA	SATA_HDD_D2R_RDRV_OUT_P
		SATA_90D	SATA	SATA_HDD_D2R_RDRV_OUT_N
		SATA_90D	SATA	SATA_HDD_R2D_RDRV_OUT_P
		SATA_90D	SATA	SATA_HDD_R2D_RDRV_OUT_N
		SATA_90D	SATA	SATA_HDD_D2R_NORDRV_P
		SATA_90D	SATA	SATA_HDD_D2R_NORDRV_N
		SATA_90D	SATA	SATA_HDD_R2D_NORDRV_P
		SATA_90D	SATA	SATA_HDD_R2D_NORDRV_N
	PCIE_AP_D2R	PCIE_90D	PCIE	CONN_PCIE_MINI_D2R_P
		PCIE_90D	PCIE	CONN_PCIE_MINI_D2R_N
	PCIE_AP_R2D	PCIE_90D	PCIE	CONN_PCIE_MINI_R2D_P
		PCIE_90D	PCIE	CONN_PCIE_MINI_R2D_N
	USB_BT	USB_90D	USB	CONN_USB2_BT_P
		USB_90D	USB	CONN_USB2_BT_N
	LVDS_IG_A_CLK	LVDS_100D	LVDS	LVDS_IG_A_CLK_F_P
		LVDS_100D	LVDS	LVDS_IG_A_CLK_F_N
	MCP_PE1_REFCLK	CLK_PCIE 100D	CLK_PCIE	PCIE_CLK100M_MINI_CONN_P
		CLK_PCIE 100D	CLK_PCIE	PCIE_CLK100M_MINI_CONN_N
USB	USB_EXT_A	USB_90D	USB	CONN_USB_EXT_A_P
USB		USB_90D	USB	CONN_USB_EXT_A_N
USB	USB_EXT_B	USB_90D	USB	CONN_USB_EXT_B_P
USB		USB_90D	USB	CONN_USB_EXT_B_N

Power Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	CPUTHMSNS_D2	THERM_1T01_55S	THERM	CPUTHMSNS D1 P	41
		THERM_1T01_55S	THERM	CPUTHMSNS D1 N	41
	CPUTHMSNS_D2	THERM_1T01_55S	THERM	CPUTHMSNS D2 P	41
		THERM_1T01_55S	THERM	CPUTHMSNS D2 N	41
	CPU_THERMD	THERM_1T01_55S	THERM	CPU_THERMD P	10 41
		THERM_1T01_55S	THERM	CPU_THERMD N	10 41
	MCPTHMSNS_D2	THERM_1T01_55S	THERM	MCPTHMSNS D2 P	
		THERM_1T01_55S	THERM	MCPTHMSNS D2 N	
	MCP_THMDIODE	THERM_1T01_55S	THERM	MCP_THMDIODE P	19 41
		THERM_1T01_55S	THERM	MCP_THMDIODE N	19 41
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_1V5 S3 P	
		SENSE_1T01_55S	SENSE	ISNS_1V5 S3 N	
		SENSE_1T01_55S	SENSE	ISNS_1V5 S3 R P	
		SENSE_1T01_55S	SENSE	ISNS_1V5 S3 R N	
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_AIRPORT P	30 46
		SENSE_1T01_55S	SENSE	ISNS_AIRPORT N	30 46
		SENSE_1T01_55S	SENSE	ISNS_AIRPORT R P	46
		SENSE_1T01_55S	SENSE	ISNS_AIRPORT R N	
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_HDD P	33 46
		SENSE_1T01_55S	SENSE	ISNS_HDD N	33 46
		SENSE_1T01_55S	SENSE	ISNS_HDD R P	46
		SENSE_1T01_55S	SENSE	ISNS_HDD R N	46
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_LCDBKLT P	46 67
		SENSE_1T01_55S	SENSE	ISNS_LCDBKLT N	46 67
		SENSE_1T01_55S	SENSE	ISNS_LCDBKLT R P	
		SENSE_1T01_55S	SENSE	ISNS_LCDBKLT R N	
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_ODD P	33
		SENSE_1T01_55S	SENSE	ISNS_ODD N	33 46
		SENSE_1T01_55S	SENSE	ISNS_ODD R P	46
		SENSE_1T01_55S	SENSE	ISNS_ODD R N	46
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_CPUVTT P	40
		SENSE_1T01_55S	SENSE	ISNS_CPUVTT N	40
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	MCPCORES0_VSEN P	72 99
		SENSE_1T01_55S	SENSE	MCPCORES0_VSEN N	72 99
			MEM_POWER	PP1V5R1V35_S3	7 8
			SR_POWER	PP3V3_S5	7 8 62
			SR_POWER	PP3V3_S0	7 8 62
			SR_POWER	PP1V5_S0	7 8 62
			GND	GND	
					

Audio Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
		DIEFFPAIR	AUDIO	AUD SPKRAMP LIN P
		DIEFFPAIR	AUDIO	AUD SPKRAMP LIN N
		DIEFFPAIR	AUDIO	AUD SPKRAMP SUBIN P
		DIEFFPAIR	AUDIO	AUD SPKRAMP SUBIN N
		DIEFFPAIR	AUDIO	AUD SPKRAMP RIN P
		DIEFFPAIR	AUDIO	AUD SPKRAMP RIN N
		DIEFFPAIR	AUDIO	SSM2315L P
		DIEFFPAIR	AUDIO	SSM2315L N
		DIEFFPAIR	AUDIO	SSM2315S P
		DIEFFPAIR	AUDIO	SSM2315S N
		DIEFFPAIR	AUDIO	SSM2315R P
		DIEFFPAIR	AUDIO	SSM2315R N
	SPK_OUT	DIEFFPAIR	AUDIO	SPKRCONN L OUT P
		DIEFFPAIR	AUDIO	SPKRCONN L OUT N
	SPK_OUT	DIEFFPAIR	AUDIO	SPKRCONN S OUT P
		DIEFFPAIR	AUDIO	SPKRCONN S OUT N
	SPK_OUT	DIEFFPAIR	AUDIO	SPKRCONN R OUT P
		DIEFFPAIR	AUDIO	SPKRCONN R OUT N
		DIEFFPAIR	AUDIO	BI MIC P
		DIEFFPAIR	AUDIO	BI MIC N
		DIEFFPAIR	AUDIO	HS MIC P
		DIEFFPAIR	AUDIO	HS MIC N

GRAPHICS NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
     	(DP_EXT_ML)	DP_96D	DISPLAYPORT	DP EXT ML P<3..0>	9 66
		DP_96D	DISPLAYPORT	DP EXT ML N<3..0>	9 66
		DP_96D	DISPLAYPORT	DP EXT ML C P<3..0>	66
		DP_96D	DISPLAYPORT	DP EXT ML C N<3..0>	66
		DP_96D	DISPLAYPORT	DP EXT ML F P<3..0>	66
		DP_96D	DISPLAYPORT	DP EXT ML F N<3..0>	66
   	(DP_EXT_AUX_CH)	DP_96D	DISPLAYPORT	DP EXT AUX CH C P	9 66
		DP_96D	DISPLAYPORT	DP EXT AUX CH C N	9 66
		DP_96D	DISPLAYPORT	DP EXT DDC DATA	65
		DP_96D	DISPLAYPORT	DP EXT DDC CLK	65

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K87 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM	NO_TYPE, BGA_P4M	MM	15.5.1

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	0.160MM	30 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OH1_SE	TOP,BOTTOM	Y	0.090 MM	0.090 MM			
55_OH1_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OH1_SE	TOP, BOTTOM	Y	0.115 MM	0.115 MM			
50_OH1_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
4B_OHM_SE	TOP, BOTTOM	Y	0.165 MM	0.100 MM			
4B_OHM_SE	*	Y	0.126 MM	0.100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHR_SE	TOP,BOTTOM	Y	0.310 MM	0.310 MM			
27P4_OHR_SE	*	Y	0.222 MM	0.222 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
70_OHM_DIFF	ISL3, ISL4, ISL6, ISL10	Y	0.151 MM	0.100 MM	=STANDARD	0.224 MM	0.224 MM
70_OHM_DIFF	TOP, BOTTOM	Y	0.185 MM	0.100 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHR_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHR_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.095 MM	0.095 MM		0.234 MM	0.234 MM
90_OHR_DIFF	TOP, BOTTOM	Y	0.112 MM	0.112 MM		0.226 MM	0.226 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
180_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
180_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.075 MM	0.075 MM		0.244 MM	0.244 MM
180_OHM_DIFF	TOP, BOTTOM	Y	0.091 MM	0.091 MM		0.238 MM	0.238 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P3MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?
BGA_P3MM	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:5:1_SPACING	"	0.15 MM	?
2:1:1_SPACING	"	0.2 MM	?
2.5:1:1_SPACING	"	0.25 MM	?
3:1:1_SPACING	"	0.3 MM	?
4:1:1_SPACING	"	0.4 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1_5X_DIELECTRIC	TOP, BOTTOM	0.165 MM	?
2X_DIELECTRIC	TOP, BOTTOM	0.149 MM	?
3X_DIELECTRIC	TOP, BOTTOM	0.219 MM	?
4X_DIELECTRIC	TOP, BOTTOM	0.289 MM	?
5X_DIELECTRIC	TOP, BOTTOM	0.359 MM	?
1_5X_DIELECTRIC	"	0.695 MM	?
2X_DIELECTRIC	"	0.126 MM	?
3X_DIELECTRIC	"	0.189 MM	?
4X_DIELECTRIC	"	0.252 MM	?
5X_DIELECTRIC	"	0.315 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
MEM_CLK	*	BGA_P1MM	BGA_P2MM
CLK_FSB	*	BGA_P1MM	BGA_P2MM
CLK_LFC	*	BGA_P1MM	BGA_P2MM
CLK_PCIE	*	BGA_P1MM	BGA_P2MM
CLK_SLOW	*	BGA_P1MM	BGA_P2MM
FSB_D5TB	FSB_D5TB	BGA_P1MM	BGA_P3MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_4BS	BGA_P1MM	STANDARD

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